FOUNDATIONS

CHAPTER 1

1.1 Introduction

The field of electronics is one of the great success stories of the 20th century. From the crude spark-gap transmitters and "cat's-whisker" detectors at its beginning, the first half-century brought an era of vacuum-tube electronics that developed considerable sophistication and found ready application in areas such as communications, navigation, instrumentation, control, and computation. The latter half-century brought "solid-state" electronics – first as discrete transistors, then as magnificent arrays of them within "integrated circuits" (ICs) – in a flood of stunning advances that shows no sign of abating. Compact and inexpensive consumer products now routinely contain many millions of transistors in VLSI (very large-scale integration) chips, combined with elegant optoelectronics (displays, lasers, and so on); they can process sounds, images, and data, and (for example) permit wireless networking and shirt-pocket access to the pooled capabilities of the Internet. Perhaps as noteworthy is the pleasant trend toward increased performance per dollar.[1] The cost of an electronic microcircuit routinely decreases to a fraction of its initial cost as the manufacturing process is perfected (see Figure 10.87 for an example). In fact, it is often the case that the panel controls and cabinet hardware of an instrument cost more than the electronics inside.

On reading of these exciting new developments in electronics, you may get the impression that you should be able to construct powerful, elegant, yet inexpensive, little gadgets to do almost any conceivable task – all you need to know is how all these miracle devices work. If you've had that feeling, this book is for you. In it we have attempted to convey the excitement and know-how of the subject of electronics.

In this chapter we begin the study of the laws, rules of thumb, and tricks that constitute the art of electronics as we see it. It is necessary to begin at the beginning – with talk of voltage, current, power, and the components that make up electronic circuits. Because you can't touch, see, smell, or hear electricity, there will be a certain amount of abstraction (particularly in the first chapter), as well as some dependence on such visualizing instruments as oscilloscopes and voltmeters. In many ways the first chapter is also the most mathematical, in spite of our efforts to keep mathematics to a minimum in order to foster a good intuitive understanding of circuit design and behavior.

In this new edition we've included some intuition-aiding approximations that our students have found helpful. And, by introducing one or two "active" components ahead of their time, we're able to jump directly into some applications that are usually impossible in a traditional textbook "passive electronics" chapter; this will keep things interesting, and even exciting.

Once we have considered the foundations of electronics, we will quickly get into the active circuits (amplifiers, oscillators, logic circuits, etc.) that make electronics the exciting field it is. The reader with some background in electronics may wish to skip over this chapter, since it assumes no prior knowledge of electronics. Further generalizations at this time would be pointless, so let's just dive right in.

1.2 Voltage, current, and resistance

1.2.1 Voltage and current

There are two quantities that we like to keep track of in electronic circuits: voltage and current. These are usually changing with time; otherwise nothing interesting is happening.

Voltage (symbol V or sometimes E). Officially, the voltage between two points is the cost in energy (work done) required to move a unit of positive charge from the more negative point (lower potential) to the more positive point (higher potential). Equivalently, it is the energy released when a unit charge moves "downhill" from the higher potential to the lower.[2] Voltage is also called

[1] A mid-century computer (the IBM 650) cost $300,000, weighed 2.7 tons, and contained 126 lamps on its control panel; in an amusing reversal, a contemporary energy-efficient lamp contains a computer of greater capability *within its base*, and costs about $10.

[2] These are the *definitions*, but hardly the way circuit designers think of voltage. With time, you'll develop a good intuitive sense of what voltage really is, in an electronic circuit. Roughly (*very* roughly) speaking, voltages are what you apply to cause currents to flow.

potential difference or *electromotive force* (EMF). The unit of measure is the *volt*, with voltages usually expressed in volts (V), kilovolts ($1\,\text{kV} = 10^3$ V), millivolts ($1\,\text{mV} = 10^{-3}$ V), or microvolts ($1\,\mu\text{V} = 10^{-6}$ V) (see the box on prefixes). A joule (J) of work is done in moving a coulomb (C) of charge through a potential difference of 1 V. (The coulomb is the unit of electric charge, and it equals the charge of approximately 6×10^{18} electrons.) For reasons that will become clear later, the opportunities to talk about nanovolts ($1\,\text{nV} = 10^{-9}$ V) and megavolts ($1\,\text{MV} = 10^6$ V) are rare.

Current (symbol I). Current is the rate of flow of electric charge past a point. The unit of measure is the ampere, or amp, with currents usually expressed in amperes (A), milliamperes ($1\,\text{mA} = 10^{-3}$ A), microamperes ($1\,\mu\text{A} = 10^{-6}$ A), nanoamperes ($1\,\text{nA} = 10^{-9}$ A), or occasionally picoamperes ($1\,\text{pA} = 10^{-12}$ A). A current of 1 amp equals a flow of 1 coulomb of charge per second. By convention, current in a circuit is considered to flow from a more positive point to a more negative point, even though the actual electron flow is in the opposite direction.

Important: from these definitions you can see that currents flow *through* things, and voltages are applied (or appear) *across* things. So you've got to say it right: always refer to the voltage *between* two points or *across* two points in a circuit. Always refer to current *through* a device or connection in a circuit.

To say something like "the voltage through a resistor . . . " is nonsense. However, we do frequently speak of the voltage *at a point* in a circuit. This is always understood to mean the voltage between that point and "ground," a common point in the circuit that everyone seems to know about. Soon you will, too.

We *generate* voltages by doing work on charges in devices such as batteries (conversion of electrochemical energy), generators (conversion of mechanical energy by magnetic forces), solar cells (photovoltaic conversion of the energy of photons), etc. We *get* currents by placing voltages across things.

At this point you may well wonder how to "see" voltages and currents. The single most useful electronic instrument is the oscilloscope, which allows you to look at voltages (or occasionally currents) in a circuit as a function of time.[3] We will deal with oscilloscopes, and also voltmeters, when we discuss signals shortly; for a preview see Appendix O, and the multimeter box later in this chapter.

[3] It has been said that engineers in other disciplines are envious of electrical engineers, because we have such a splendid visualization tool.

In real circuits we connect things together with wires (metallic conductors), each of which has the same voltage on it everywhere (with respect to ground, say).[4] We mention this now so that you will realize that an actual circuit doesn't have to look like its schematic diagram, because wires can be rearranged.

Here are some simple rules about voltage and current:

1. The sum of the currents into a point in a circuit equals the sum of the currents out (conservation of charge). This is sometimes called Kirchhoff's current law (KCL). Engineers like to refer to such a point as a *node*. It follows that, for a series circuit (a bunch of two-terminal things all connected end-to-end), the current is the same everywhere.

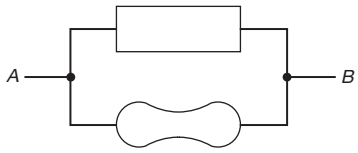


Figure 1.1. Parallel connection.

2. Things hooked in parallel (Figure 1.1) have the same voltage across them. Restated, the sum of the "voltage drops" from A to B via one path through a circuit equals the sum by any other route, and is simply the voltage between A and B. Another way to say it is that the sum of the voltage drops around any closed circuit is zero. This is Kirchhoff's voltage law (KVL).
3. The power (energy per unit time) consumed by a circuit device is

$$P = VI \tag{1.1}$$

This is simply (energy/charge) × (charge/time). For V in volts and I in amps, P comes out in watts. A watt is a joule per second (1W = 1 J/s). So, for example, the current flowing through a 60W lightbulb running on 120 V is 0.5 A.

Power goes into heat (usually), or sometimes mechanical work (motors), radiated energy (lamps, transmitters), or stored energy (batteries, capacitors, inductors). Managing the heat load in a complicated system (e.g., a large computer, in which many kilowatts of electrical energy are converted to heat, with the energetically insignificant byproduct of a few pages of computational results) can be a crucial part of the system design.

[4] In the domain of high frequencies or low impedances, that isn't strictly true, and we will have more to say about this later, and in Chapter *1x*. For now, it's a good approximation.

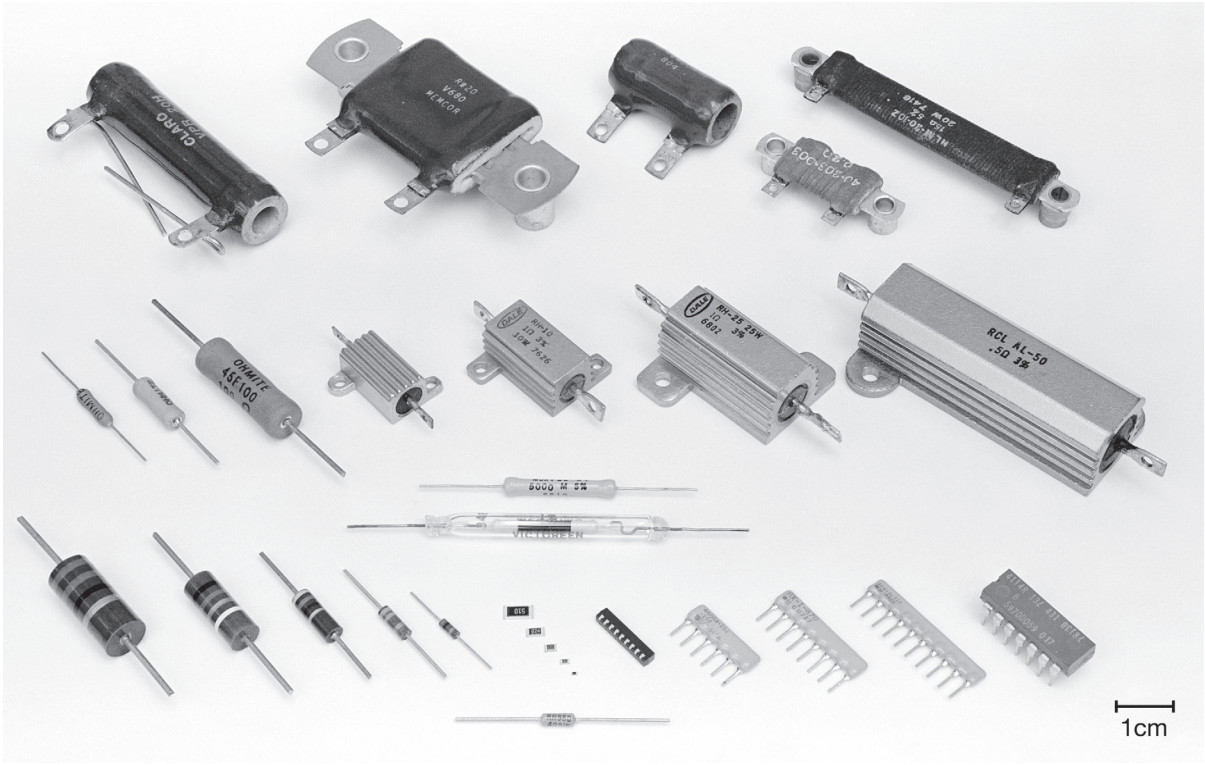


Figure 1.2. A selection of common resistor types. Top row, left to right (wirewound ceramic power resistors): 20W vitreous enamel with leads, 20W with mounting studs, 30W vitreous enamel, 5W and 20W with mounting studs. Middle row (wirewound power resistors): 1W, 3W, and 5W axial ceramic; 5W, 10W, 25W, and 50W conduction-cooled ("Dale-type"). Bottom row: 2W, 1W, $\frac{1}{2}$W, $\frac{1}{4}$W, and $\frac{1}{8}$W carbon composition; surface-mount thick-film (2010, 1206, 0805, 0603, and 0402 sizes); surface-mount resistor array; 6-, 8-, and 10-pin single in-line package arrays; dual in-line package array. The resistor at bottom is the ubiquitous RN55D $\frac{1}{4}$W, 1% metal-film type; and the pair of resistors above are Victoreen high-resistance types (glass, 2 GΩ; ceramic, 5 GΩ).

Soon, when we deal with periodically varying voltages and currents, we will have to generalize the simple equation $P = VI$ to deal with *average* power, but it's correct as a statement of *instantaneous* power just as it stands.

Incidentally, don't call current "amperage"; that's strictly bush league.[5] The same caution will apply to the term "ohmage"[6] when we get to resistance in the next section.

1.2.2 Relationship between voltage and current: resistors

This is a long and interesting story. It is the heart of electronics. Crudely speaking, the name of the game is to make and use gadgets that have interesting and useful I-versus-V characteristics. Resistors (I simply proportional to V), capacitors (I proportional to rate of change of V), diodes (I flows in only one direction), thermistors (temperature-dependent resistor), photoresistors (light-dependent resistor), strain gauges (strain-dependent resistor), etc., are examples. Perhaps more interesting still are *three-terminal* devices, such as transistors, in which the current that can flow between a pair of terminals is controlled by the voltage applied to a third terminal. We will gradually get into some of these exotic devices; for now, we will start with the most mundane (and most widely used) circuit element, the resistor (Figure 1.3).



Figure 1.3. Resistor.

A. Resistance and resistors

It is an interesting fact that the current through a metallic conductor (or other partially conducting material) is proportional to the voltage across it. (In the case of wire

[5] Unless you're a power engineer working with giant 13 kV transformers and the like – those guys are allowed to say amperage.

[6] . . . also, Dude, "ohmage" is not the preferred nomenclature: *resistance*, please.

PREFIXES

Multiple	*Prefix*	*Symbol*	*Derivation*
10^{24}	yotta	Y	end-1 of Latin alphabet, hint of Greek *iota*
10^{21}	zetta	Z	end of Latin alphabet, hint of Greek *zeta*
10^{18}	exa	E	Greek *hexa* (six: power of 1000)
10^{15}	peta	P	Greek *penta* (five: power of 1000)
10^{12}	tera	T	Greek *teras* (monster)
10^{9}	giga	G	Greek *gigas* (giant)
10^{6}	mega	M	Greek *megas* (great)
10^{3}	kilo	k	Greek *khilioi* (thousand)
10^{-3}	milli	m	Latin *milli* (thousand)
10^{-6}	micro	μ	Greek *mikros* (small)
10^{-9}	nano	n	Greek *nanos* (dwarf)
10^{-12}	pico	p	from Italian/Spanish *piccolo/pico* (small)
10^{-15}	femto	f	Danish/Norwegian *femten* (fifteen)
10^{-18}	atto	a	Danish/Norwegian *atten* (eighteen)
10^{-21}	zepto	z	end of Latin alphabet, mirrors *zetta*
10^{-24}	yocto	y	end-1 of Latin alphabet, mirrors *yotta*

These prefixes are universally used to scale units in science and engineering. Their etymological derivations are a matter of some controversy and should not be considered historically reliable. When abbreviating a unit with a prefix, the symbol for the unit follows the prefix without space. Be careful about uppercase and lowercase letters (especially m and M) in both prefix and unit: 1 mW is a milliwatt, or one-thousandth of a watt; 1 MHz is a megahertz or 1 million hertz. In general, units are spelled with lowercase letters, even when they are derived from proper names. The unit name is not capitalized when it is spelled out and used with a prefix, only when abbreviated. Thus: hertz and kilohertz, but Hz and kHz; watt, milliwatt, and megawatt, but W, mW, and MW.

conductors used in circuits, we usually choose a thick-enough gauge of wire so that these "voltage drops" will be negligible.) This is by no means a universal law for all objects. For instance, the current through a neon bulb is a highly nonlinear function of the applied voltage (it is zero up to a critical voltage, at which point it rises dramatically). The same goes for a variety of interesting special devices – diodes, transistors, lightbulbs, etc. (If you are interested in understanding why metallic conductors behave this way, read §§4.4–4.5 in Purcell and Morin's splendid text *Electricity and Magnetism*).

A resistor is made out of some conducting stuff (carbon, or a thin metal or carbon film, or wire of poor conductivity), with a wire or contacts at each end. It is characterized by its resistance:

$$R = V/I; \tag{1.2}$$

R is in ohms for V in volts and I in amps. This is known as Ohm's law. Typical resistors of the most frequently used type (metal-oxide film, metal film, or carbon film) come in values from 1 ohm (1 Ω) to about 10 megohms (10 MΩ). Resistors are also characterized by how much power they can safely dissipate (the most commonly used ones are rated at 1/4 or 1/8 W), their physical size,[7] and by other parameters such as tolerance (accuracy), temperature coefficient, noise, voltage coefficient (the extent to which R depends on applied V), stability with time, inductance, etc. See the box on resistors, Chapter *1x*, and Appendix C for further details. Figure 1.2 shows a collection of resistors, with most of the available morphologies represented.

Roughly speaking, resistors are used to convert a

[7] The sizes of *chip resistors* and other components intended for surface mounting are specified by a four-digit size code, in which each pair of digits specifies a dimension in units of 0.010″ (0.25 mm). For example, an 0805 size resistor is 2 mm×1.25 mm, or 80 mils×50 mils (1 mil is 0.001″); the height must be separately specified. To add confusion to this simple scheme, the four-digit size code may instead be *metric* (sometimes without saying so!), in units of 0.1 mm: thus an "0805" (English) is also a "2012" (metric).

RESISTORS

Resistors are truly ubiquitous. There are almost as many types as there are applications. Resistors are used in amplifiers as loads for active devices, in bias networks, and as feedback elements. In combination with capacitors they establish time constants and act as filters. They are used to set operating currents and signal levels. Resistors are used in power circuits to reduce voltages by dissipating power, to measure currents, and to discharge capacitors after power is removed. They are used in precision circuits to establish currents, to provide accurate voltage ratios, and to set precise gain values. In logic circuits they act as bus and line terminators and as "pullup" and "pull-down" resistors. In high-voltage circuits they are used to measure voltages and to equalize leakage currents among diodes or capacitors connected in series. In radiofrequency (RF) circuits they set the bandwidth of resonant circuits, and they are even used as coil forms for inductors.

Resistors are available with resistances from 0.0002 Ω through 10^{12} Ω, standard power ratings from 1/8 watt through 250 watts, and accuracies from 0.005% through 20%. Resistors can be made from metal films, metal-oxide films, or carbon films; from carbon-composition or ceramic-composition moldings; from metal foil or metal wire wound on a form; or from semiconductor elements similar to field-effect transistors (FETs). The most commonly used resistor type is formed from a carbon, metal, or oxide film, and comes in two widely used "packages": the cylindrical *axial-lead* type (typified by the generic RN55D 1% 1/4 W metal-film resistor),[8] and the much smaller *surface-mount* "chip resistor." These common types come in 5%, 2%, and 1% tolerances, in a standard set of values ranging from 1 Ω to 10 MΩ. The 1% types have 96 values per decade, whereas the 2% and 5% types have 48 and 24 values per decade (see Appendix C). Figure 1.2 illustrates most of the common resistor packages.

Resistors are so easy to use and well behaved that they're often taken for granted. They're not perfect, though, and you should be aware of some of their limitations so that you won't be surprised someday. The principal defects are variations in resistance with temperature, voltage, time, and humidity. Other defects relate to inductance (which may be serious at high frequencies), the development of thermal hot spots in power applications, or electrical noise generation in low-noise amplifiers. We treat these in the advanced Chapter *1x*.

voltage to a current, and vice versa. This may sound awfully trite, but you will soon see what we mean.

B. Resistors in series and parallel

From the definition of R, some simple results follow:

1. The resistance of two resistors in series (Figure 1.4) is

$$R = R_1 + R_2. \tag{1.3}$$

   By putting resistors in series, you always get a *larger* resistor.

2. The resistance of two resistors in parallel (Figure 1.5) is

$$R = \frac{R_1 R_2}{R_1 + R_2} \quad \text{or} \quad R = \frac{1}{\dfrac{1}{R_1} + \dfrac{1}{R_2}}. \tag{1.4}$$

By putting resistors in parallel, you always get a *smaller* resistor. Resistance is measured in ohms (Ω), but in practice we frequently omit the Ω symbol when referring to resistors that are more than 1000 Ω (1 kΩ). Thus, a 4.7 kΩ resistor is often referred to as a 4.7k resistor, and a 1 MΩ resistor as a 1M resistor (or 1 meg).[9] If these preliminaries bore you, please have patience – we'll soon get to numerous amusing applications.



Figure 1.4. Resistors in series.

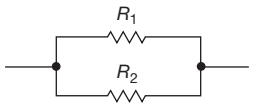


Figure 1.5. Resistors in parallel.

Exercise 1.1. You have a 5k resistor and a 10k resistor. What is their combined resistance (a) in series and (b) in parallel?

Exercise 1.2. If you place a 1 ohm resistor across a 12 volt car battery, how much power will it dissipate?

Exercise 1.3. Prove the formulas for series and parallel resistors.

[8] Conservatively rated at 1/8 watt in its RN55 military grade ("MIL-spec"), but rated at 1/4 watt in its CMF-55 industrial grade.

[9] A popular "international" alternative notation replaces the decimal point with the unit multiplier, thus 4k7 or 1M0. A 2.2 Ω resistor becomes 2R2. There is an analogous scheme for capacitors and inductors.

Exercise 1.4. Show that several resistors in parallel have resistance

$$R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots} \tag{1.5}$$

Beginners tend to get carried away with complicated algebra in designing or trying to understand electronics. Now is the time to begin learning intuition and shortcuts. Here are a couple of good tricks:

Shortcut #1 A large resistor in series (parallel) with a small resistor has the resistance of the larger (smaller) one, roughly. So you can "trim" the value of a resistor up or down by connecting a second resistor in series or parallel: to trim *up*, choose an available resistor value below the target value, then add a (much smaller) series resistor to make up the difference; to trim *down*, choose an available resistor value above the target value, then connect a (much larger) resistor in parallel. For the latter you can approximate with proportions – to lower the value of a resistor by 1%, say, put a resistor 100 times as large in parallel.[10]

Shortcut #2 Suppose you want the resistance of 5k in parallel with 10k. If you think of the 5k as two 10k's in parallel, then the whole circuit is like three 10k's in parallel. Because the resistance of n equal resistors in parallel is 1/nth the resistance of the individual resistors, the answer in this case is 10k/3, or 3.33k. This trick is handy because it allows you to analyze circuits quickly in your head, without distractions. We want to encourage mental designing, or at least "back-of-the-envelope" designing, for idea brainstorming.

Some more home-grown philosophy: there is a tendency among beginners to want to compute resistor values and other circuit component values to many significant places, particularly with calculators and computers that readily oblige. There are two reasons you should try to avoid falling into this habit: (a) the components themselves are of finite precision (resistors typically have tolerances of ±5% or ±1%; for capacitors it's typically ±10% or ±5%; and the parameters that characterize transistors, say, frequently are known only to a factor of 2); (b) one mark of a good circuit design is insensitivity of the finished circuit to precise values of the components (there are exceptions, of course). You'll also learn circuit intuition more quickly if you get into the habit of doing approximate calculations in your head, rather than watching meaningless numbers pop up on a calculator display. We believe strongly that reliance on formulas and equations early in your electronic circuit education is a fine way to prevent you from understanding what's really going on.

In trying to develop intuition about resistance, some people find it helpful to think about *conductance*, $G = 1/R$. The current through a device of conductance G bridging a voltage V is then given by $I = GV$ (Ohm's law). A small resistance is a large conductance, with correspondingly large current under the influence of an applied voltage. Viewed in this light, the formula for parallel resistors is obvious: when several resistors or conducting paths are connected across the same voltage, the total current is the sum of the individual currents. Therefore the net conductance is simply the sum of the individual conductances, $G = G_1 + G_2 + G_3 + \cdots$, which is the same as the formula for parallel resistors derived earlier.

Engineers are fond of defining reciprocal units, and they have designated as the unit of conductance the siemens (S$= 1/\Omega$), also known as the mho (that's ohm spelled backward, given the symbol ℧). Although the concept of conductance is helpful in developing intuition, it is not used widely;[11] most people prefer to talk about resistance instead.

C. Power in resistors

The power dissipated by a resistor (or any other device) is $P = IV$. Using Ohm's law, you can get the equivalent forms $P = I^2R$ and $P = V^2/R$.

Exercise 1.5. Show that it is not possible to exceed the power rating of a 1/4 watt resistor of resistance greater than 1k, no matter how you connect it, in a circuit operating from a 15 volt battery.

Exercise 1.6. Optional exercise: New York City requires about 10^{10} watts of electrical power, at 115 volts[12] (this is plausible: 10 million people averaging 1 kilowatt each). A heavy power cable might be an inch in diameter. Let's calculate what will happen if we try to supply the power through a cable 1 foot in diameter made of pure copper. Its resistance is 0.05 $\mu\Omega$ (5×10^{-8} ohms) per foot. Calculate (a) the power lost per foot from "I^2R losses," (b) the length of cable over which you will lose all 10^{10} watts, and (c) how hot the cable will get, if you know the physics involved ($\sigma = 6 \times 10^{-12}$W/K^4 cm^2). If you have done your computations correctly, the result should seem preposterous. What is the solution to this puzzle?

[10] With an error, in this case, of just 0.01%.

[11] Although the elegant *Millman's theorem* has its admirers: it says that the output voltage from a set of resistors (call them R_i) that are driven from a set of corresponding input voltages (V_i) and connected together at the output is $V_{\text{out}}=(\sum V_iG_i)/\sum G_i$, where the G_i are the conductances ($G_i=1/R_i$).

[12] Although the "official" line voltage is 120 V ±5%, you'll sometimes see 110 V, 115 V, or 117 V. This loose language is OK (and we use it in this book), because (a) the median voltage at the wall plug is 3 to 5 volts lower, when powering stuff; and (b) the *minimum* wall-plug voltage is 110 V. See ANSI standard C84.1.

D. Input and output

Nearly all electronic circuits accept some sort of applied *input* (usually a voltage) and produce some sort of corresponding *output* (which again is often a voltage). For example, an audio amplifier might produce a (varying) output voltage that is 100 times as large as a (similarly varying) input voltage. When describing such an amplifier, we imagine measuring the output voltage for a given applied input voltage. Engineers speak of the *transfer function* **H**, the ratio of (measured) output divided by (applied) input; for the audio amplifier above, **H** is simply a constant (**H** = 100). We'll get to amplifiers soon enough, in the next chapter. However, with only resistors we can already look at a very important circuit fragment, the *voltage divider* (which you might call a "de-amplifier").

1.2.3 Voltage dividers

We now come to the subject of the voltage divider, one of the most widespread electronic circuit fragments. Show us any real-life circuit and we'll show you half a dozen voltage dividers. To put it very simply, a voltage divider is a circuit that, given a certain voltage input, produces a predictable fraction of the input voltage as the output voltage. The simplest voltage divider is shown in Figure 1.6.

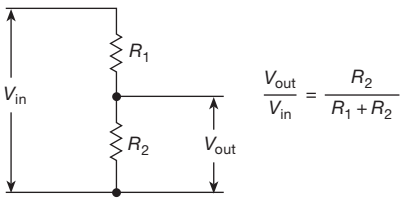


Figure 1.6. Voltage divider. An applied voltage V_{in} results in a (smaller) output voltage V_{out}.

An important word of explanation: when engineers draw a circuit like this, it's generally assumed that the V_{in} on the left is a voltage that you are applying to the circuit, and that the V_{out} on the right is the resulting output voltage (produced by the circuit) that you are measuring (or at least are interested in). You are supposed to know all this (a) because of the convention that signals generally flow from left to right, (b) from the suggestive names ("in," "out") of the signals, and (c) from familiarity with circuits like this. This may be confusing at first, but with time it becomes easy.

What is V_{out}? Well, the current (same everywhere, assuming no "load" on the output; i.e., nothing connected across the output) is

$$I = \frac{V_{in}}{R_1 + R_2}.$$

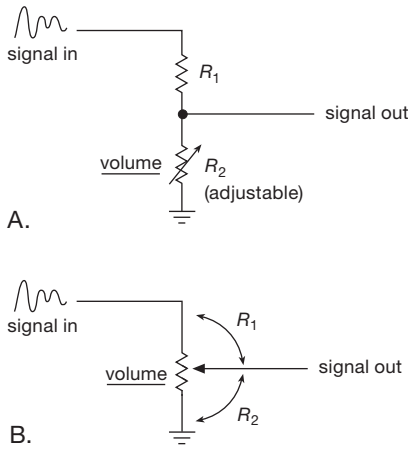


Figure 1.7. An adjustable voltage divider can be made from a fixed and variable resistor, or from a potentiometer. In some contemporary circuits you'll find instead a long series chain of equal-value resistors, with an arrangement of electronic switches that lets you choose any one of the junctions as the output; this sounds much more complicated – but it has the advantage that you can adjust the voltage ratio *electrically* (rather than mechanically).

(We've used the definition of resistance and the series law.) Then, for R_2,

$$V_{out} = IR_2 = \frac{R_2}{R_1 + R_2} V_{in}. \tag{1.6}$$

Note that the output voltage is always less than (or equal to) the input voltage; that's why it's called a divider. You could get amplification (more output than input) if one of the resistances were negative. This isn't as crazy as it sounds; it is possible to make devices with negative "incremental" resistances (e.g., the component known as a *tunnel diode*) or even true negative resistances (e.g., the negative-impedance converter that we will talk about later in the book, §6.2.4B). However, these applications are rather specialized and need not concern you now.

Voltage dividers are often used in circuits to generate a particular voltage from a larger fixed (or varying) voltage. For instance, if V_{in} is a varying voltage and R_2 is an adjustable resistor (Figure 1.7A), you have a "volume control"; more simply, the combination R_1R_2 can be made from a single variable resistor, or *potentiometer* (Figure 1.7B). This and similar applications are common, and potentiometers come in a variety of styles, some of which are shown in Figure 1.8.

The humble voltage divider is even more useful, though, as a way of *thinking* about a circuit: the input voltage and upper resistance might represent the output of an amplifier, say, and the lower resistance might represent the input of

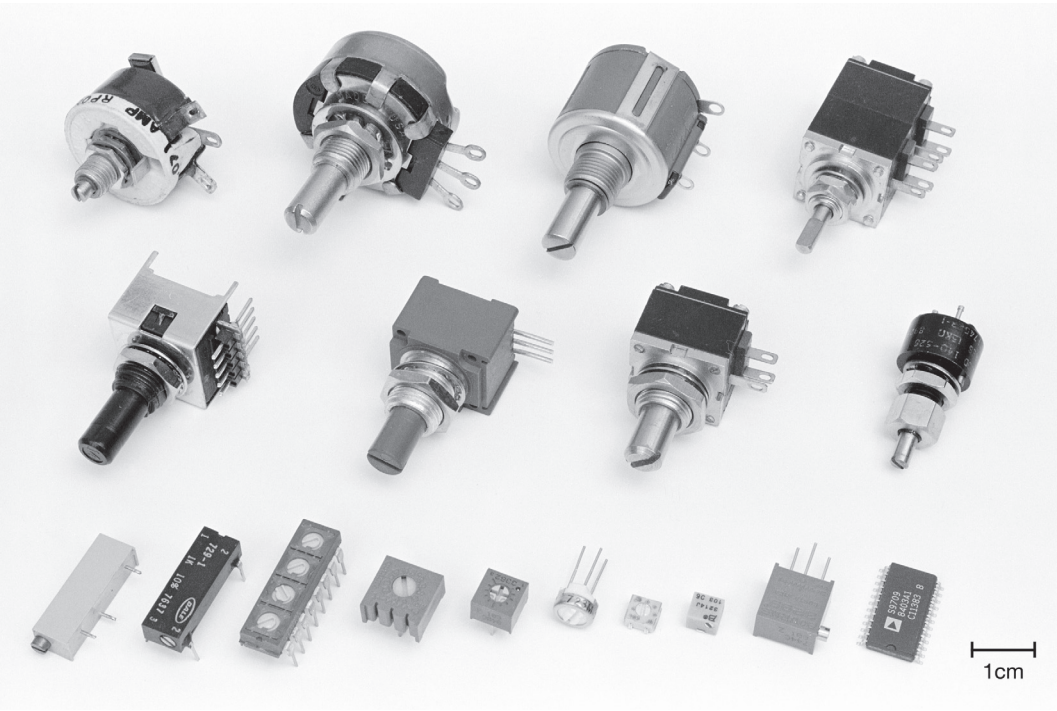


Figure 1.8. Most of the common potentiometer styles are shown here. Top row, left to right (panel mount): power wirewound, "type AB" 2W carbon composition, 10-turn wirewound/plastic hybrid, ganged dual pot. Middle row (panel mount): optical encoder (continuous rotation, 128 cycles per turn), single-turn cermet, single-turn carbon, screw-adjust single-turn locking. Front row (board-mount trimmers): multiturn side-adjust (two styles), quad single-turn, 3/8″ (9.5 mm) square single-turn, 1/4″ (6.4 mm) square single-turn, 1/4″ (6.4 mm) round single-turn, 4 mm square single-turn surface mount, 4 mm square multiturn surface mount, 3/8″ (9.5 mm) square multiturn, quad nonvolatile 256-step integrated pot (E^2POT) in 24-pin small-outline IC.

the following stage. In this case the voltage-divider equation tells you how much signal gets to the input of that last stage. This will all become clearer after you know about a remarkable fact (Thévenin's theorem) that will be discussed later. First, though, a short aside on voltage sources and current sources.

1.2.4 Voltage sources and current sources

A perfect *voltage source* is a two-terminal "black box" that maintains a fixed voltage drop across its terminals, regardless of load resistance. This means, for instance, that it must supply a current $I = V/R$ when a resistance R is attached to its terminals. A real voltage source can supply only a finite maximum current, and in addition it generally behaves like a perfect voltage source with a small resistance in series. Obviously, the smaller this series resistance, the better. For example, a standard 9 volt alkaline battery behaves approximately like a perfect 9 volt voltage source in series with a 3 Ω resistor, and it can provide a maximum current (when shorted) of 3 amps (which, however, will kill the battery in a few minutes). A voltage source "likes" an open-circuit load and "hates" a short-circuit load, for obvious reasons. (The meaning of "open circuit" and "short circuit" sometimes confuse the beginner: an open circuit has nothing connected to it, whereas a short circuit is a piece of wire bridging the output.) The symbols used to indicate a voltage source are shown in Figure 1.9.

A perfect *current source* is a two-terminal black box that maintains a constant current through the external circuit, regardless of load resistance or applied voltage. To do this it must be capable of supplying any necessary voltage across its terminals. Real current sources (a much-neglected subject in most textbooks) have a limit to the voltage they can provide (called the *output-voltage compliance,* or just *compliance*), and in addition they do not provide absolutely constant output current. A current source "likes" a short-circuit load and "hates" an open-circuit load. The symbols used to indicate a current source are shown in Figure 1.10.

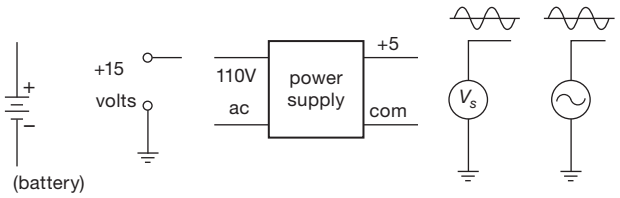


Figure 1.9. Voltage sources can be either steady (dc) or varying (ac).

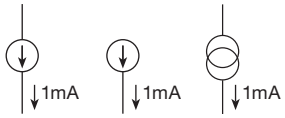


Figure 1.10. Current-source symbols.

A battery is a real-life approximation to a voltage source (there is no analog for a current source). A standard D-size flashlight cell, for instance, has a terminal voltage of 1.5 V, an equivalent series resistance of about 0.25 Ω, and a total energy capacity of about 10,000 watt–seconds (its characteristics gradually deteriorate with use; at the end of its life, the voltage may be about 1.0 V, with an internal series resistance of several ohms). It is easy to construct voltage sources with far better characteristics, as you will learn when we come to the subject of feedback; this is a major topic of Chapter 9. Except in the important class of devices intended for portability, the use of batteries in electronic devices is rare.

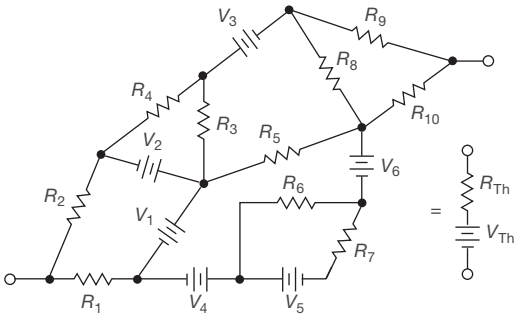


Figure 1.11. The Thévenin equivalent circuit.

1.2.5 Thévenin equivalent circuit

Thévenin's theorem states[12] that any two-terminal network of resistors and voltage sources is equivalent to a single resistor R in series with a single voltage source V. This is remarkable. Any mess of batteries and resistors can be mimicked with one battery and one resistor (Figure 1.11). (Incidentally, there's another theorem, Norton's theorem, that says you can do the same thing with a current source in parallel with a resistor.)

How do you figure out the Thévenin equivalent $R_{\rm Th}$ and $V_{\rm Th}$ for a given circuit? Easy! $V_{\rm Th}$ is the open-circuit voltage of the Thévenin equivalent circuit; so if the two circuits behave identically, it must also be the open-circuit voltage of the given circuit (which you get by calculation, if you know what the circuit is, or by measurement, if you don't). Then you find $R_{\rm Th}$ by noting that the short-circuit current of the equivalent circuit is $V_{\rm Th}/R_{\rm Th}$. In other words,

$$\begin{aligned} V_{\rm Th} &= V\ (\text{open circuit}),\\ R_{\rm Th} &= \frac{V\ (\text{open circuit})}{I\ (\text{short circuit})}. \end{aligned} \tag{1.7}$$

Let's apply this method to the voltage divider, which must have a Thévenin equivalent:

1. The open-circuit voltage is

$$V = V_{\rm in}\frac{R_2}{R_1+R_2}.$$

2. The short-circuit current is

$$V_{\rm in}/R_1.$$

So the Thévenin equivalent circuit is a voltage source,

$$V_{\rm Th} = V_{\rm in}\frac{R_2}{R_1+R_2}, \tag{1.8}$$

in series with a resistor,

$$R_{\rm Th} = \frac{R_1R_2}{R_1+R_2}. \tag{1.9}$$

(It is not a coincidence that this happens to be the parallel resistance of R_1 and R_2. The reason will become clear later.)

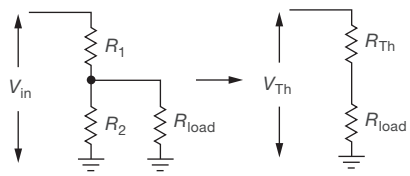


Figure 1.12. Thévenin equivalent of a voltage divider.

From this example it is easy to see that a voltage divider is not a very good battery, in the sense that its output voltage drops severely when a load is attached. As an example, consider Exercise 1.10. You now know everything you need to know to calculate exactly how much the output will

[12] We provide a proof, for those who are interested, in Appendix D.

MULTIMETERS

There are numerous instruments that let you measure voltages and currents in a circuit. The oscilloscope is the most versatile; it lets you "see" voltages versus time at one or more points in a circuit. Logic probes and logic analyzers are special-purpose instruments for troubleshooting digital circuits. The simple multimeter provides a good way to measure voltage, current, and resistance, often with good precision; however, it responds slowly, and thus it cannot replace the oscilloscope where changing voltages are of interest. Multimeters are of two varieties: those that indicate measurements on a conventional scale with a moving pointer, and those that use a digital display.

The traditional (and now largely obsolete) VOM (volt-ohm-milliammeter) multimeter uses a meter movement that measures current (typically 50 μA full scale). (See a less-design-oriented electronics book for pretty pictures of the innards of meter movements; for our purposes, it suffices to say that it uses coils and magnets.) To measure *voltage*, the VOM puts a resistor in series with the basic movement. For instance, one kind of VOM will generate a 1 V (full-scale) range by putting a 20k resistor in series with the standard 50 μA movement; higher voltage ranges use correspondingly larger resistors. Such a VOM is specified as 20,000 Ω/V, meaning that it looks like a resistor whose value is 20k multiplied by the full-scale voltage of the particular range selected. Full scale on any voltage range is 1/20,000 amps, or 50 μA. It should be clear that one of these voltmeters disturbs a circuit less on a higher range, since it looks like a higher resistance (think of the voltmeter as the lower leg of a voltage divider, with the Thévenin resistance of the circuit you are measuring as the upper resistor). Ideally, a voltmeter should have infinite input resistance.

Most contemporary multimeters use electronic amplification and have an input resistance of 10 MΩ to 1000 MΩ when measuring voltage; they display their results digitally, and are known collectively as digital multimeters (DMMs). A word of caution: sometimes the input resistance of these meters is very high on the most sensitive ranges, dropping to a lower resistance for the higher ranges. For instance, you might typically have an input resistance of 10^9 Ω on the 0.2 V and 2 V ranges, and 10^7 Ω on all higher ranges. Read the specifications carefully! However, for most circuit measurements these high input resistances will produce negligible loading effects. In any case, it is easy to calculate how serious the effect is by using the voltage-divider equation. Typically, multimeters provide voltage ranges from a volt (or less) to a kilovolt (or more), full scale.

A multimeter usually includes *current*-measuring capability, with a set of switchable ranges. Ideally, a current-measuring meter should have zero resistance[13] in order not to disturb the circuit under test, since it must be put in series with the circuit. In practice, you tolerate a few tenths of a volt drop (sometimes called "voltage burden") with both VOMs and digital multimeters. For either kind of meter, selecting a current range puts a small resistor across the meter's input terminals, typically of resistance value to create a voltage drop of 0.1 V to 0.25 V for the chosen full-scale current; the voltage drop is then converted to a corresponding current indication.[14] Typically, multimeters provide current ranges from 50 μA (or less) to an amp (or more), full scale.

Multimeters also have one or more batteries in them to power the resistance measurement. By supplying a small current and measuring the voltage drop, they measure resistance, with several ranges to cover values from 1 Ω (or less) to 10 MΩ (or more).

Important: don't try to measure "the current of a voltage source," by sticking the meter across the wall plug; the same applies for ohms. This is a leading cause of blown-out meters.

Exercise 1.7. What will a 20,000 Ω/V meter read, on its 1 V scale, when attached to a 1 V source with an internal resistance of 10k? What will it read when attached to a 10k–10k voltage divider driven by a "stiff" (zero source resistance) 1 V source?

Exercise 1.8. A 50 μA meter movement has an internal resistance of 5k. What shunt resistance is needed to convert it to a 0–1 A meter? What series resistance will convert it to a 0–10 V meter?

Exercise 1.9. The very high internal resistance of *digital* multimeters, in their voltage-measuring ranges, can be used to measure extremely low *currents* (even though the DMM may not offer a low current range explicitly). Suppose, for example, you want to measure the small current that flows through a 1000 MΩ "leakage" resistance (that term is used to describe a small current that ideally should be absent entirely, for example through the insulation of an underground cable). You have available a standard DMM, whose 2 V dc range has 10 MΩ internal resistance, and you have available a dc source of +10 V. How can you use what you've got to measure accurately the leakage resistance?

drop for a given load resistance: use the Thévenin equivalent circuit, attach a load, and calculate the new output, noting that the new circuit is nothing but a voltage divider (Figure 1.12).

Exercise 1.10. For the circuit shown in Figure 1.12, with V_{in}=30 V and $R_1 = R_2 = 10$k, find (a) the output voltage with no load attached (the open-circuit voltage); (b) the output voltage with a 10k load (treat as a voltage divider, with R_2 and R_{load} combined into a single resistor); (c) the Thévenin equivalent circuit; (d) the same as in part (b), but using the Thévenin equivalent circuit [again, you wind up with a voltage divider; the answer should agree with the result in part (b)]; (e) the power dissipated in each of the resistors.

A. Equivalent source resistance and circuit loading

As we have just seen, a voltage divider powered from some fixed voltage is equivalent to some smaller voltage source

[13] This is the opposite of an ideal voltage-measuring meter, which should present infinite resistance across its input terminals.

[14] A special class of current meters known as *electrometers* operate with very small voltage burdens (as little at 0.1 mV) by using the technique of feedback, something we'll learn about in Chapters 2 and 4.

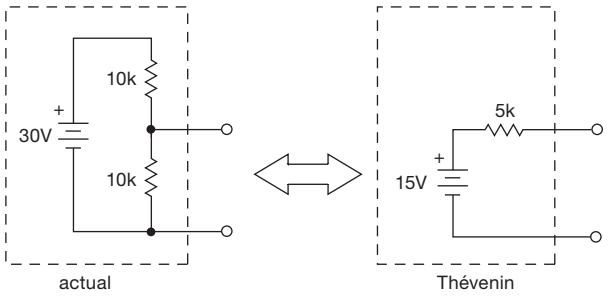


Figure 1.13. Voltage divider example.

in series with a resistor. For example, the output terminals of a 10k–10k voltage divider driven by a perfect 30 volt battery are precisely equivalent to a perfect 15 volt battery in series with a 5k resistor (Figure 1.13). Attaching a load resistor causes the voltage divider's output to drop, owing to the finite *source resistance* (Thévenin equivalent resistance of the voltage divider output, viewed as a source of voltage). This is often undesirable. One solution to the problem of making a stiff voltage source ("stiff" is used in this context to describe something that doesn't bend under load) might be to use much smaller resistors in a voltage divider. Occasionally this brute-force approach is useful. However, it is usually best to construct a voltage source, or power supply, as it's commonly called, using active components like transistors or operational amplifiers, which we will treat in Chapters 2–4. In this way you can easily make a voltage source with internal (Thévenin equivalent) resistance as small as milliohms (thousandths of an ohm), without the large currents and dissipation of power characteristic of a low-resistance voltage divider delivering the same performance. In addition, with an active power supply it is easy to make the output voltage adjustable. These topics are treated extensively in Chapter 9.

The concept of equivalent internal resistance applies to all sorts of sources, not just batteries and voltage dividers. Signal sources (e.g., oscillators, amplifiers, and sensing devices) all have an equivalent internal resistance. Attaching a load whose resistance is less than or even comparable to the internal resistance will reduce the output considerably. This undesirable reduction of the open-circuit voltage (or signal) by the load is called "circuit loading." Therefore you should strive to make $R_{\text{load}} \gg R_{\text{internal}}$, because a high-resistance load has little attenuating effect on the source (Figure 1.14).[15] We will see numerous circuit examples in the chapters ahead. This high-resistance condition ideally characterizes measuring instruments such as voltmeters and oscilloscopes.

A word on language: you frequently hear things like "the resistance looking into the voltage divider" or "the output sees a load of so-and-so many ohms," as if circuits had eyes. It's OK (in fact, it's a rather good way of keeping straight which resistance you're talking about) to say what part of the circuit is doing the "looking."[16]

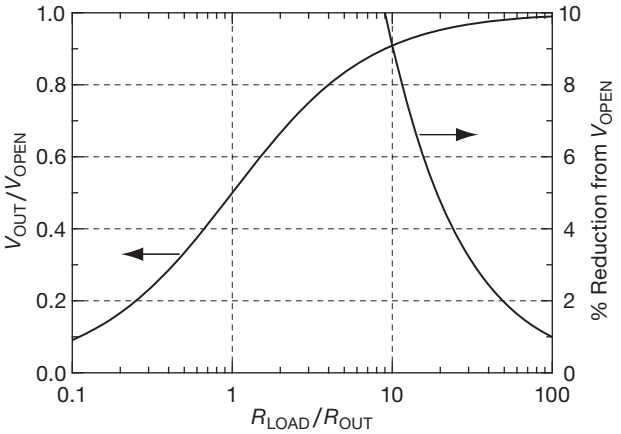


Figure 1.14. To minimize the attenuation of a signal source below its open-circuit voltage, keep the load resistance large compared with the output resistance.

B. Power transfer

Here is an interesting problem: what load resistance will result in maximum power being transferred to the load for a given source resistance? (The terms *source resistance, internal resistance*, and *Thévenin equivalent resistance* all mean the same thing.) It is easy to see that either R_{load}=0 or R_{load}=∞ results in zero power transferred, because R_{load}=0 means that V_{load}=0 and $I_{\text{load}}=V_{\text{source}}/R_{\text{source}}$, so that $P_{\text{load}}=V_{\text{load}}I_{\text{load}}$=0. But R_{load}=∞ means that $V_{\text{load}}=V_{\text{source}}$ and I_{load}=0, so that again P_{load}=0. There has to be a maximum in between.

Exercise 1.11. Show that $R_{\text{load}} = R_{\text{source}}$ maximizes the power in the load for a given source resistance. Note: skip this exercise if you don't know calculus, and take it on faith that the answer is true.

[15] There are two important exceptions to this general principle: (1) a current source has a high (ideally infinite) internal resistance and should drive a load of relatively low load resistance; (2) when dealing with radio frequencies and transmission lines, you must "match impedances" (i.e., set $R_{\text{load}}=R_{\text{internal}}$) in order to prevent reflection and loss of power. See Appendix H on transmission lines.

[16] The urge to anthropomorphize runs deep in the engineering and scientific community, despite warnings like "don't anthropomorphize computers … they don't like it."

Lest this example leave the wrong impression, we would like to emphasize again that circuits are ordinarily designed so that the load resistance is much greater than the source resistance of the signal that drives the load.

1.2.6 Small-signal resistance

We often deal with electronic devices for which I is not proportional to V; in such cases there's not much point in talking about resistance, since the ratio V/I will depend on V, rather than being a nice constant, independent of V. For these devices it is sometimes useful to know instead the *slope* of the V–I curve, in other words, the ratio of a small change in applied voltage to the resulting change in current through the device, $\Delta V/\Delta I$ (or dV/dI). This quantity has the units of resistance (ohms) and substitutes for resistance in many calculations. It is called the small-signal resistance, incremental resistance, or dynamic resistance.

A. Zener diodes

As an example, consider the *zener diode*, which has the I–V curve shown in Figure 1.15. Zeners are used to create a constant voltage inside a circuit somewhere, simply done by providing them with a (roughly constant) current derived from a higher voltage within the circuit.[17] For example, the zener diode in Figure 1.15 will convert an applied current in the range shown to a corresponding (but fractionally narrower) range of voltages. It is important to know how the resulting zener voltage will change with applied current; this is a measure of its "regulation" against changes in the driving current provided to it. Included in the specifications of a zener will be its dynamic resistance, given at a certain current. For example, a zener might have a dynamic resistance of 10 Ω at 10 mA, at its specified zener voltage of 5 V. Using the definition of dynamic resistance, we find that a 10% change in applied current will therefore result in a change in voltage of

$$\Delta V = R_{\rm dyn}\Delta I = 10 \times 0.1 \times 0.01 = 10\,{\rm mV}$$

or

$$\Delta V/V = 0.002 = 0.2\%,$$

thus demonstrating good voltage-regulating ability. In this sort of application you frequently get the zener current through a resistor from a higher voltage available somewhere in the circuit, as in Figure 1.16.

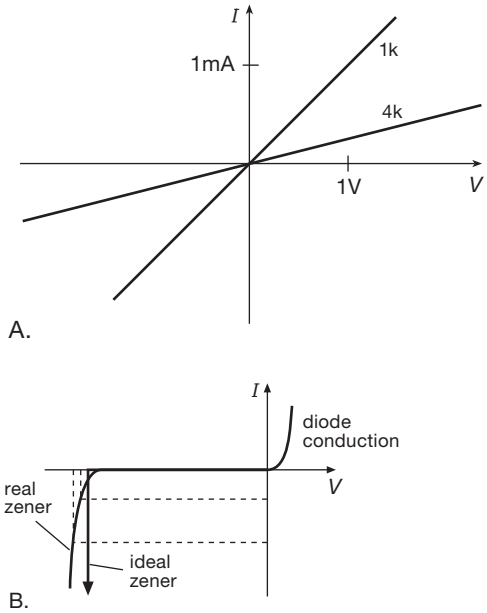


Figure 1.15. *I–V* curves: A. Resistor (linear). B. Zener diode (nonlinear).

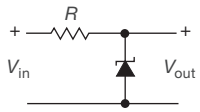


Figure 1.16. Zener regulator.

Then,

$$I = \frac{V_{\rm in} - V_{\rm out}}{R}$$

and

$$\Delta I = \frac{\Delta V_{\rm in} - \Delta V_{\rm out}}{R},$$

so

$$\Delta V_{\rm out} = R_{\rm dyn}\Delta I = \frac{R_{\rm dyn}}{R}(\Delta V_{\rm in} - \Delta V_{\rm out})$$

and finally

$$\Delta V_{\rm out} = \frac{R_{\rm dyn}}{R + R_{\rm dyn}}\Delta V_{\rm in}.$$

Aha – the voltage-divider equation, again! Thus, for *changes* in voltage, the circuit behaves like a voltage divider, with the zener replaced by a resistor equal to its dynamic resistance at the operating current. This is the

[17] Zeners belong to the more general class of *diodes* and *rectifiers*, important devices that we'll see later in the chapter (§1.6), and indeed throughout the book. The ideal diode (or rectifier) acts as a perfect conductor for current flow in one direction, and a perfect insulator for current flow in the reverse direction; it is a "one-way valve" for current.

utility of incremental resistance. For instance, suppose in the preceding circuit we have an input voltage ranging between 15 and 20 V, and we use a 1N4733 (5.1 V, 1W zener diode) in order to generate a stable 5.1 V power supply. We choose $R = 300\,\Omega$, for a maximum zener current of 50 mA: (20 V−5.1 V)/300 Ω. We can now estimate the output-voltage regulation (variation in output voltage), knowing that this particular zener has a specified maximum dynamic resistance of 7.0 Ω at 50 mA. The zener current varies from 50 mA to 33 mA over the input-voltage range; this 17 mA change in current then produces a voltage change at the output of $\Delta V = R_{\rm dyn}\Delta I$, or 0.12 V.

It's a useful fact, when dealing with zener diodes, that the dynamic resistance of a zener diode varies roughly in inverse proportion to current. It's worth knowing, also, that there are ICs designed to substitute for zener diodes; these "two-terminal voltage references" have superior performance – much lower dynamic resistance (less than 1 Ω, even at currents as small as 0.1 mA; that's a thousand times better than the zener we just used), and excellent temperature stability (better than 0.01%/C). We will see more of zeners and voltage references in §§2.2.4 and 9.10.

In real life, a zener will provide better regulation if driven by a current source, which has, by definition, $R_{\rm incr}=\infty$ (the same current, regardless of voltage). But current sources are more complex, and therefore in practice we often resort to the humble resistor. When thinking about zeners, it's worth remembering that low-voltage units (e.g., 3.3 V) behave rather poorly, in terms of constancy of voltage versus current (Figure 1.17); if you think you need a low voltage zener, use a two-terminal reference instead (§9.10).

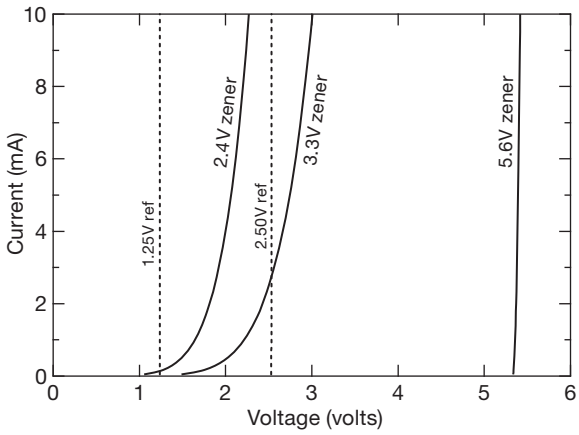


Figure 1.17. Low-voltage zeners are pretty dismal, as seen in these measured *I* vs. *V* curves (for three members of the 1N5221–67 series), particularly in contrast to the excellent measured performance of a pair of "IC voltage references" (LM385Z-1.2 and LM385Z-2.5, see §9.10 and Table 9.7). However, zener diodes in the neighborhood of 6 V (such as the 5.6 V 1N5232B or 6.2 V 1N5234B) exhibit admirably steep curves, and are useful parts.

1.2.7 An example: "It's too hot!"

Some people like to turn the thermostat way up, annoying other people who like their houses cool. Here's a little gadget (Figure 1.18) that lets folks of the latter persuasion know when to complain – it lights up a red light-emitting diode (LED) indicator when the room is warmer than 30°C (86°F). It also shows how to use the humble voltage divider (and even humbler Ohm's law), and how to deal with an LED, which behaves like a zener diode (and is sometimes used as such).

The triangular symbol is a *comparator*, a handy device (discussed in §12.3) that switches its output according to the relative voltages at its two input terminals. The temperature sensing device is R_4, which decreases in resistance by about 4%/°C, and which is 10kΩ at 25°C. So we've made it the lower leg of a voltage divider (R_3R_4), whose output is compared with the temperature-insensitive divider R_1R_2. When it's hotter than 30°C, point "X" is at a lower voltage than point "Y," so the comparator pulls its output to ground.

At the output there's an LED, which behaves electrically like a 1.6 V zener diode; and when current is flowing, it lights up. Its lower terminal is then at 5 V−1.6 V, or +3.4 V. So we've added a series resistor, sized to allow 5 mA when the comparator output is at ground: R_5=3.4 V/5 mA, or 680 Ω.

If you wanted to, you could make the setpoint adjustable by replacing R_2 with a 5k pot in series with a 5k fixed resistor. We'll see later that it's also a good idea to add some *hysteresis*, to encourage the comparator to be decisive. Note that this circuit is insensitive to the exact power-supply voltage because it compares *ratios*. Ratiometric techniques are good; we'll see them again later.

1.3 Signals

A later section in this chapter will deal with capacitors, devices whose properties depend on the way the voltages and currents in a circuit are *changing*. Our analysis of dc circuits so far (Ohm's law, Thévenin equivalent circuits, etc.) still holds, even if the voltages and currents are changing in time. But for a proper understanding of alternating-current (ac) circuits, it is useful to have in mind certain common

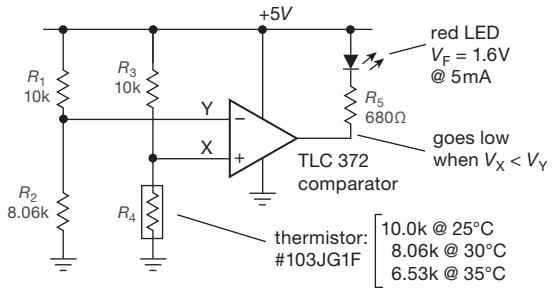


Figure 1.18. The LED lights up when it's hotter than 30°C. The comparator (which we'll see later, in Chapters 4 and 12) pulls its output to ground when the voltage at "X" is less than the voltage at "Y." R_4 is a thermistor, which is a resistor with a deliberate negative temperature coefficient; that is, its resistance decreases with increasing temperature – by about 4%/°C.

types of *signals*, voltages that change in time in a particular way.

1.3.1 Sinusoidal signals

Sinusoidal signals are the most popular signals around; they're what you get out of the wall plug. If someone says something like "take a 10 μV signal at 1 MHz," they mean a sinewave. Mathematically, what you have is a voltage described by

$$V = A\sin 2\pi ft \tag{1.10}$$

where A is called the amplitude and f is the frequency in hertz (cycles per second). A sinewave looks like the wave shown in Figure 1.19. Sometimes it is important to know the value of the signal at some arbitrary time $t = 0$, in which case you may see a *phase* ϕ in the expression:

$$V = A\sin(2\pi ft + \phi).$$

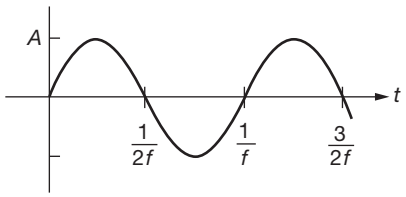


Figure 1.19. Sinewave of amplitude A and frequency f.

The other variation on this simple theme is the use of *angular frequency*, which looks like this:

$$V = A\sin\omega t.$$

Here ω is the angular frequency, measured in radians per second. Just remember the important relation $\omega = 2\pi f$ and you won't go wrong.

The great merit of sinewaves (and the cause of their perennial popularity) is the fact that they are the solutions to certain linear differential equations that happen to describe many phenomena in nature as well as the properties of linear circuits. A linear circuit has the property that its output, when driven by the sum of two input signals, equals the sum of its individual outputs when driven by each input signal in turn; i.e., if $\mathscr{O}(A)$ represents the output when driven by signal A, then a circuit is linear if $\mathscr{O}(A+B) = \mathscr{O}(A) + \mathscr{O}(B)$. A linear circuit driven by a sinewave always responds with a sinewave, although in general the phase and amplitude are changed. No other periodic signal can make this statement. It is standard practice, in fact, to describe the behavior of a circuit by its *frequency response*, by which we mean the way the circuit alters the amplitude of an applied sinewave as a function of frequency. A stereo amplifier, for instance, should be characterized by a "flat" frequency response over the range 20 Hz to 20 kHz, at least.

The sinewave frequencies we usually deal with range from a few hertz to a few tens of megahertz. Lower frequencies, down to 0.0001 Hz or lower, can be generated with carefully built circuits, if needed. Higher frequencies, up to say 2000 MHz (2 GHz) and above, can be generated, but they require special transmission-line techniques. Above that, you're dealing with microwaves, for which conventional wired circuits with lumped-circuit elements become impractical, and exotic waveguides or "striplines" are used instead.

1.3.2 Signal amplitudes and decibels

In addition to its amplitude, there are several other ways to characterize the magnitude of a sinewave or any other signal. You sometimes see it specified by *peak-to-peak amplitude* (pp amplitude), which is just what you would guess, namely, twice the amplitude. The other method is to give the *root-mean-square amplitude* (rms amplitude), which is $V_{\rm rms} = (1/\sqrt{2})A = 0.707A$ (this is for sinewaves only; the ratio of pp to rms will be different for other waveforms). Odd as it may seem, this is the usual method, because rms voltage is what's used to compute power. The nominal voltage across the terminals of a wall socket (in the United States) is 120 volts rms, 60 Hz. The *amplitude* is 170 volts (339 volts pp).[18]

[18] Occasionally you'll encounter devices (e.g., mechanical moving-pointer meters) that respond to the *average* magnitude of an ac signal.

A. Decibels

How do you compare the relative amplitudes of two signals? You could say, for instance, that signal X is twice as large as signal Y. That's fine, and useful for many purposes. But because we often deal with ratios as large as a million, it is better to use a logarithmic measure, and for this we present the decibel (it's one-tenth as large as something called a bel, which no one ever uses). By definition, the ratio of two signals, in decibels (dB), is

$$\mathrm{dB} = 10 \log_{10} \frac{P_2}{P_1}, \tag{1.11}$$

where P_1 and P_2 represent the *power* in the two signals. We are often dealing with signal *amplitudes*, however, in which case we can express the ratio of two signals having the same waveform as

$$\mathrm{dB} = 20 \log_{10} \frac{A_2}{A_1}, \tag{1.12}$$

where A_1 and A_2 are the two signal amplitudes. So, for instance, one signal of twice the amplitude of another is $+6$ dB relative to it, since $\log_{10} 2 = 0.3010$. A signal 10 times as large is $+20$ dB; a signal one-tenth as large is -20 dB.

Although decibels are ordinarily used to specify the ratio of two signals, they are sometimes used as an absolute measure of amplitude. What is happening is that you are assuming some reference signal level and expressing any other level in decibels relative to it. There are several standard levels (which are unstated, but understood) that are used in this way; the most common references are (a) 0 dBV (1 V rms); (b) 0 dBm (the voltage corresponding to 1 mW into some assumed load impedance, which for radiofrequencies is usually 50 Ω, but for audio is often 600 Ω; the corresponding 0 dBm amplitudes, when loaded by those impedances, are then 0.22 V rms and 0.78 V rms); and (c) the small noise voltage generated by a resistor at room temperature (this surprising fact is discussed in §8.1.1). In addition to these, there are reference amplitudes used for measurements in other fields of engineering and science. For instance, in acoustics, 0 dB SPL (sound pressure level) is a wave whose rms pressure is 20 μPa (that's 2×10^{-10} atm); in audio communications, levels can be stated in dBrnC (relative noise reference weighted in frequency by "curve C"). When stating amplitudes this way, it is best to be specific about the 0 dB reference amplitude; say something like "an amplitude of 27 decibels relative to 1 V rms," or abbreviate "27 dB re 1 V rms," or define a term like "dBV."[19]

Exercise 1.12. Determine the voltage and power ratios for a pair of signals with the following decibel ratios: (a) 3 dB, (b) 6 dB, (c) 10 dB, (d) 20 dB.

Exercise 1.13. We might call this amusing exercise "Desert Island dBs": in the table below we've started entering some values for power ratios corresponding to the first dozen integral dBs, using the results for parts (a) and (c) of the last exercise. Your job is to complete the table, without recourse to a calculator. A possibly helpful hint: starting at 10 dB, go down the table in steps of 3 dB, then up in a step of 10 dB, then down again. Finally, get rid of yucky numbers like 3.125 (and its near relatives) by noticing that it's charmingly close to π.

dB	ratio (P/P_0)
0	1
1	
2	
3	2
4	
5	
6	4
7	
8	
9	8
10	10
11	

1.3.3 Other signals

A. Ramp

The ramp is a signal that looks like the one shown in Figure 1.20A. It is simply a voltage rising (or falling) at a constant rate. That can't go on forever, of course, even in science fiction movies. It is sometimes approximated by a finite ramp (Figure 1.20B) or by a periodic ramp (known as a *sawtooth*, Figure 1.20C).

B. Triangle

The triangle wave is a close cousin of the ramp; it is simply a symmetrical ramp (Figure 1.21).

C. Noise

Signals of interest are often mixed with *noise*; this is a catch-all phrase that usually applies to random noise of thermal origin. Noise voltages can be specified by their

For a sinewave the relationship is $V_{\mathrm{avg}} = V_{\mathrm{rms}}/1.11$. However, such meters are usually calibrated so that they indicate the rms sinewave amplitude. For signals other than sinewaves their indication is in error; be sure to use a "true rms" meter if you want the right answer.

[19] One of the authors, when asked by his nontechnical spouse how much we spent on that big plasma screen, replied "36 dB$."

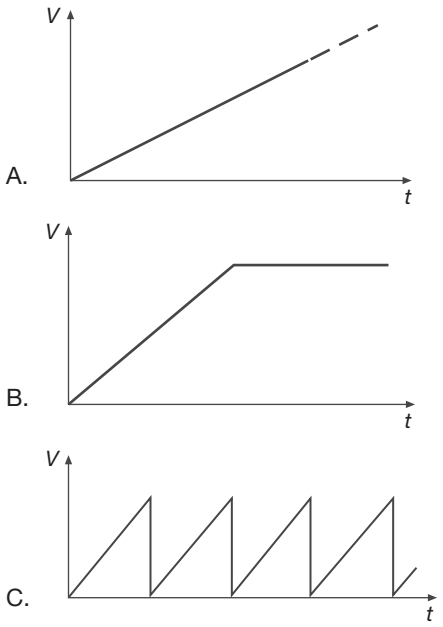


Figure 1.20. A: Voltage-ramp waveform. B: Ramp with limit. C: Sawtooth wave.

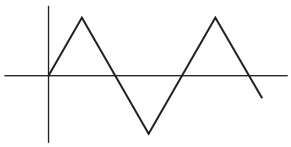


Figure 1.21. Triangle wave.

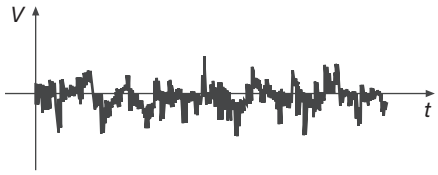


Figure 1.22. Noise.

frequency spectrum (power per hertz) or by their amplitude distribution. One of the most common kind of noise is *band-limited white Gaussian noise*, which means a signal with equal power per hertz in some band of frequencies and that exhibits a Gaussian (bell-shaped) distribution of amplitudes when many instantaneous measurements of its amplitude are made. This kind of noise is generated by a resistor (Johnson noise or Nyquist noise), and it plagues sensitive measurements of all kinds. On an oscilloscope it appears as shown in Figure 1.22. We will discuss noise and low-noise techniques in considerable detail in Chapter 8.

D. Square wave

A square wave is a signal that varies in time as shown in Figure 1.23. Like the sinewave, it is characterized by amplitude and frequency (and perhaps phase). A linear circuit driven by a square wave rarely responds with a square wave. For a square wave, the peak amplitude and the rms amplitude are the same.

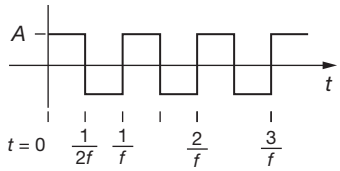


Figure 1.23. Square wave.

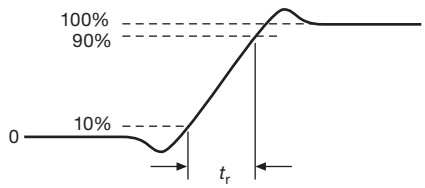


Figure 1.24. Rise time of a step waveform.

The edges of a square wave are not perfectly square; in typical electronic circuits the *rise time* t_r ranges from a few nanoseconds to a few microseconds. Figure 1.24 shows the sort of thing usually seen. The rise time is conventionally defined as the time required for the signal to go from 10% to 90% of its total transition.

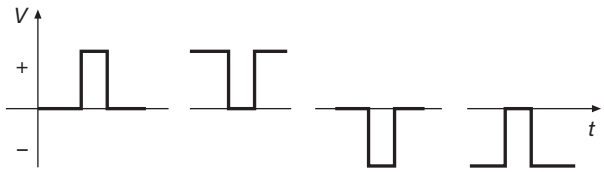


Figure 1.25. Positive- and negative-going pulses of both polarities.

E. Pulses

A pulse is a signal that looks like the objects shown in Figure 1.25. It is defined by amplitude and pulse width. You can generate a train of periodic (equally spaced) pulses, in which case you can talk about the frequency, or pulse repetition rate, and the "duty cycle," the ratio of pulse width to repetition period (duty cycle ranges from zero to 100%). Pulses can have positive or negative polarity; in addition, they can be "positive-going" or "negative-going." For

instance, the second pulse in Figure 1.25 is a negative-going pulse of positive polarity.

F. Steps and spikes

Steps and spikes are signals that are talked about a lot but are not so often used. They provide a nice way of describing what happens in a circuit. If you could draw them, they would look something like the example in Figure 1.26. The step function is part of a square wave; the spike is simply a jump of vanishingly short duration.

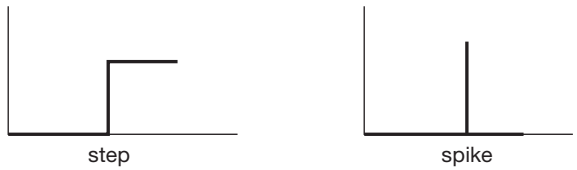


Figure 1.26. Steps and spikes.

1.3.4 Logic levels

Pulses and square waves are used extensively in digital electronics, in which predefined voltage levels represent one of two possible states present at any point in the circuit. These states are called simply HIGH and LOW, and correspond to the 1 (true) and 0 (false) states of Boolean logic (the algebra that describes such two-state systems).

Precise voltages are not necessary in digital electronics. You need only to distinguish which of the two possible states is present. Each digital logic family therefore specifies legal HIGH and LOW states. For example, the "74LVC" digital logic family runs from a single +3.3 V supply, with output levels that are typically 0 V (LOW) and 3.3 V (HIGH), and an input decision threshold of 1.5 V. However, actual outputs can be as much as 0.4 V away from ground or from +3.3 V without malfunction. We'll have much more to say about logic levels in Chapters 10 through 12.

1.3.5 Signal sources

Often the source of a signal is some part of the circuit you are working on. But for test purposes a flexible signal source is invaluable. They come in three flavors: signal generators, pulse generators, and function generators.

A. Signal generators

Signal generators are sinewave oscillators, usually equipped to give a wide range of frequency coverage, with provision for precise control of amplitude (using a resistive divider network called an *attenuator*). Some units let you *modulate* (i.e., vary in time) the output amplitude ("AM" for "amplitude modulated") or frequency ("FM" for "frequency modulated"). A variation on this theme is the *sweep generator*, a signal generator that can sweep its output frequency repeatedly over some range. These are handy for testing circuits whose properties vary with frequency in a particular way, e.g., "tuned circuits" or filters. Nowadays these devices, as well as most test instruments, are available in configurations that allow you to program the frequency, amplitude, etc., from a computer or other digital instrument.

For many signal generators the signal source is a *frequency synthesizer*, a device that generates sinewaves whose frequencies can be set precisely. The frequency is set digitally, often to eight significant figures or more, and is internally synthesized from a precise standard (a standalone quartz-crystal oscillator or rubidium frequency standard, or a GPS-derived oscillator) by digital methods we will discuss later (§13.13.6). Typical of synthesizers is the programmable SG384 from Stanford Research Systems, with a frequency range of 1 μHz to 4 GHz, an amplitude range of -110 dBm to $+16.5$ dBm (0.7 μV to 1.5 V, rms), and various modulation modes such as AM, FM, and ΦM; it costs about $4,600. You can get synthesized sweep generators, and you can get synthesizers that produce other waveforms (see *Function Generators*, below). If your requirement is for no-nonsense accurate frequency generation, you can't beat a synthesizer.

B. Pulse generators

Pulse generators make only pulses, but what pulses! Pulse width, repetition rate, amplitude, polarity, rise time, etc., may all be adjustable. The fastest ones go up to gigahertz pulse rates. In addition, many units allow you to generate pulse pairs, with settable spacing and repetition rate, or even programmable patterns (they are sometimes called pattern generators). Most contemporary pulse generators are provided with logic-level outputs for easy connection to digital circuitry. As with signal generators, these come in the programmable variety.

C. Function generators

In many ways function generators are the most flexible signal sources of all. You can make sine, triangle, and square waves over an enormous frequency range (0.01 Hz to 30 MHz is typical), with control of amplitude and dc offset (a constant-dc voltage added to the signal). Many of them have provision for frequency sweeping, often in

several modes (linear or logarithmic frequency variation versus time). They are available with pulse outputs (although not with the flexibility you get with a pulse generator), and some of them have provision for modulation.

Traditional function generators used analog circuitry, but contemporary models generally are synthesized digital function generators, exhibiting all the flexibility of a function generator along with the stability and accuracy of a frequency synthesizer. In addition, they let you program an "arbitrary" waveform, specifying the amplitude at a set of equally spaced points. An example is the Tektronix AFG3102, with a lower frequency limit of 1 *micro*hertz, which can make sine and square waves to 100 MHz, pulses and "noise" to 50 MHz, and arbitrary waveforms (up to 128k points) to 50 MHz. It has modulation (five kinds), sweep (linear and log), and burst modes (1 to 10^6 cycles), and everything is programmable, including frequency, pulse width and rise times, modulation, and amplitude (20 mV to 10 Vpp); it even includes some bizarre built-in waveforms such as $\sin(x)/x$, exponential rise and fall, Gaussian, and Lorentzian. It has two independent outputs and costs about $5k. For general use, if you can have only one signal source, the function generator is for you.

1.4 Capacitors and ac circuits

Once we enter the world of *changing* voltages and currents, or "signals," we encounter two very interesting circuit elements that are useless in purely dc circuits: capacitors and inductors. As you will see, these humble devices, combined with resistors, complete the triad of passive linear circuit elements that form the basis of nearly all circuitry.[20] Capacitors, in particular, are essential in nearly every circuit application. They are used for waveform generation, filtering, and blocking and bypass applications. They are used in integrators and differentiators. In combination with inductors, they make possible sharp filters for separating desired signals from background. You will see some of these applications as we continue with this chapter, and there will be numerous interesting examples in later chapters.

Let's proceed, then, to look at capacitors in detail. Portions of the treatment that follows are necessarily mathematical in nature; the reader with little mathematical preparation may find the math review in Appendix A helpful. In any case, an understanding of the details is less important in the long run than an understanding of the results.

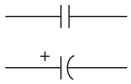


Figure 1.27. Capacitors. The curved electrode indicates the negative terminal of a polarized capacitor, or the "outer foil" of a wrapped-film capacitor.

1.4.1 Capacitors

A capacitor (Figure 1.27) (the old-fashioned name was *condenser*) is a device that has two wires sticking out of it and has the property

$$Q = CV. \tag{1.13}$$

Its basic form is a pair of closely-spaced metal plates, separated by some insulating material, as in the rolled-up "axial-film capacitor" of Figure 1.28. A capacitor of C farads with V volts across its terminals has Q coulombs of stored charge on one plate and $-Q$ on the other. The capacitance is proportional to the area and inversely proportional to the spacing. For the simple parallel-plate capacitor, with separation d and plate area A (and with the spacing d much less than the dimensions of the plates), the capacitance C is given by

$$C = 8.85\times10^{-14}\,\varepsilon A/d \text{ F}, \tag{1.14}$$

where ε is the dielectric constant of the insulator, and the dimensions are measured in centimeters. It takes a lot of area, and tiny spacing, to make the sort of capacitances commonly used in circuits.[21] For example, a pair of 1 cm^2 plates separated by 1 mm is a capacitor of slightly less than 10^{-12} F (a picofarad); you'd need 100,000 of them just to create the 0.1 μF capacitor of Figure 1.28 (which is nothing special; we routinely use capacitors with many microfarads of capacitance). Ordinarily you don't need to calculate capacitances, because you buy a capacitor as an electronic component.

To a first approximation, capacitors are devices that might be considered simply frequency-dependent resistors.

[20] Readers of the scientific journal *Nature* (London) were greeted, in 2008, with an article titled "The missing memristor found" (D. B. Strukov et al., **453**, 80, 2008), purporting to have found a heretofore missing "fourth fundamental [passive circuit] element." We are skeptical. However the controversy is ultimately resolved, it should be noted that the memristor is a *non*linear element; there are only three *linear* passive 2-terminal circuit elements.

[21] And it doesn't hurt to have a high dielectric constant, as well: air has ε=1, but plastic films have ε=2.1 (polypropylene) or 3.1 (polyester). And certain ceramics are popular among capacitor makers: ε=45 ("C0G" type) or 3000 ("X7R" type).

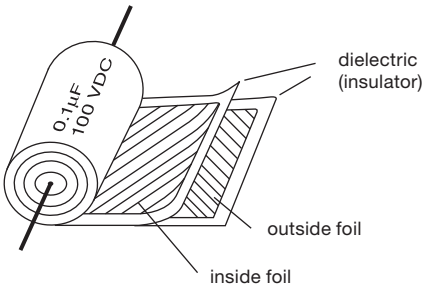


Figure 1.28. You get a lot of area by rolling up a pair of metallized plastic films. And it's great fun unrolling one of these axial-lead Mylar capacitors (ditto for the old-style golf balls with their lengthy wound-up rubber band).

They allow you to make frequency-dependent voltage dividers, for instance. For some applications (bypass, coupling) this is almost all you need to know, but for other applications (filtering, energy storage, resonant circuits) a deeper understanding is needed. For example, ideal capacitors cannot dissipate power, even though current can flow through them, because the voltage and current are 90° out of phase.

Before launching into the details of capacitors in the following dozen pages (including some necessary mathematics that describes their behavior in time and in frequency), we wish to emphasize those first two applications – bypass and coupling – because they are the most common uses of capacitors, and they are easy to understand at the simplest level. We'll see these in detail later (§§1.7.1C and 1.7.16A), but no need to wait – it's easy, and intuitive. Because a capacitor looks like an open circuit at dc, it lets you couple a varying signal while blocking its average dc level. This is a *blocking* capacitor (also called a *coupling* capacitor), as in Figure 1.93. Likewise, because a capacitor looks like a short circuit at high frequencies, it suppresses ("bypasses") signals where you don't want them, for example on the dc voltages that power your circuits, as in Figure 8.80A (where capacitors are suppressing signals on the +5 V and −5 V dc supply voltages, and also on the base terminal of transistor Q_2).[22] Demographically, these two applications account for the vast majority of capacitors that are wired into the world's circuits.

Taking the derivative of the defining equation 1.13, you get

$$I = C\frac{dV}{dt}. \tag{1.15}$$

So a capacitor is more complicated than a resistor: the current is not simply proportional to the voltage, but rather to the rate of change of voltage. If you change the voltage across a farad by 1 volt per second, you are supplying an amp. Conversely, if you supply an amp, its voltage changes by 1 volt per second. A farad is an enormous capacitance, and you usually deal in microfarads (μF), nanofarads (nF), or picofarads (pF).[23] For instance, if you supply a current of 1 mA to 1μF, the voltage will rise at 1000 volts per second. A 10 ms pulse of this current will increase the voltage across the capacitor by 10 volts (Figure 1.29).

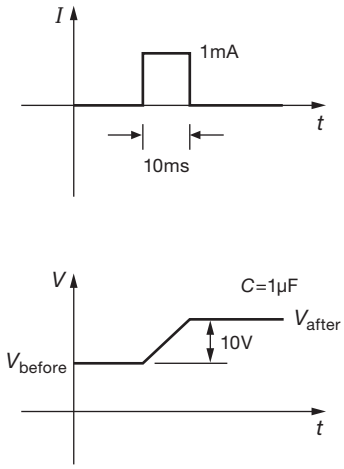


Figure 1.29. The voltage across a capacitor changes when a current flows through it.

When you charge up a capacitor, you're supplying energy. The capacitor doesn't get hot; instead, it stores the energy in its internal electric fields. It's an easy exercise to discover for yourself that the amount of stored energy in a charged capacitor is just

$$U_C = \frac{1}{2}CV^2, \tag{1.16}$$

where U_C is in joules for C in farads and V in volts. This is an important result; we'll see it often.

Exercise 1.14. Take the energy challenge: imagine charging up a capacitor of capacitance C, from 0 V to some final voltage V_f. If you do it right, the result won't depend on how you get there,

[22] Ironically, these essential bypass capacitors are so taken for granted that they are usually omitted from schematic diagrams (a practice we follow in this book). Don't make the mistake of omitting them also from your actual circuits!

[23] To make matters confusing to the uninitiated, the units are often omitted on capacitor values specified in schematic diagrams. You have to figure it out from the context.

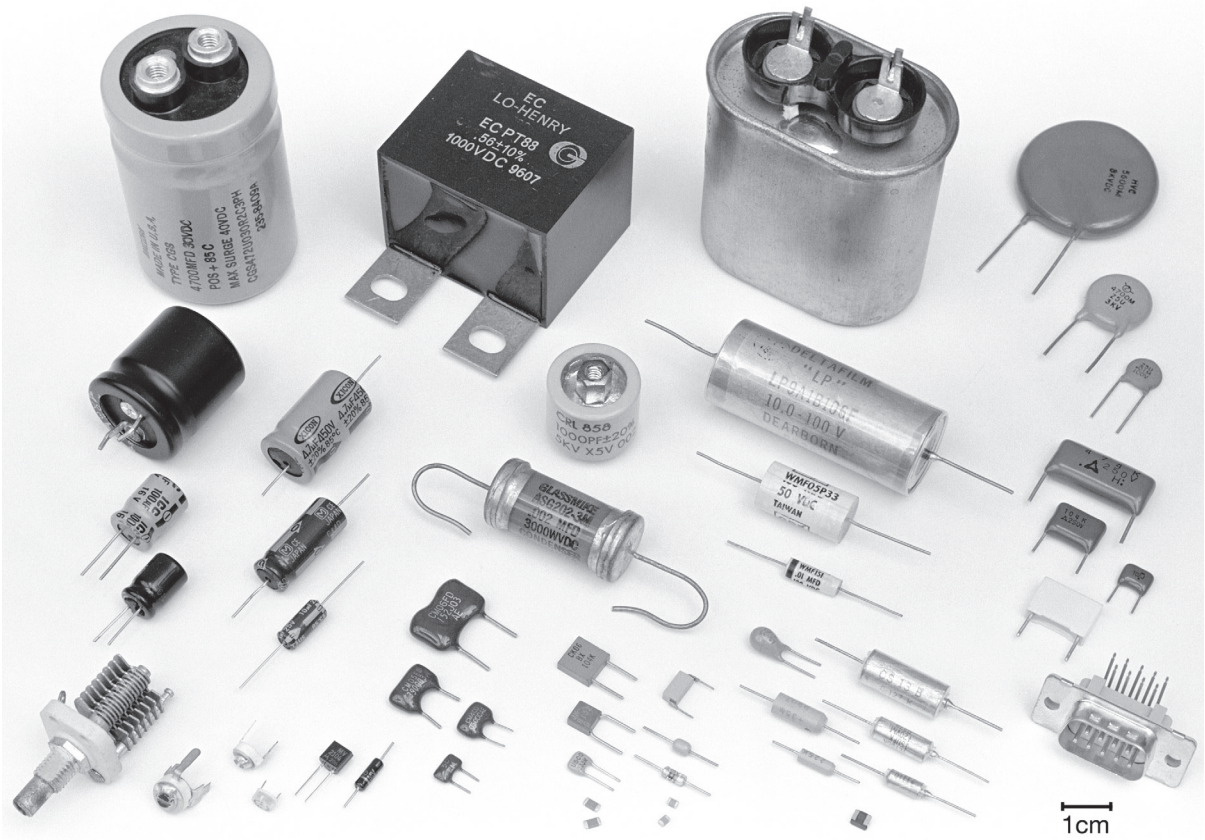


Figure 1.30. Capacitors masquerade as anything they like! Here is a representative collection. In the lower left are small-value variable capacitors (one air, three ceramic), with large-value polarized aluminum electrolytics above them (the three on the left have *radial leads*, the three on the right have *axial leads*, and the specimen with screw terminals at top is often called a *computer electrolytic*). Next in line across the top is a low-inductance film capacitor (note the wide strap terminals), then an oil-filled paper capacitor, and last, a set of disc ceramic capacitors running down the right. The four rectangular objects below are film capacitors (polyester, polycarbonate, or polypropylene). The D-subminiature connector seems misplaced – but it is a *filtered* connector, with a 1000 pF capacitor from each pin to the shell. To its left is a group of seven polarized tantalum electrolytics (five with axial leads, one radial, and one surface-mount). The three capacitors above them are axial-film capacitors. The ten capacitors at bottom center are all ceramic types (four with radial leads, two axial, and four surface-mount *chip capacitors*); above them are high-voltage capacitors – an axial-glass capacitor, and a ceramic *transmitting capacitor* with screw terminals. Finally, below them and to the left are four mica capacitors and a pair of diode-like objects known as *varactors*, which are voltage-variable capacitors made from a diode junction.

so you don't need to assume constant current charging (though you're welcome to do so). At any instant the rate of flow of energy into the capacitor is VI (joules/s); so you need to integrate $dU = VIdt$ from start to finish. Take it from there.

Capacitors come in an amazing variety of shapes and sizes (Figure 1.30 shows examples of most of them); with time, you will come to recognize their more common incarnations. For the smallest capacitances you may see examples of the basic parallel-plate (or cylindrical piston) construction. For greater capacitance, you need more area and closer spacing; the usual approach is to plate some conductor onto a thin insulating material (the dielectric), for instance, aluminized plastic film rolled up into a small cylindrical configuration. Other popular types are thin ceramic wafers (ceramic chip capacitors), metal foils with oxide insulators (electrolytic capacitors), and metallized mica. Each of these types has unique properties; for a brief rundown, see the section on capacitors in Chapter *1x*. In general, ceramic and polyester types are used for most noncritical circuit applications; capacitors with polycarbonate, polystyrene, polypropylene, Teflon, or glass dielectric are

used in demanding applications; tantalum capacitors are used where greater capacitance is needed; and aluminum electrolytics are used for power-supply filtering.

A. Capacitors in parallel and series

The capacitance of several capacitors in parallel is the sum of their individual capacitances. This is easy to see: put voltage V across the parallel combination; then

$$\begin{aligned} C_{\text{total}}V &= Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \cdots \\ &= C_1V + C_2V + C_3V + \cdots \\ &= (C_1 + C_2 + C_3 + \cdots)V \end{aligned}$$

or

$$C_{\text{total}} = C_1 + C_2 + C_3 + \cdots. \tag{1.17}$$

For capacitors in series, the formula is like that for resistors in parallel:

$$C_{\text{total}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots} \tag{1.18}$$

or (two capacitors only)

$$C_{\text{total}} = \frac{C_1C_2}{C_1 + C_2}.$$

Exercise 1.15. Derive the formula for the capacitance of two capacitors in series. *Hint*: because there is no external connection to the point where the two capacitors are connected together, they must have equal stored charges.

The current that flows in a capacitor during charging ($I = C\,dV/dt$) has some unusual features. Unlike resistive current, it's not proportional to voltage, but rather to the rate of change (the "time derivative") of voltage. Furthermore, unlike the situation in a resistor, the power ($V \times I$) associated with capacitive current is not turned into heat, but is stored as energy in the capacitor's internal electric field. You get all that energy back when you discharge the capacitor. We'll see another way to look at these curious properties when we talk about *reactance*, beginning in §1.7.

1.4.2 *RC* circuits: *V* and *I* versus time

When dealing with ac circuits (or, in general, any circuits that have changing voltages and currents), there are two possible approaches. You can talk about V and I versus time, or you can talk about amplitude versus signal frequency. Both approaches have their merits, and you find yourself switching back and forth according to which description is most convenient in each situation. We begin our study of ac circuits in the *time domain*. Starting with §1.7, we will tackle the *frequency domain*.

What are some of the features of circuits with capacitors? To answer this question, let's begin with the simple *RC* circuit (Figure 1.31). Application of the capacitor rules gives

$$C\frac{dV}{dt} = I = -\frac{V}{R}. \tag{1.19}$$

This is a differential equation, and its solution is

$$V = Ae^{-t/RC}. \tag{1.20}$$

So a charged capacitor placed across a resistor will discharge as in Figure 1.32. Intuition serves well here: the current that flows is (from the resistor equation) proportional to the remaining voltage; but the slope of the discharge is (from the capacitor equation) proportional to that current. So the discharge curve has to be a function whose derivative is proportional to its value, i.e., an exponential.

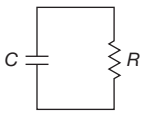


Figure 1.31. The simplest *RC* circuit.

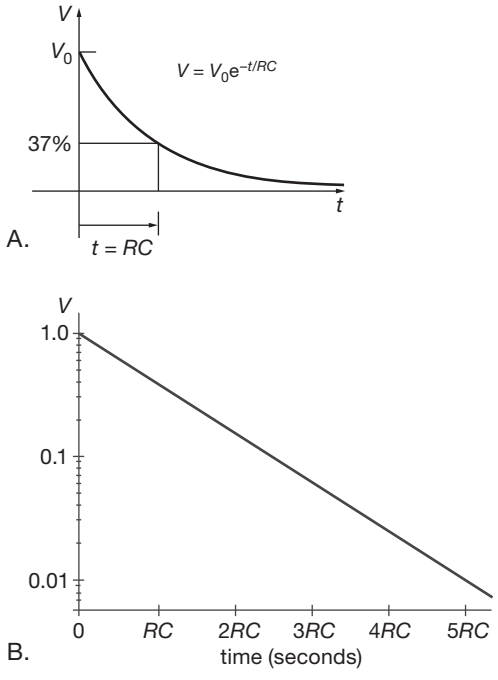


Figure 1.32. *RC* discharge waveform, plotted with (A) linear and (B) logarithmic voltage axes.

A. Time constant

The product RC is called the *time constant* of the circuit. For R in ohms and C in farads, the product RC is in seconds. A microfarad across 1.0k has a time constant of 1 ms; if the capacitor is initially charged to 1.0 V, the initial current is 1.0 mA.

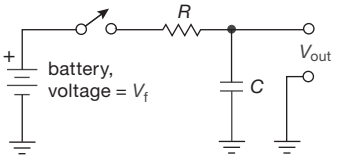


Figure 1.33. *RC* charging circuit.

Figure 1.33 shows a slightly different circuit. At time $t = 0$, someone connects the battery. The equation for the circuit is then

$$I = C\frac{dV}{dt} = \frac{V_f - V_{out}}{R},$$

with the solution

$$V_{out} = V_f + A\,e^{-t/RC}.$$

(Please don't worry if you can't follow the mathematics. What we are doing is getting some important results, which you should remember. Later we will use the results often, with no further need for the mathematics used to derive them. For readers whose knowledge of math is somewhat, uh, *rusty*, the brief review in Appendix A may prove helpful.) The constant A is determined by initial conditions (Figure 1.34): $V = 0$ at $t = 0$; therefore, $A = -V_f$, and

$$V_{out} = V_f\,(1 - e^{-t/RC}). \tag{1.21}$$

Once again there's good intuition: as the capacitor charges up, the slope (which is proportional to current, because it's a capacitor) is proportional to the *remaining* voltage (because that's what appears across the resistor, producing the current); so we have a waveform whose slope decreases proportionally to the vertical distance it has still to go – an exponential.

You can turn around the last equation to figure out the time required to reach a voltage V on the way to the final voltage V_f. Try it! (Refer to Appendix A if you need help with logarithms.) You should get

$$t = RC\log_e\left(\frac{V_f}{V_f - V}\right) \tag{1.22}$$

B. Decay to equilibrium

Eventually (when $t \gg RC$), V reaches V_f. (Presenting the "5RC rule of thumb": a capacitor charges or decays to within 1% of its final value in five time constants.) If we then change the battery voltage to some other value (say, 0 V), V will decay toward that new value with an exponential $e^{-t/RC}$. For example, replacing the battery's step input from 0 to $+V_f$ with a square-wave input $V_{in}(t)$ would produce the output shown in Figure 1.35.

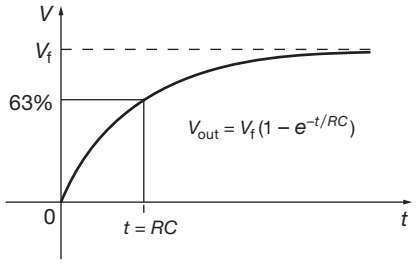


Figure 1.34. *RC* charging waveform.

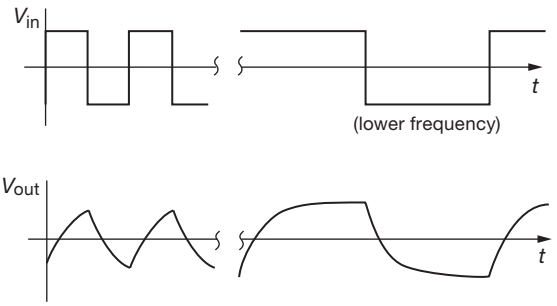


Figure 1.35. Output (lower waveforms) across a capacitor, when driven by square waves through a resistor.

Exercise 1.16. Show that the rise time (the time required for going from 10% to 90% of its final value) of this signal is 2.2RC.

You might ask the obvious next question: what about $V(t)$ for arbitrary $V_{in}(t)$? The solution involves an inhomogeneous differential equation and can be solved by standard methods (which are, however, beyond the scope of this book). You would find

$$V(t) = \frac{1}{RC}\int_{-\infty}^{t} V_{in}(\tau)e^{-(t-\tau)/RC}d\tau.$$

That is, the RC circuit averages past history at the input with a weighting factor of

$$e^{-\Delta t/RC}.$$

In practice, you seldom ask this question. Instead, you deal in the *frequency domain*, in which you ask how much of each frequency component present in the input gets through. We will get to this important topic soon (§1.7). Before we do, though, there are a few other interesting

circuits we can analyze simply with this time-domain approach.

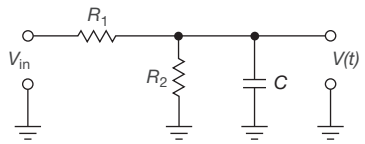


Figure 1.36. Looks complicated, but it's not! (Thévenin to the rescue.)

C. Simplification by Thévenin equivalents

We could go ahead and analyze more complicated circuits by similar methods, writing down the differential equations and trying to find solutions. For most purposes it simply isn't worth it. This is as complicated an *RC* circuit as we will need. Many other circuits can be reduced to it; take, for example, the circuit in Figure 1.36. By just using the Thévenin equivalent of the voltage divider formed by R_1 and R_2, you can find the output $V(t)$ produced by a step input for V_{in}.

Exercise 1.17. In the circuit shown in Figure 1.36, $R_1 = R_2 = 10$k, and $C = 0.1\,\mu$F. Find $V(t)$ and sketch it.

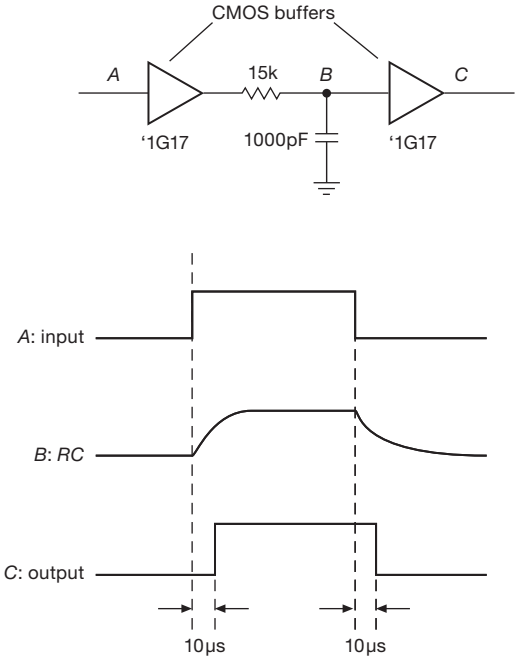


Figure 1.37. Producing a delayed digital waveform with the help of an *RC* and a pair of LVC-family logic buffers (tiny parts with a huge part number: SN74LVC1G17DCKR!).

D. A circuit example: time-delay circuit

Let's take a short detour to try out these theoretical ideas on a couple of real circuits. Textbooks usually avoid such pragmatism, especially in early chapters, but we think it's fun to apply electronics to practical applications. We'll need to introduce a few "black-box" components to get the job done, but you'll learn about them in detail later, so don't worry.

We have already mentioned logic levels, the voltages that digital circuits live on. Figure 1.37 shows an application of capacitors to produce a delayed pulse. The triangular symbols are "CMOS[24] buffers." They give a HIGH output if the input is HIGH (more than one-half the dc power-supply voltage used to power them), and vice versa. The first buffer provides a replica of the input signal, but with low source resistance, to prevent input loading by the *RC* (recall our earlier discussion of circuit loading in §1.2.5A). The *RC* output has the characteristic decays and causes the output buffer to switch 10 μs after the input transitions (an *RC* reaches 50% output after a time $t = 0.7RC$). In an actual application you would have to consider the effect of the buffer input threshold deviating from one-half the supply voltage, which would alter the delay and change the output pulse width. Such a circuit is sometimes used to delay a pulse so that something else can happen first. In designing circuits you try not too often to rely on tricks like this, but they're occasionally handy.

E. Another circuit example: "One Minute of Power"

Figure 1.38 shows another example of what can be done with simple *RC* timing circuits. The triangular symbol is a *comparator*, something we'll treat in detail later, in Chapters 4 and 10; all you need to know, for now, is that (a) it is an IC (containing a bunch of resistors and transistors), (b) it is powered from some positive dc voltage that you connect to the pin labeled "V_+," and (c) it drives its output (the wire sticking out to the right) either to V_+ or to ground, depending on whether the input labeled "+" is more or less positive than the input labeled "−," respectively. (These are called the *non-inverting* and the *inverting* inputs, respectively.) It doesn't draw any current from its inputs, but it happily drives loads that require up to 20 mA or so. And a comparator is decisive: its output is either "HIGH" (at V_+) or "LOW" (ground).

Here's how the circuit works: the voltage divider R_3R_4 holds the (−) input at 37% of the supply voltage, in this case about +1.8 V; let's call that the "reference voltage."

[24] Complementary metal-oxide semiconductor, the dominant form of digital logic, as we'll see from Chapter 10 onward.

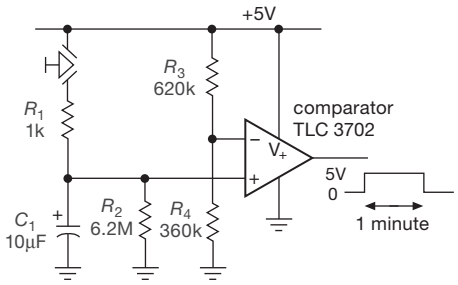


Figure 1.38. *RC* timing circuit: one push → one minute!.

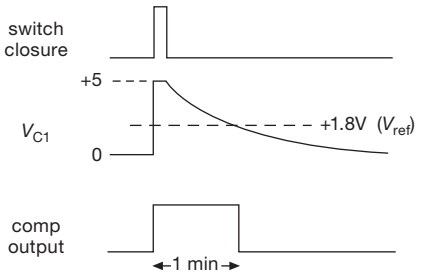


Figure 1.39. Producing a delayed digital waveform for the circuit of Figure 1.38. The voltage V_{C1} has a rise time of $R_1C_1 \approx 10$ ms.

So if the circuit has been sitting there for a while, C_1 is fully discharged, and the comparator's output is at ground. When you push the START button momentarily, C_1 charges quickly (10 ms time constant) to +5 V, which makes the comparator's output switch to +5 V; see Figure 1.39. After the button is released, the capacitor discharges exponentially toward ground, with a time constant of $\tau = R_2C_1$, which we've set to be 1 minute. At that time its voltage crosses the reference voltage, so the comparator's output switches rapidly back to ground. (Note that we've conveniently chosen the reference voltage to be a fraction $1/e$ of V_+, so it takes exactly one time constant τ for that to happen. For R_2 we used the closest standard value to 6 MΩ; see Appendix C.) The bottom line is that the output spends 1 minute at +5 V, after the button is pushed.

We'll add a few details shortly, but first let's use the output to do some interesting things, which are shown in Figures 1.40A–D. You can make a self-stopping flashlight key fob by connecting its output to an LED; you need to put a resistor in series, to set the current (we'll say much more about this later). If you prefer to make some noise, you could connect a *piezo beeper* to beep continuously (or intermittently) for a minute (this might be an end-of-cycle signal for a clothes dryer). Another possibility is to attach a small electromechanical *relay*, which is just an electrically operated mechanical switch, to provide a pair of contacts that can activate pretty much any load you care to switch on and off. The use of a relay has the important property that the load – the circuit being switched by the relay – is electrically isolated from the +5 V and ground of the timing circuit itself.

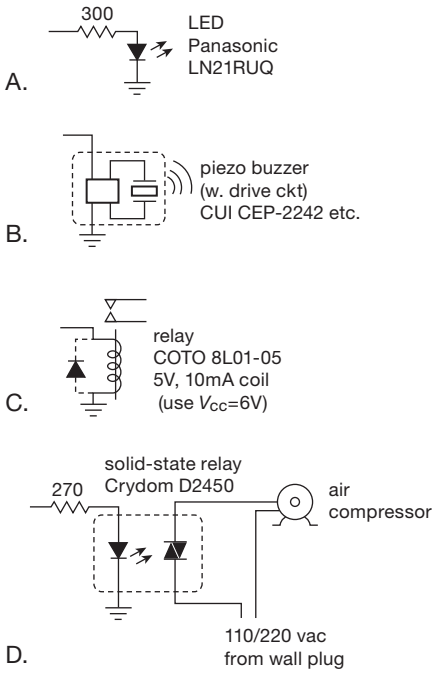


Figure 1.40. Driving interesting stuff from the output of the timer circuit in Figure 1.38.

Finally, for turning serious industrial machinery on and off, you would probably use a hefty *solid-state relay* (SSR, §12.7), which has within it an infrared LED coupled to an ac switching device known as a *triac*. When activated, the triac acts as an excellent mechanical switch, capable of switching many amperes, and (like the electromechanical relay) is fully isolated electrically from its input circuit. The example shows this thing hooked to an air compressor, so your friends will get a minute's worth of air to inflate their tires at your home "gas station" (literally!) after they drop a quarter into your coin-initiated timer. You could do an analogous thing with a coin-operated hot shower (but, hey, we get only *one minute*?!).

Some details: (a) in the circuit of Figure 1.38 you could omit R_1 and the circuit would still work, but there would be a large transient current when the discharged capacitor was initially connected across the +5 V supply (recall $I = C\,dV/dt$: here you would be trying to produce 5 V of "dV" in roughly 0 s of "dt"). By adding a series resistor R_1 you limit the peak current to a modest 5 mA while charging

the capacitor fast enough ($>99\%$ in $5\,RC$ time constants, or 0.05 s). (b) The comparator output would likely bounce around a bit (see Figure 4.31), just as the (+) input crosses the reference voltage in its leisurely exponential promenade toward ground, owing to unavoidable bits of electrical noise. To fix this problem you usually see the circuit arranged so that some of the output is coupled back to the input in a way that reinforces the switching (this is officially called *hysteresis*, or *positive feedback*; we'll see it in Chapters 4 and 10). (c) In electronic circuits it's always a good idea to *bypass* the dc supply by connecting one or more capacitors between the dc "rail" and ground. The capacitance is noncritical – values of 0.1 μF to 10 μF are commonly used; see §1.7.16A.

Our simple examples above all involved turning some load on and off. But there are other uses for an electronic *logic* signal, like the output of the comparator, that is in one of two possible binary states, called HIGH and LOW (in this case +5 V and ground), 1 and 0, or TRUE and FALSE. For example, such a signal can enable or disable the operation of some other circuit. Imagine that the opening of a car door triggers our 1 min HIGH output, which then enables a keypad to accept a security code so you can start the car. After a minute, if you haven't managed to type the magic code, it shuts off, thus ensuring a certain minimum of operator sobriety.

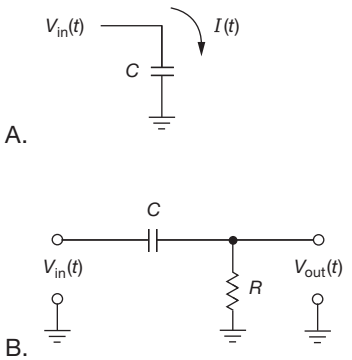


Figure 1.41. Differentiators. A. Perfect (except it has no output terminal). B. Approximate (but at least it has an output!).

1.4.3 Differentiators

You can make a simple circuit that differentiates an input signal; that is, $V_{out} \propto dV_{in}/dt$. Let's take it in two steps.

1. First look at the (impractical) circuit in Figure 1.41A: The input voltage $V_{in}(t)$ produces a current through the capacitor of $I_{cap} = C\,dV_{in}/dt$. That's just what we want – if we could somehow use the current through C as our "output"! But we can't.[25]

2. So we add a small resistor from the low side of the capacitor to ground, to act as a "current-sensing" resistor (Figure 1.41B). The good news is that we now have an output proportional to the current through the capacitor. The bad news is that the circuit is no longer a perfect mathematical differentiator. That's because the voltage across C (whose derivative produces the current we are sensing with R) is no longer equal to V_{in}; it now equals the difference between V_{in} and V_{out}. Here's how it goes: the voltage across C is $V_{in} - V_{out}$, so

$$I = C\frac{d}{dt}(V_{in} - V_{out}) = \frac{V_{out}}{R}.$$

If we choose R and C small enough so that $dV_{out}/dt \ll dV_{in}/dt$, then

$$C\frac{dV_{in}}{dt} \approx \frac{V_{out}}{R}$$

or

$$V_{out}(t) \approx RC\frac{d}{dt}V_{in}(t).$$

That is, we get an output proportional to the rate of change of the input waveform.

To keep $dV_{out}/dt \ll dV_{in}/dt$, we make the product RC small, taking care not to "load" the input by making R too small (at the transition the change in voltage across the capacitor is zero, so R is the load seen by the input). We will have a better criterion for this when we look at things in the frequency domain (§1.7.10). If you drive this circuit with a square wave, the output will be as shown in Figure 1.42.

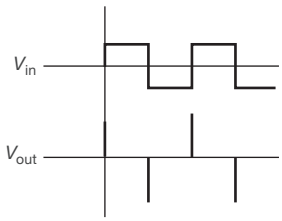


Figure 1.42. Output waveform (bottom) from differentiator driven by a square wave.

Differentiators are handy for detecting *leading edges* and *trailing edges* in pulse signals, and in digital circuitry you sometimes see things like those depicted in Figure 1.43. The RC differentiator generates spikes at the transitions of the input signal, and the output buffer converts

[25] Devotees of the cinema will be reminded of Dr. Strangelove's outburst: "The whole point of a doomsday machine is lost … if you keep it *secret*!"

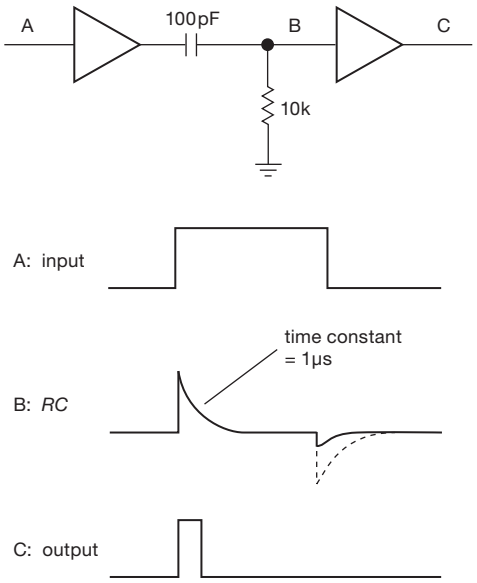


Figure 1.43. Leading-edge detector.

the spikes to short square-topped pulses. In practice, the negative spike will be small because of a diode (a handy device discussed in §1.6) built into the buffer.

To inject some real-world realism here, we hooked up and made some measurements on a differentiator that we configured for high-speed signals. For this we used $C=1$ pF and $R=50\,\Omega$ (the latter is the world-wide standard for high-speed circuits, see Appendix H), we drove it with a 5 V step of settable slew rate (i.e., dV/dt). Figure 1.44 shows both input and output waveforms, for three choices of $dV_{\rm in}/dt$. At these speeds (note the horizontal scale: 4 *nano*seconds per division!) circuits often depart from ideal performance, as can be seen in the fastest risetime. The two slower steps show reasonable behavior; that is, a flat-top output waveform during the input's upward ramp; check for yourself that the output amplitude is correctly predicted by the formula.

A. Unintentional capacitive coupling

Differentiators sometimes crop up unexpectedly, in situations where they're not welcome. You may see signals like those shown in Figure 1.45. The first case is caused by a square wave somewhere in the circuit coupling capacitively to the signal line you're looking at; that might indicate a missing resistor termination on your signal line. If not, you must either reduce the source resistance of the signal line or find a way to reduce capacitive coupling from the offending square wave. The second case is typical of what you might see when you look at a square wave, but have a

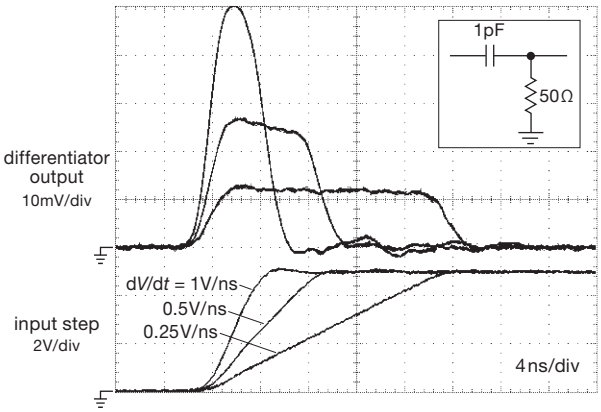


Figure 1.44. Three fast step waveforms, differentiated by the *RC* network shown. For the fastest waveform (10^9 volts per second!), imperfections in the components and measuring instruments cause deviation from the ideal.

broken connection somewhere, usually at the scope probe. The very small capacitance of the broken connection combines with the scope input resistance to form a differentiator. Knowing that you've got a differentiated "something" can help you find the trouble and eliminate it.

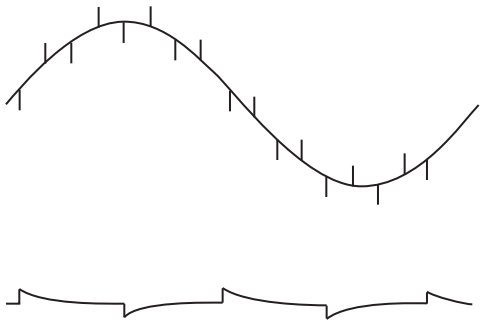


Figure 1.45. Two examples of unintentional capacitive coupling.

1.4.4 Integrators

If *RC* circuits can take derivatives, why not integrals? As before, let's take it in two steps.

1. Imagine that we have an input signal that is a time-varying *current* versus time, $I_{\rm in}(t)$ (Figure 1.46A).[26] That input current is precisely the current through the capacitor, so $I_{\rm in}(t) = C\,dV(t)/dt$, and therefore $V(t) = \int I_{\rm in}(t)\,dt$.

[26] We're used to thinking of signals as time-varying *voltages*; but we'll see how we can convert such signals to proportional time-varying *currents*, by using "voltage-to-current converters" (with the fancier name "transconductance amplifiers").

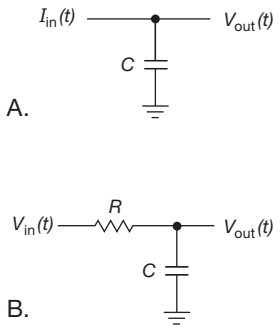


Figure 1.46. Integrator. A. Perfect (but requires a *current* input signal). B. Approximate (see text).

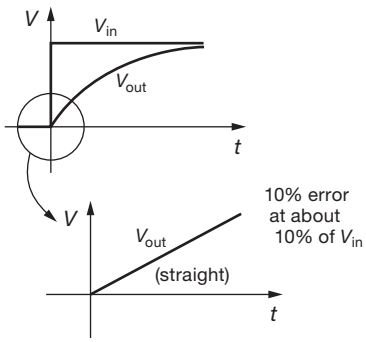


Figure 1.47. Integrator approximation is good when $V_{out} \ll V_{in}$.

That's just what we wanted! Thus a simple capacitor, with one side grounded, is an integrator, *if* we have an input signal in the form of a current $I_{in}(t)$. Most of the time we don't, though.

2. So we connect a resistor in series with the more usual input *voltage* signal $V_{in}(t)$, to convert it to a current (Figure 1.46B). The good news is that it works, sort of. The bad news is that the circuit is no longer a perfect integrator. That's because the current through C (whose integral produces the output voltage) is no longer proportional to V_{in}; it is now proportional to the difference between V_{in} and V. Here's how it goes: the voltage across R is $V_{in}-V$, so

$$I = C\frac{dV}{dt} = \frac{V_{in}-V}{R}.$$

If we manage to keep $V \ll V_{in}$, by keeping the product RC large,[27] then

$$C\frac{dV}{dt} \approx \frac{V_{in}}{R}$$

or

$$V(t) = \frac{1}{RC}\int^{t} V_{in}(t)\,dt + \text{constant}.$$

That is, we get an output proportional to the integral over time of the input waveform. You can see how the approximation works for a square-wave input: $V(t)$ is then the exponential charging curve we saw earlier (Figure 1.47). The first part of the exponential is a ramp, the integral of a constant; as we increase the time constant RC, we pick off a smaller part of the exponential, i.e., a better approximation to a perfect ramp.

Note that the condition $V \ll V_{in}$ is the same as saying that I is proportional to V_{in}, which was our first integrator circuit. A large voltage across a large resistance approximates a current source and, in fact, is frequently used as one.

Later, when we get to operational amplifiers and feedback, we will be able to build integrators without the restriction $V_{out} \ll V_{in}$. They will work over large frequency and voltage ranges with negligible error.

The integrator is used extensively in analog computation. It is a useful subcircuit that finds application in control systems, feedback, analog–digital conversion, and waveform generation.

A. Ramp generators

At this point it is easy to understand how a ramp generator works. This nice circuit is extremely useful, for example in timing circuits, waveform and function generators, analog oscilloscope sweep circuits, and analog-digital conversion circuitry. The circuit uses a constant current to charge a capacitor (Figure 1.48). From the capacitor equation $I = C(dV/dt)$, you get $V(t) = (I/C)t$. The output waveform is as shown in Figure 1.49. The ramp stops when the current source "runs out of voltage," i.e., reaches the limit of its compliance. On the same figure is shown the curve for a simple RC, with the resistor tied to a voltage source equal to the compliance of the current source, and with R chosen so that the current at zero output voltage is the same as that of the current source. (Real current sources generally have output compliances limited by the power-supply voltages used in making them, so the comparison is realistic.) In the next chapter, which deals with transistors, we will design some current sources, with some refinements to follow in the chapters on operational amplifiers (op-amps) and FETs. Exciting things to look forward to!

Exercise 1.18. A current of 1 mA charges a 1 μF capacitor. How long does it take the ramp to reach 10 volts?

[27] Just as with the differentiator, we'll have another way of framing this criterion in §1.7.10.

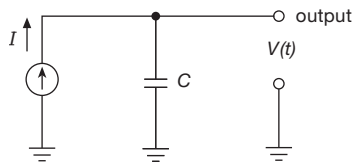


Figure 1.48. A constant-current source charging a capacitor generates a ramp voltage waveform.

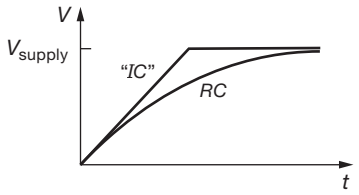


Figure 1.49. Constant-current charging (with finite compliance) versus *RC* charging.

1.4.5 Not quite perfect...

Real capacitors (the kind you can see, and touch, and pay money for) generally behave according to theory; but they have some additional "features" that can cause problems in some demanding applications. For example, all capacitors exhibit *some series resistance* (which may be a function of frequency), and *some series inductance* (see the next section), along with some frequency-dependent parallel resistance. Then there's a "memory" effect (known as *dielectric absorption*), which is rarely discussed in polite society: if you charge a capacitor up to some voltage V_0 and hold it there for a while, and then discharge it to 0 V, then when you remove the short across its terminals it will tend to drift back a bit toward V_0.

Don't worry about this stuff, for now. We'll treat in detail these effects (and other oddities of real-world components) in the advanced topics Chapter *1x*.

1.5 Inductors and transformers

1.5.1 Inductors

If you understand capacitors, you shouldn't have great trouble with inductors (Figure 1.50). They're closely related to capacitors: the rate of current change in an inductor is proportional to the voltage applied across it (for a capacitor it's the other way around – the rate of *voltage* change is proportional to the *current* through it). The defining equation for an inductor is

$$V = L\frac{dI}{dt}, \tag{1.23}$$

where L is called the *inductance* and is measured in henrys (or mH, μH, nH, etc.). Putting a constant voltage across an inductor causes the current to rise as a ramp (compare with a capacitor, in which a constant *current* causes the *voltage* to rise as a ramp); 1 V across 1 H produces a current that increases at 1 amp per second.

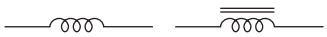


Figure 1.50. Inductors. The parallel-bar symbol represents a *core* of magnetic material.

Just as with capacitors, the energy invested in ramping up the current in an inductor is stored internally, here in the form of magnetic fields. And the analogous formula is

$$U_L = \frac{1}{2}LI^2, \tag{1.24}$$

where U_L is in joules (watt seconds) for L in henrys and I in amperes. As with capacitors, this is an important result, one which lies at the core of switching power conversion (exemplified by those little black "wall-warts" that provide power to all manner of consumer electronic gadgets). We'll see lots more of this in Chapter 9.

The symbol for an inductor looks like a coil of wire; that's because, in its simplest form, that's all it is. Its somewhat peculiar behavior comes about because inductors are magnetic devices, in which two things are going on: current flowing through the coil creates a magnetic field aligned along the coil's axis; and then changes in that field produce a voltage (sometimes called "back EMF") in a way that tries to cancel out those changes (an effect known as Lenz's law). The inductance L of a coil is simply the ratio of magnetic flux passing through the coil divided by the current through the coil that produces that flux (multiplied by an overall constant). Inductance depends on the coil geometry (e.g., diameter and length) and the properties of any magnetic material (the "core") that may be used to confine the magnetic field. That's all you need to understand why the inductance of a coil of given geometry is proportional to the square of the number of turns.

Exercise 1.19. Explain why $L \propto n^2$ for an inductor consisting of a coil of n turns of wire, maintaining fixed diameter and length as n is varied.

We'll get into some more detail in the Chapter *1x*. But it's worth displaying a semi-empirical formula for the approximate inductance L of a coil of diameter d and length l, in which the n^2 dependence is on display:

$$L \approx K\frac{d^2n^2}{18d + 40l} \quad \mu\text{H},$$

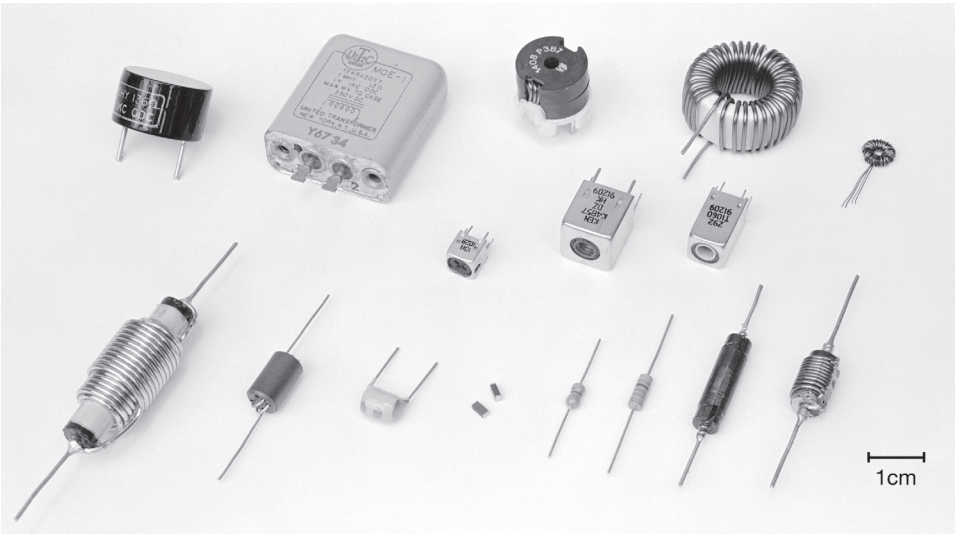


Figure 1.51. Inductors. Top row, left to right: encapsulated toroid, hermetically-sealed toroid, board-mount pot core, bare toroid (two sizes). Middle row: slug-tuned ferrite-core inductors (three sizes). Bottom row: high-current ferrite-core choke, ferrite-bead choke, dipped radial-lead ferrite-core inductor, surface-mount ferrite chokes, molded axial-lead ferrite-core chokes (two styles), lacquered ferrite-core inductors (two styles).

where $K = 1.0$ or 0.395 for dimensions in inches or centimeters, respectively. This is known as Wheeler's formula and is accurate to 1% as long as $l > 0.4d$.

As with capacitive current, inductive current is not simply proportional to voltage (as in a resistor). Furthermore, unlike the situation in a resistor, the power associated with inductive current ($V \times I$) is not turned into heat, but is stored as energy in the inductor's magnetic field (recall that for a capacitor the power associated with capacitive current is likewise not dissipated as heat, but is stored as energy in the capacitor's electric field). You get all that energy back when you interrupt the inductor's current (with a capacitor you get all the energy back when you discharge the voltage to zero).

The basic inductor is a coil, which may be just a loop with one or more turns of wire; or it may be a coil with some length, known as a solenoid. Variations include coils wound on various core materials, the most popular being iron (or iron alloys, laminations, or powder) and ferrite (a gray, nonconductive, brittle magnetic material). These are all ploys to multiply the inductance of a given coil by the "permeability" of the core material. The core may be in the shape of a rod, a toroid (doughnut), or even more bizarre shapes, such as a "pot core" (which has to be seen to be understood; the best description we can come up with is a doughnut mold split horizontally in half, if doughnuts were made in molds). See Figure 1.51 for some typical geometries.

Inductors find heavy use in radiofrequency (RF) circuits, serving as RF "chokes" and as parts of tuned circuits (§1.7.14). A pair of closely coupled inductors forms the interesting object known as a transformer. We will talk briefly about them shortly.

An inductor is, in a real sense, the opposite of a capacitor.[28] You will see how that works out later in the chapter when we deal with the important subject of *impedance*.

A. A look ahead: some magic with inductors

Just to give a taste of some of the tricks that you can do with inductors, take a look at Figure 1.52. Although we'll understand these circuits a lot better when we go at them in Chapter 9, it's possible to see what's going on with what we know already. In Figure 1.52A the left-hand side of inductor L is alternately switched between a dc input voltage V_{in} and ground, at some rapid rate, spending equal times

[28] In practice, however, capacitors are much more widely used in electronic circuits. That is because practical inductors depart significantly from ideal performance – by having winding resistance, core losses, and self-capacitance – whereas practical capacitors are nearly perfect (more on this in Chapter *1x*). Inductors are indispensable, however, in *switching power converters*, as well as in tuned *LC* circuits for RF applications.

connected to each (a "50% duty cycle"). But the defining equation $V = L\,dI/dt$ requires that the *average* voltage across an inductor must be zero, otherwise the magnitude of its average current is rising without limit. (This is sometimes called the *volt-second balance* rule.) From this it follows that the average output voltage is half the input voltage (make sure you understand why). In this circuit C_2 acts as a storage capacitor for steadying the output voltage (more on this later, and in Chapter 9).

Producing an output that is half the voltage of an input is not very exciting; after all, a simple voltage divider can do that. But, unlike a voltage divider, this circuit does not waste any energy; apart from non-idealities of the components, it is 100% efficient. And in fact this circuit is widely used in power conversion; it's called a "synchronous buck converter."

But look now at Figure 1.52B, which is just a turned-around version of Figure 1.52A. This time, volt-second balance requires that the output voltage be *twice* the input voltage. You can't do *that* with a voltage divider! Once again, the output capacitor (C_1 this time) serves to hold the output voltage steady by storing charge. This configuration is called a "synchronous boost converter."

These and other switching converters are discussed extensively in Chapter 9, where Table 9.5 lists some fifty representative types.

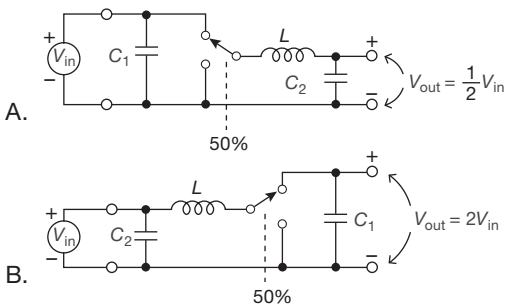


Figure 1.52. Inductors let you do neat tricks, such as *increasing* a dc input voltage.

1.5.2 Transformers

A transformer is a device consisting of two closely coupled coils (called primary and secondary). An ac voltage applied to the primary appears across the secondary, with a voltage multiplication proportional to the turns ratio of the transformer, and with a current multiplication inversely proportional to the turns ratio. Power is conserved. Figure 1.53 shows the circuit symbol for a laminated-core transformer (the kind used for 60 Hz ac power conversion).

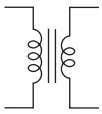


Figure 1.53. Transformer.

Transformers are quite efficient (output power is very nearly equal to input power); thus, a step-up transformer gives higher voltage at lower current. Jumping ahead for a moment, a transformer of turns ratio n increases the impedance by n^2. There is very little primary current if the secondary is unloaded.

Power transformers (meant for use from the 115 V powerline) serve two important functions in electronic instruments: they change the ac line voltage to a useful (usually lower) value that can be used by the circuit, and they "isolate" the electronic device from actual connection to the powerline, because the windings of a transformer are electrically insulated from each other. They come in an enormous variety of secondary voltages and currents: outputs as low as 1 volt or so up to several thousand volts, current ratings from a few milliamps to hundreds of amps. Typical transformers for use in electronic instruments might have secondary voltages from 10 to 50 volts, with current ratings of 0.1 to 5 amps or so. A related class of transformers is used in electronic power conversion, in which plenty of power is flowing, but typically as pulse or square waveforms, and at much higher frequencies (50 kHz to 1 MHz is typical).

Transformers for signals at audio frequencies and radio frequencies are also available. At radio frequencies you sometimes use tuned transformers if only a narrow range of frequencies is present. There is also an interesting class of transmission-line transformers. In general, transformers for use at high frequencies must use special core materials or construction to minimize core losses, whereas low-frequency transformers (e.g., ac powerline transformers) are burdened instead by large and heavy cores. The two kinds of transformers are in general not interchangeable.

A. Problems, problems...

This simple "first-look" description ignores interesting – and important – issues. For example, there are inductances associated with the transformer, as suggested by its circuit symbol: an effective parallel inductance (called the *magnetizing inductance*) and an effective series inductance (called the *leakage inductance*). Magnetizing inductance causes a primary current even with no secondary load; more significantly, it means that you cannot make a "dc

transformer." And leakage inductance causes a voltage drop that depends on load current, as well as bedeviling circuits that have fast pulses or edges. Other departures from ideal performance include winding resistance, core losses, capacitance, and magnetic coupling to the outside world. Unlike capacitors (which behave nearly ideally in most circuit applications), the deficiencies of inductors have significant effects in real-world circuit applications. We'll deal with these in Chapter *1x* and Chapter 9.

1.6 Diodes and diode circuits

We are not done with capacitors and inductors! We have dealt with them in the *time domain* (*RC* circuits, exponential charge and discharge, differentiators and integrators, and so on), but we have not yet tackled their behavior in the *frequency domain*.

We will get to that soon enough. But this is a good time to take a break from *"RLC"* and put our knowledge to use with some clever and useful circuits. We begin by introducing a new device, the *diode*. It's our first example of a nonlinear device, and you can do nifty things with it.

1.6.1 Diodes

The circuit elements we've discussed so far (resistors, capacitors, and inductors) are all *linear*, meaning that a doubling of the applied signal (a voltage, say) produces a doubling of the response (a current, say). This is true even for the reactive devices (capacitors and inductors). These components are also *passive*, as opposed to *active* devices, the latter exemplified by transistors, which are semiconductor devices that control the flow of power. And they are all two-terminal devices, which is self-explanatory.



Figure 1.54. Diode.

The diode (Figure 1.54) is an important and useful two-terminal passive *nonlinear* device. It has the *V–I* curve shown in Figure 1.55. (In keeping with the general philosophy of this book, we will not attempt to describe the solid-state physics that makes such devices possible.)

The diode's arrow (the anode terminal) points in the direction of forward current flow. For example, if the diode is in a circuit in which a current of 10 mA is flowing from anode to cathode, then (from the graph) the anode is approximately 0.6 V more positive than the cathode; this is called the "forward voltage drop." The reverse current, which is measured in the nanoamp range for a general-purpose diode (note the hugely different scales in the graph for forward and reverse current), is almost never of any consequence until you reach the reverse breakdown voltage (also called the peak inverse voltage, PIV), typically 75 volts for a general-purpose diode like the 1N4148. (Normally you never subject a diode to voltages large enough to cause reverse breakdown; the exception is the zener diode we mentioned earlier.) Frequently, also, the forward voltage drop of about 0.5 to 0.8 V is of little concern, and the diode can be treated as a good approximation to an ideal one-way conductor. There are other important characteristics that distinguish the thousands of diode types available, e.g., maximum forward current, capacitance, leakage current, and reverse recovery time; Table 1.1 includes a few popular diodes, to give a sense of the capabilities of these little devices.

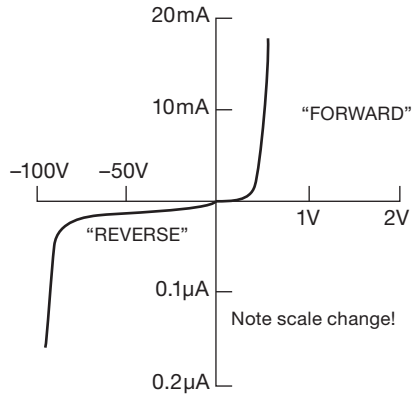


Figure 1.55. Diode *V–I* curve.

Before jumping into some circuits with diodes, we should point out two things: (a) a diode doesn't have a resistance (it doesn't obey Ohm's law). (b) If you put some diodes in a circuit, it won't have a Thévenin equivalent.

1.6.2 Rectification

A rectifier changes ac to dc; this is one of the simplest and most important applications of diodes (which are sometimes called rectifiers). The simplest circuit is shown in Figure 1.56. The "ac" symbol represents a source of ac voltage; in electronic circuits it is usually provided by a transformer, powered from the ac powerline. For a sine-wave input that is much larger than the forward drop (about 0.6 V for silicon diodes, the usual type), the output will look like that in Figure 1.57. If you think of the diode as a one-way conductor, you won't have any trouble

Table 1.1 Representative Diodes

Part #	V_R (max) (V)	I_R (typ, 25°C) (A @ V)	V_F @ I_F (mV)	V_F @ I_F (mA)	Capacitance (pF @ V_R)	SMT[a] p/n	Comments
Silicon							
PAD5	45	0.25pA 20V	800	1	0.5pF 5V	SSTPAD5	metal + glass can
1N4148	75	10nA 20V	750	10	0.9pF 0V	1N4148W	jellybean sig diode
1N4007	1000	50nA 800V	800	250	12pF 10V	DL4007	1N4004 lower V
1N5406	600	<10µA 600V	1.0V	10A	18pF 10V	none	heat through leads
Schottky[b]							
1N6263	60	7nA 20V	400	1	0.6pF 10V	1N6263W	see also 1N5711
1N5819	40	10µA 32V	400	1000	150pF 1V	1N5819HW	jellybean
1N5822	40	40µA 32V	480	3000	450pF 1V	none	power Schottky
MBRP40045	45	500µA 40V	540	400A	3500pF 10V	you jest!	Moby dual Schottky

Notes: (a) SMT, surface-mount technology. (b) Schottky diodes have lower forward voltage and zero reverse-recovery time, but more capacitance.

understanding how the circuit works. This circuit is called a *half-wave rectifier*, because only half of the input waveform is used.

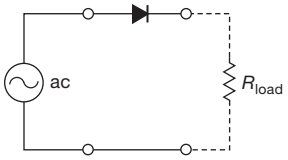


Figure 1.56. Half-wave rectifier.

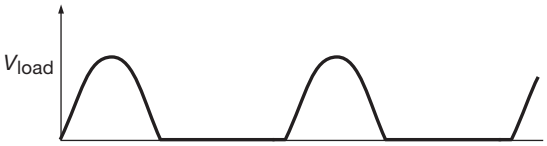


Figure 1.57. Half-wave output voltage (unfiltered).

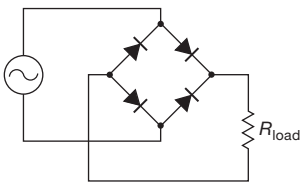


Figure 1.58. Full-wave bridge rectifier.

Figure 1.58 shows another rectifier circuit, a "full-wave bridge." Figure 1.59 shows the voltage across the load; note that the entire input waveform is used. The gaps at zero voltage occur because of the diodes' forward voltage drop. In this circuit, two diodes are always in series with the input; when you design low-voltage power supplies, for which the diode drop becomes significant, you have to remember that.[29]

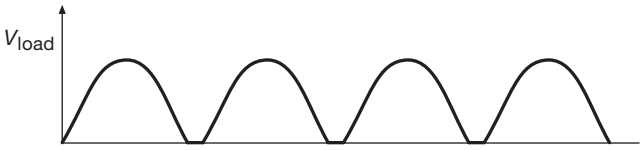


Figure 1.59. Full-wave output voltage (unfiltered).

1.6.3 Power-supply filtering

The preceding rectified waveforms aren't good for much as they stand. They're "dc" only in the sense that they don't change polarity. But they still have a lot of "ripple" (periodic variations in voltage about the steady value) that has to be smoothed out in order to generate genuine dc. This we do by attaching a relatively large value capacitor (Figure 1.60); it charges up to the peak output voltage during the diode conduction, and its stored charge ($Q = CV$) provides the output current in between charging cycles. Note that the diodes prevent the capacitor from discharging back through the ac source. In this application you should think of the capacitor as an energy storage device, with stored energy $U = \frac{1}{2}CV^2$ (recall §1.4.1; for C in farads and V in volts, U comes out in joules, or equivalently, watt seconds).

The capacitor value is chosen so that

$$R_{load}C \gg 1/f,$$

[29] The diode drop can be eliminated with *active switching* (or *synchronous switching*, a technique in which the diodes are replaced by transistor switches, actuated in synchronism with the input ac waveform (see §9.5.3B).

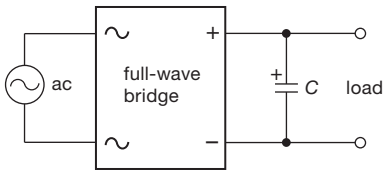


Figure 1.60. Full-wave bridge with output storage ("filter") capacitor.

(where f is the ripple frequency, here 120 Hz) in order to ensure small ripple by making the time constant for discharge much longer than the time between recharging. We make this vague statement clearer now.

A. Calculation of ripple voltage

It is easy to calculate the approximate ripple voltage, particularly if it is small compared with the dc (see Figure 1.61). The load causes the capacitor to discharge somewhat between cycles (or half-cycles, for full-wave rectification). If you assume that the load current stays constant (it will, for small ripple), you have

$$\Delta V = \frac{I}{C}\Delta t \qquad \left(\text{from } I = C\frac{dV}{dt}\right). \tag{1.25}$$

Just use $1/f$ (or $1/2f$ for full-wave rectification) for Δt (this estimate is a bit on the safe side, because the capacitor begins charging again in less than a half-cycle). You get[30]

$$\Delta V = \frac{I_{\text{load}}}{fC} \qquad \text{(half-wave)},$$

$$\Delta V = \frac{I_{\text{load}}}{2fC} \qquad \text{(full-wave)}.$$

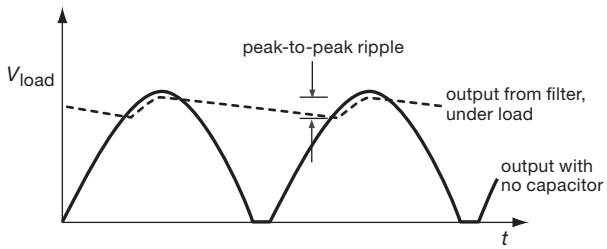


Figure 1.61. Power-supply ripple calculation.

If you wanted to do the calculation without any approximation, you would use the exact exponential discharge formula. You would be misguided in insisting on that kind of accuracy, though, for two reasons. (a) The discharge is an exponential only if the load is a resistance; many loads are not. In fact, the most common load, a *voltage regulator*, looks like a constant-current load. (b) Power supplies are built with capacitors with typical tolerances of 20% or more. Realizing the manufacturing spread, you design conservatively, allowing for the worst-case combination of component values.

In this case, viewing the initial part of the discharge as a ramp is in fact quite accurate, especially if the ripple is small, and in any case it errs in the direction of conservative design – it overestimates the ripple.

Exercise 1.20. Design a full-wave bridge rectifier circuit to deliver 10 V dc with less than 0.1 V (pp) ripple into a load drawing up to 10 mA. Choose the appropriate ac input voltage, assuming 0.6 V diode drops. Be sure to use the correct ripple frequency in your calculation.

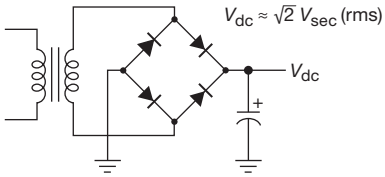


Figure 1.62. Bridge rectifier circuit. The polarity marking and curved electrode indicate a polarized capacitor, which must not be allowed to charge with the opposite polarity.

1.6.4 Rectifier configurations for power supplies

A. Full-wave bridge

A dc power supply with the bridge circuit we just discussed looks as shown in Figure 1.62. In practice, you generally buy the bridge as a prepackaged module. The smallest ones come with maximum current ratings of 1 A average, with a selection of rated minimum breakdown voltages going from 100 V to 600 V, or even 1000 V. Giant bridge rectifiers are available with current ratings of 25 A or more.

B. Center-tapped full-wave rectifier

The circuit in Figure 1.63 is called a center-tapped full-wave rectifier. The output voltage is half what you get if you use a bridge rectifier. It is not the most efficient circuit in terms of transformer design, because each half of the secondary is used only half the time. To develop some intuition on this subtle point, consider two different configurations that produce the same rectified dc output voltage: (a) the circuit of Figure 1.63, and (b) the same transformer, this time with its secondary cut at the center tap and

[30] While teaching electronics we've noticed that students love to memorize these equations! An informal poll of the authors showed that two out of two engineers don't memorize them. Please don't waste brain cells that way – instead, learn how to derive them.

rewired with the two halves in parallel, the resultant combined secondary winding connected to a full-wave bridge. Now, to deliver the same output power, each half winding in (a), during its conduction cycle, must supply the same current as the parallel pair in (b). But the power dissipated in the winding resistances goes like I^2R, so the power lost to heating in the transformer secondary windings reduced by a factor of 2 for the bridge configuration (b).

Here's another way to see the problem: imagine we use the same transformer as in (a), but for our comparison circuit we replace the pair of diodes with a bridge, as in Figure 1.62, and we leave the center tap unconnected. Now, to deliver the same output *power*, the current through the winding during that time is twice what it would be for a true full-wave circuit. To expand on this subtle point: heating in the windings, calculated from Ohm's law, is I^2R, so you have four times the heating for half the time, or twice the average heating of an equivalent full-wave bridge circuit. You would have to choose a transformer with a current rating 1.4 (square root of 2) times as large compared with the (better) bridge circuit; besides costing more, the resulting supply would be bulkier and heavier.

Exercise 1.21. This illustration of I^2R heating may help you understand the disadvantage of the center-tapped rectifier circuit. What fuse rating (minimum) is required for passing the current waveform shown in Figure 1.64, which has 1 amp average current? *Hint*: a fuse "blows out" by melting a metallic link (I^2R heating), for steady currents larger than its rating. Assume for this problem that the thermal time constant of the fusible link is much longer than the time scale of the square wave, i.e., that the fuse responds to the value of I^2 averaged over many cycles.

C. Split supply

A popular variation of the center-tapped full-wave circuit is shown in Figure 1.65. It gives you split supplies (equal plus and minus voltages), which many circuits need. It is an efficient circuit, because both halves of the input waveform are used in each winding section.

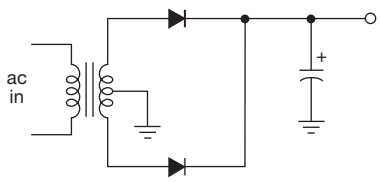


Figure 1.63. Full-wave rectifier using center-tapped transformer.

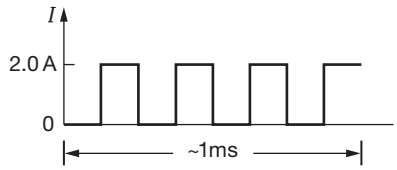


Figure 1.64. Illustrating greater I^2R heating with discontinuous current flow.

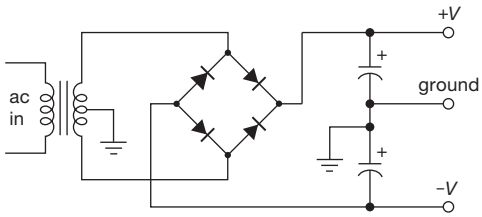


Figure 1.65. Dual-polarity (split) supply.

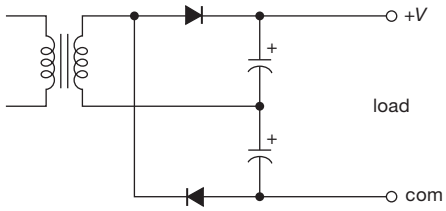


Figure 1.66. Voltage doubler.

D. Voltage multipliers

The circuit shown in Figure 1.66 is called a voltage doubler. Think of it as two half-wave rectifier circuits in series. It is officially a full-wave rectifier circuit because both halves of the input waveform are used – the ripple frequency is twice the ac frequency (120 Hz for the 60 Hz line voltage in the United States).

Variations of this circuit exist for voltage triplers, quadruplers, etc. Figure 1.67 shows doubler, tripler, and quadrupler circuits that let you ground one side of the transformer. You can extend this scheme as far as you want, producing what's called a Cockcroft–Walton generator; these are used in arcane applications (such as particle accelerators) and in everyday applications (such as image intensifiers, air ionizers, laser copiers, and even bug zappers) that require a high dc voltage but hardly any current.

1.6.5 Regulators

By choosing capacitors that are sufficiently large, you can reduce the ripple voltage to any desired level. This brute-force approach has three disadvantages.

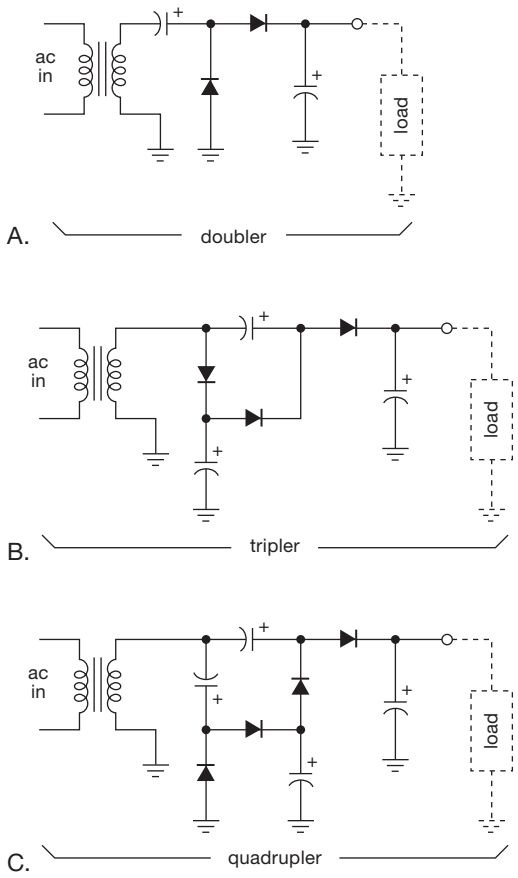


Figure 1.67. Voltage multipliers; these configurations don't require a floating voltage source.

- The required capacitors may be prohibitively bulky and expensive.
- The very short interval of current flow during each cycle[31] (only very near the top of the sinusoidal waveform) produces more I^2R heating.
- Even with the ripple reduced to negligible levels, you still have variations of output voltage that are due to other causes, e.g., the dc output voltage will be roughly proportional to the ac input voltage, giving rise to fluctuations caused by input line voltage variations. In addition, changes in load current will still cause the output voltage to change because of the finite internal resistances of the transformer, diode, etc. In other words, the Thévenin equivalent circuit of the dc power supply has $R > 0$.

A better approach to power-supply design is to use enough capacitance to reduce ripple to low levels (perhaps 10% of the dc voltage), then use an active *feedback circuit* to eliminate the remaining ripple. Such a feedback circuit "looks at" the output, making changes in a controllable series resistor (a transistor) as necessary to keep the output voltage constant (Figure 1.68). This is known as a "linear regulated dc power supply."[32]

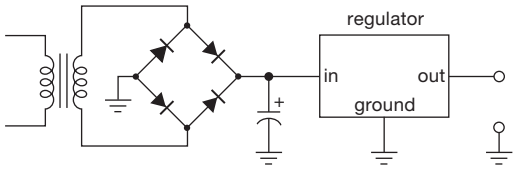


Figure 1.68. Regulated dc power supply.

These voltage regulators are used almost universally as power supplies for electronic circuits. Nowadays complete voltage regulators are available as inexpensive ICs (priced under $1). A power supply built with a voltage regulator can be made easily adjustable and self-protecting (against short circuits, overheating, etc.), with excellent properties as a voltage source (e.g., internal resistance measured in milliohms). We will deal with regulated dc power supplies in Chapter 9.

1.6.6 Circuit applications of diodes

A. Signal rectifier

There are other occasions when you use a diode to make a waveform of one polarity only. If the input waveform isn't a sinewave, you usually don't think of it as a rectification in the sense of a power supply. For instance, you might want a train of pulses corresponding to the rising edge of a square wave. The easiest way is to rectify the differentiated wave (Figure 1.69). Always keep in mind the 0.6 V(approximately) forward drop of the diode. This circuit, for instance, gives no output for square waves smaller than 0.6 V pp. If this is a problem, there are various tricks to circumvent this limitation. One possibility is to use *hot carrier diodes* (Schottky diodes), with a forward drop of about 0.25 V.

A possible *circuit solution* to this problem of finite diode drop is shown in Figure 1.70. Here D_1 compensates D_2's forward drop by providing 0.6 V of *bias* to hold D_2 at the threshold of conduction. Using a diode (D_1) to provide the bias (rather than, say, a voltage divider) has several

[31] Called the *conduction angle*.

[32] A popular variant is the regulated *switching* power converter. Although its operation is quite different in detail, it uses the same feedback principle to maintain a constant output voltage. See Chapter 9 for much more on both techniques.

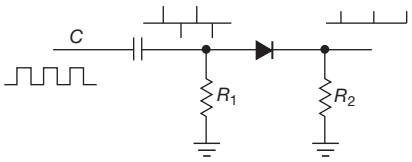


Figure 1.69. Signal rectifier applied to differentiator output.

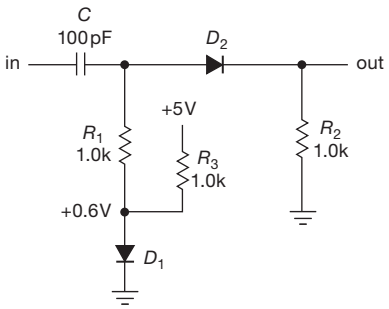


Figure 1.70. Compensating the forward voltage drop of a diode signal rectifier.

advantages: (a) there is nothing to adjust, (b) the compensation will be nearly perfect, and (c) changes of the forward drop (e.g., with changing temperature) will be compensated properly. Later we will see other instances of matched-pair compensation of forward drops in diodes, transistors, and FETs. It is a simple and powerful trick.

B. Diode gates

Another application of diodes, which we will recognize later under the general heading of *logic*, is to pass the higher of two voltages without affecting the lower. A good example is *battery backup*, a method of keeping something running (e.g, the "real-time clock" chip in a computer, which keeps a running count of date and time) that must continue running even when the device is switched off. Figure 1.71 shows a circuit that does the job. The battery does nothing until the +5 V power is switched off; then it takes over without interruption.

C. Diode clamps

Sometimes it is desirable to limit the range of a signal (i.e., prevent it from exceeding certain voltage limits) somewhere in a circuit. The circuit shown in Figure 1.72 will accomplish this. The diode prevents the output from exceeding about $+5.6$ V, with no effect on voltages less than that (including negative voltages); the only limitation is that the input must not go so negative that the reverse breakdown voltage of the diode is exceeded (e.g., -75 V for a 1N4148). The series resistor limits the diode current during

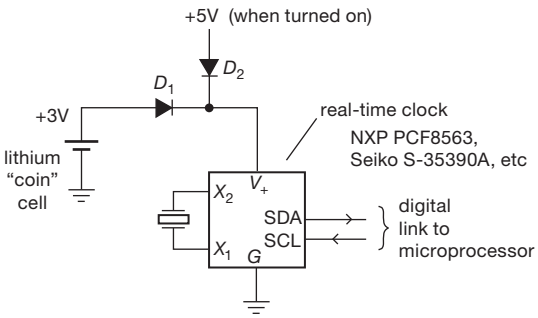


Figure 1.71. Diode OR gate: battery backup. The real-time clock chips are specified to operate properly with supply voltages from +1.8 V to +5.5 V. They draw a paltry 0.25 μA, which calculates to a 1-million-hour life (a hundred years) from a standard CR2032 coin cell!

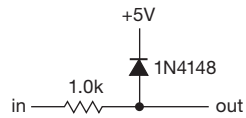


Figure 1.72. Diode voltage clamp.

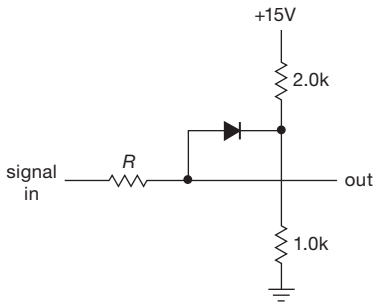


Figure 1.73. Voltage divider providing clamping voltage.

clamping action; however, a side effect is that it adds 1 kΩ of series resistance (in the Thévenin sense) to the signal, so its value is a compromise between maintaining a desirable low source (Thévenin) resistance and a desirable low clamping current. Diode clamps are standard equipment on all inputs in contemporary CMOS digital logic. Without them, the delicate input circuits are easily destroyed by static electricity discharges during handling.

Exercise 1.22. Design a symmetrical clamp, i.e., one that confines a signal to the range -5.6 to $+5.6$ V.

A voltage divider can provide the reference voltage for a clamp (Figure 1.73). In this case you must ensure that the resistance looking into the voltage divider (R_{vd}) is small compared with R because what you have looks as shown

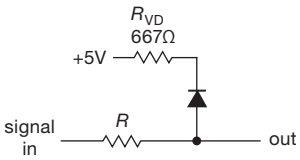


Figure 1.74. Clamping to voltage divider: equivalent circuit.

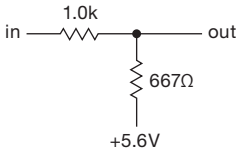


Figure 1.75. Poor clamping: voltage divider not stiff enough.

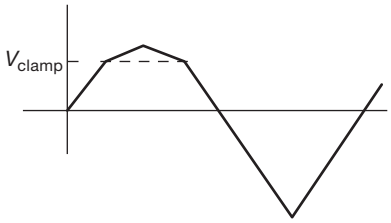


Figure 1.76. Clamping waveform for circuit of Figure 1.73.

in Figure 1.74 when the voltage divider is replaced with its Thévenin equivalent circuit. When the diode conducts (input voltage exceeds clamp voltage), the output is really just the output of a voltage divider, with the Thévenin equivalent resistance of the voltage reference as the lower resistor (Figure 1.75). So, for the values shown, the output of the clamp for a triangle-wave input would look as shown in Figure 1.76. The problem is that the voltage divider doesn't provide a stiff reference, in the language of electronics. A stiff voltage source is one that doesn't bend easily, i.e., it has low internal (Thévenin) resistance.

In practice, the problem of finite impedance of the voltage-divider reference can be easily solved by use of a transistor or an op-amp. This is usually a better solution than using very small resistor values, because it doesn't consume large currents, yet it provides a voltage reference with a Thévenin resistance of a few ohms or less. Furthermore, there are other ways to construct a clamp, using an op-amp as part of the clamp circuit. You will see these methods in Chapter 4.

Alternatively, a simple way to stiffen the clamp circuit of Figure 1.73, for time-varying signals only, is to add a so-called *bypass capacitor* across the lower (1 kΩ) resistor. To understand this fully we need to know about capacitors in the frequency domain, a subject we'll take up shortly. For now we'll simply say that you can put a capacitor across the 1k resistor, and its stored charge acts to maintain that point at constant voltage. For example, a 15 μF capacitor to ground would make the divider look as if it had a Thévenin resistance of less than 10 Ω for frequencies above 1 kHz. (You could similarly add a bypass capacitor across D_1 in Figure 1.70.) As we'll learn, the effectiveness of this trick decreases at low frequencies, and it does nothing at dc.

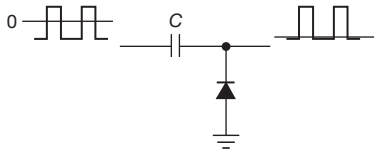


Figure 1.77. dc restoration.

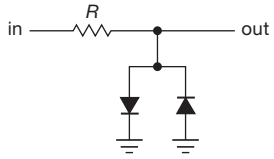


Figure 1.78. Diode limiter.

One interesting clamp application is "dc restoration" of a signal that has been ac coupled (capacitively coupled). Figure 1.77 shows the idea. This is particularly important for circuits whose inputs look like diodes (e.g., a transistor with grounded emitter, as we'll see in the next chapter); otherwise an ac-coupled signal will just fade away, as the coupling capacitor charges up to the signal's peak voltage.

D. Limiter

One last clamp circuit is shown in Figure 1.78. This circuit limits the output "swing" (again, a common electronics term) to one diode drop in either polarity, roughly ±0.6 V. That might seem awfully small, but if the next stage is an amplifier with large voltage amplification, its input will always be near 0 V; otherwise the output is in "saturation" (e.g., if the next stage has a gain of 1000 and operates from ±15 V supplies, its input must stay in the range ±15 mV in order for its output not to saturate). Figure 1.79 shows what a limiter does to oversize sinewaves and spikes. This clamp circuit is often used as input protection for a high-gain amplifier.

E. Diodes as nonlinear elements

To a good approximation the forward current through a diode is proportional to an exponential function of the

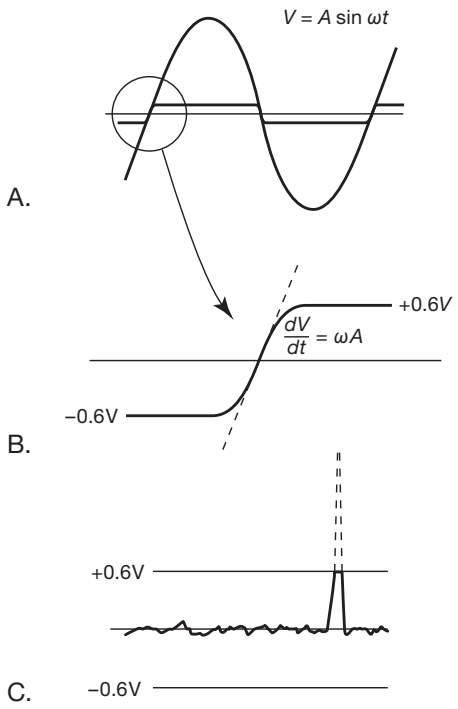


Figure 1.79. A. Limiting large-amplitude sinewaves; B. details; and C. spikes.

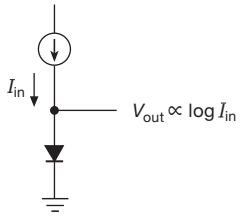


Figure 1.80. Exploiting the diode's nonlinear V–I curve: logarithmic converter.

voltage across it at a given temperature (for a discussion of the exact law, see §2.3.1). So you can use a diode to generate an output voltage proportional to the logarithm of a current (Figure 1.80). Because V hovers in the region of 0.6 V, with only small voltage changes that reflect input current variations, you can generate the input current with a resistor if the input voltage is much larger than a diode drop (Figure 1.81).

In practice, you may want an output voltage that isn't offset by the 0.6 V diode drop. In addition, it would be nice to have a circuit that is insensitive to changes in temperature (a silicon diode's voltage drop decreases approximately 2 mV/°C). The method of diode drop compensation is helpful here (Figure 1.82). R_1 makes D_2 conduct, holding

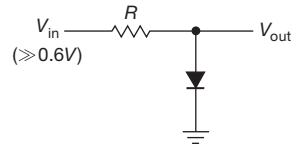


Figure 1.81. Approximate log converter.

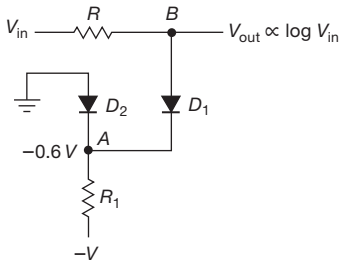


Figure 1.82. Diode drop compensation in the logarithmic converter.

point A at about -0.6 V. Point B is then near ground (making I_{in} accurately proportional to V_{in}, incidentally). As long as the two (identical) diodes are at the same temperature, there is good cancellation of the forward drops, except, of course, for the difference owing to input current through D_1, which produces the desired output. In this circuit, R_1 should be chosen so that the current through D_2 is significantly larger than the maximum input current in order to keep D_2 in conduction.

In Chapter *2x* we will examine better ways of constructing logarithmic converter circuits, along with careful methods of temperature compensation. With such methods it is possible to construct logarithmic converters accurate to a few percent over six decades or more of input current. A better understanding of diode and transistor characteristics, along with an understanding of op-amps, is necessary first. This section is meant to serve only as an introduction for things to come.

1.6.7 Inductive loads and diode protection

What happens if you open a switch that is providing current to an inductor? Because inductors have the property

$$V = L dI/dt,$$

it is not possible to turn off the current suddenly, because that would imply an infinite voltage across the inductor's terminals. What happens instead is that the voltage across the inductor rises abruptly and keeps rising until it forces current to flow. Electronic devices controlling inductive

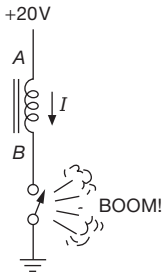


Figure 1.83. Inductive "kick."

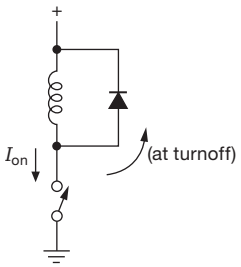


Figure 1.84. Blocking inductive kick.

loads can be easily damaged, especially the component that "breaks down" in order to satisfy the inductor's craving for continuity of current. Consider the circuit in Figure 1.83. The switch is initially closed, and current is flowing through the inductor (which might be a relay, as described later). When the switch is opened, the inductor tries to keep current flowing from *A* to *B*, as it had been. In other words, it tries to make current flow out of *B*, which it does by forcing *B* to a high positive voltage (relative to *A*). In a case like this, in which there's no connection to terminal *B*, it may go 1000 V positive before the switch contact "blows over." This shortens the life of the switch and also generates impulsive interference that may affect other circuits nearby. If the switch happens to be a transistor, it would be an understatement to say that its life is shortened; its life is *ended*.

The best solution usually is to put a diode across the inductor, as in Figure 1.84. When the switch is on, the diode is back-biased (from the dc drop across the inductor's winding resistance). At turn-off the diode goes into conduction, putting the switch terminal a diode drop above the positive supply voltage. The diode must be able to handle the initial diode current, which equals the steady current that had been flowing through the inductor; something like a 1N4004 is fine for nearly all cases.

The only disadvantage of this simple protection circuit is that it lengthens the decay of current through the inductor, because the rate of change of inductor current

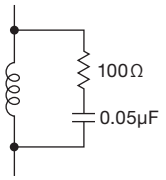


Figure 1.85. *RC* "snubber" for suppressing inductive kick.

is proportional to the voltage across it. For applications in which the current must decay quickly (high-speed actuators or relays, camera shutters, magnet coils, etc.), it may be better to put a resistor across the inductor, choosing its value so that $V_{\text{supply}} + IR$ is less than the maximum allowed voltage across the switch. For the fastest decay with a given maximum voltage, a zener with series diode (or other voltage-clamping device) can be used instead, giving a linear-like ramp-down of current rather than an exponential decay (see discussion in Chapter *1x*).

For inductors driven from ac (transformers, ac relays), the diode protection just described will not work, because the diode will conduct on alternate half-cycles when the switch is closed. In that case a good solution is an *RC* "snubber" network (Figure 1.85). The values shown are typical for small inductive loads driven from the ac powerline. Such a snubber should be included in all instruments that run from the ac powerline, because the power transformer is inductive.[33]

An alternative to the *RC* snubber is the use of a bidirectional zener-like voltage-clamping element. Among these the most common are the bidirectional "TVS" (transient voltage suppressor) zener and the metal-oxide varistor ("MOV"); the latter is an inexpensive device that looks something like a disc ceramic capacitor and behaves electrically like a bidirectional zener diode. Both classes are designed for transient voltage protection, are variously available at voltage ratings from 10 to 1000 volts, and can handle transient currents up to thousands of amperes (see Chapter *9x*). Including a transient suppressor (with appropriate fusing) across the ac powerline terminals makes good sense in a piece of electronic equipment, not only to prevent inductive spike interference to other nearby instruments but also to prevent occasional large powerline spikes from damaging the instrument itself.

1.6.8 Interlude: inductors as friends

Lest we leave the impression that inductance and inductors are things only to be feared, let's look at the circuit in

[33] As explained in §9.5.1, you should choose a capacitor rated for "across-the-line" service.

Figure 1.86. The goal is to charge up the capacitor from a source of dc voltage V_{in}. In the top circuit (Figure 1.86A) we've done it the conventional way, with a series resistor to limit the peak current demanded from the voltage source. OK, it does work – but it has a drawback that can be serious, namely that half the energy is lost as heat in the resistor. By contrast, in the circuit with the inductor (Figure 1.86B) no energy is lost (assuming ideal components); and, as a bonus, the capacitor gets charged to twice the input voltage. The output-voltage waveform is a sinusoidal half-cycle at the resonant frequency $f = 1/2\pi\sqrt{LC}$, a topic we'll see soon (§1.7.14).[34,35]

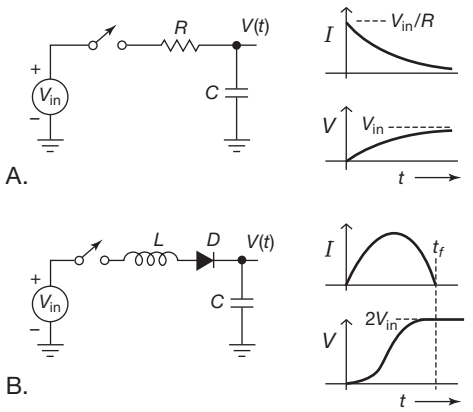


Figure 1.86. Resonant charging is lossless (with ideal components) compared with the 50% efficiency of resistive charging. Charging is complete after t_f, equal to a half-cycle of the resonant frequency. The series diode terminates the cycle, which would otherwise continue to oscillate between 0 and $2V_{in}$.

[34] A mechanical analogy may be helpful here. Imagine dropping packages onto a conveyor belt that is moving at speed v; the packages are accelerated to that speed by friction, with 50% efficiency, finally reaching the belt speed v, at which speed they ride into the sunset. That's resistive charging. Now we try something completely different, namely, we rig up a conveyor belt with little catchers attached by springs to the belt; and alongside it we have a second belt, running at twice the speed ($2v$). Now when we drop a package onto the first conveyor it compresses a spring, then rebounds at $2v$; and it makes a soft landing onto the second conveyor. No energy is lost (ideal springs), and the package rides off into the sunset at $2v$. That's reactive charging.

[35] Resonant charging is used for the high-voltage supply in flashlamps and stroboscopes, with the advantages of (a) full charge between flashes (spaced no closer than t_f), and (b) no current immediately after discharge (see waveforms), thus permitting the flashlamp to "quench" after each flash.

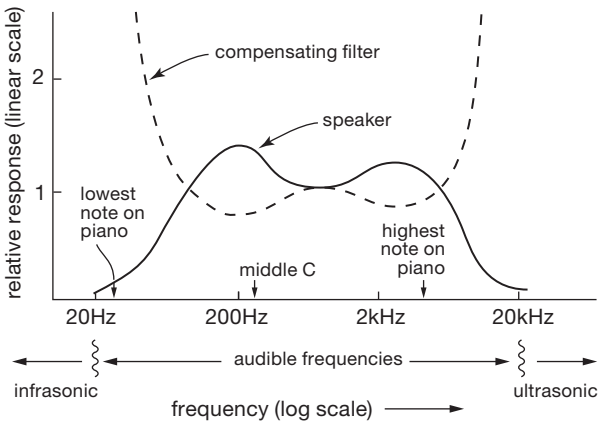


Figure 1.87. Example of frequency analysis: "boom box" loudspeaker equalization. The lowest and highest piano notes, called A0 and C8, are at 27.5 Hz and 4.2 kHz; they are four octaves below A440 and four octaves above middle C, respectively.

1.7 Impedance and reactance

Warning: this section is somewhat mathematical; you may wish to skip over the mathematics, but be sure to pay attention to the results and graphs.

Circuits with capacitors and inductors are more complicated than the resistive circuits we talked about earlier, in that their behavior depends on frequency: a "voltage divider" containing a capacitor or inductor will have a frequency-dependent division ratio. In addition, circuits containing these components (known collectively as *reactive* components) "corrupt" input waveforms such as square waves, as we saw earlier.

However, both capacitors and inductors are *linear* devices, meaning that the amplitude of the output waveform, whatever its shape, increases exactly in proportion to the input waveform's amplitude. This linearity has many consequences, the most important of which is probably the following: *the output of a linear circuit, driven with a sinewave at some frequency f, is itself a sinewave at the same frequency (with, at most, changed amplitude and phase).*

Because of this remarkable property of circuits containing resistors, capacitors, and inductors (and, later, linear amplifiers), it is particularly convenient to analyze any such circuit by asking how the output voltage (amplitude and phase) depends on the input voltage *for sinewave input at a single frequency*, even though this may not be the intended use. A graph of the resulting *frequency response*, in which the ratio of output to input is plotted for each sinewave frequency, is useful for thinking about many kinds of

waveforms. As an example, a certain "boom-box" loudspeaker might have the frequency response shown in Figure 1.87, in which the "output" in this case is of course sound pressure, not voltage. It is desirable for a speaker to have a "flat" response, meaning that the graph of sound pressure versus frequency is constant over the band of audible frequencies. In this case the speaker's deficiencies can be corrected by the introduction of a passive filter with the inverse response (as shown) within the amplifiers of the radio.

As we will see, it is possible to generalize Ohm's law, replacing the word "resistance" with "impedance," in order to describe any circuit containing these linear passive devices (resistors, capacitors, and inductors). You could think of the subject of impedance (generalized resistance) as Ohm's law for circuits that include capacitors and inductors.

Some terminology: impedance (**Z**) is the "generalized resistance"; inductors and capacitors, for which the voltage and current are always 90° out of phase, are *reactive*; they have *reactance* (X). Resistors, with voltage and current always in phase, are *resistive*; they have *resistance* (R). In general, in a circuit that combines resistive and reactive components, the voltage and current at some place will have some in-between phase relationship, described by a complex impedance: impedance = resistance + reactance, or $\mathbf{Z} = R + jX$ (more about this later).[36] However, you'll see statements like "the impedance of the capacitor at this frequency is . . . " The reason you don't have to use the word "reactance" in such a case is that impedance covers everything. In fact, you frequently use the word "impedance" even when you know it's a resistance you're talking about; you say "the source impedance" or "the output impedance" when you mean the Thévenin equivalent resistance of some source. The same holds for "input impedance."

In all that follows, we will be talking about circuits driven by sinewaves at a single frequency. Analysis of circuits driven by complicated waveforms is more elaborate, involving the methods we used earlier (differential equations) or decomposition of the waveform into sinewaves (Fourier analysis). Fortunately, these methods are seldom necessary.

1.7.1 Frequency analysis of reactive circuits

Let's start by looking at a capacitor driven by a sinewave voltage source $V(t) = V_0 \sin \omega t$ (Figure 1.88). The current is

$$I(t) = C\frac{dV}{dt} = C\omega V_0 \cos \omega t,$$

i.e., a current of amplitude $\omega C V_0$, with its phase leading the input voltage by 90°. If we consider amplitudes only, and disregard phases, the current is

$$I = \frac{V}{1/\omega C}.$$

(Recall that $\omega = 2\pi f$.) It behaves like a frequency-dependent resistance $R = 1/\omega C$, but in addition the current is 90° out of phase with the voltage (Figure 1.89).

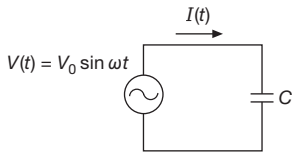


Figure 1.88. A sinusoidal ac voltage drives a capacitor.

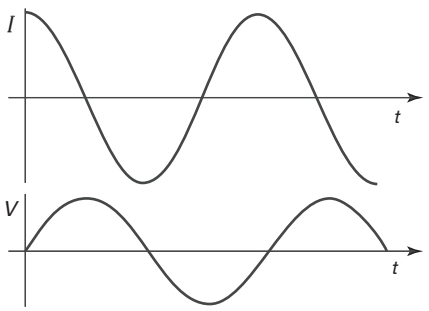


Figure 1.89. Current in a capacitor leads the sinusoidal voltage by 90°.

For example, a 1 μF capacitor put across the 115 volts (rms) 60 Hz powerline draws a current of rms amplitude:

$$I = \frac{115}{1/(2\pi \times 60 \times 10^{-6})} = 43.4\,\text{mA (rms)}.$$

Soon enough we will complicate matters by explicitly worrying about *phase shifts* and the like – and that will get us into some complex algebra that terrifies beginners (often) and mathophobes (always). Before we do that, though, this is a good time to develop intuition about the frequency-dependent behavior of some basic and important circuits that use capacitors, ignoring for the time being the troublesome fact that, when driven with a sinusoidal signal, currents and voltages in a capacitor are not in phase.

As we just saw, the ratio of *magnitudes* of voltage to

[36] But, in a nutshell, the magnitude of **Z** gives the ratio of amplitudes of voltage to current, and the polar angle of **Z** gives the phase angle between current and voltage.

current, in a capacitor driven at a frequency ω, is just

$$\frac{|V|}{|I|} = \frac{1}{\omega C} ,$$

which we can think of as a sort of "resistance" – the magnitude of the current is proportional to the magnitude of the applied voltage. The official name for this quantity is *reactance*, with the symbol X, thus X_C for the reactance of a capacitor.[37] So, for a capacitor,

$$X_C = \frac{1}{\omega C} . \tag{1.26}$$

This means that a larger capacitance has a smaller reactance. And this makes sense, because, for example, if you double the value of a capacitor, it takes twice as much current to charge and discharge it through the same voltage swing in the same time (recall $I = C\,dV/dt$). For the same reason the reactance decreases as you increase the frequency – doubling the frequency (while holding V constant) doubles the rate of change of voltage and therefore requires that you double the current, thus half the reactance.

So, roughly speaking, we can think of a capacitor as a "frequency-dependent resistor." Sometimes that's good enough, sometimes it isn't. We'll look at a few circuits in which this simplified view gets us reasonably good results, and provides nice intuition; later we'll fix it up, using the correct complex algebra, to get a precise result. (Keep in mind that the results we are about to get are approximate; we're *lying* to you – but it's a small lie, and anyway we'll tell the truth later. In the meantime we'll use the weird symbol $\asymp$ instead of $=$ in all such "approximate equations," and we'll flag the equation as approximate.)

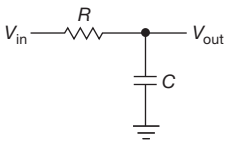


Figure 1.90. Lowpass filter.

A. *RC* lowpass filter (approximate)

The circuit in Figure 1.90 is called a *lowpass filter*, because it passes low frequencies and blocks high frequencies. If you think of it as a frequency-dependent voltage divider, this makes sense: the lower leg of the divider (the capacitor) has a decreasing reactance with increasing frequency, so the ratio of V_{out}/V_{in} decreases accordingly:

[37] Later we'll see *inductors*, which also have a 90° phase shift (though of the opposite sign), and so are likewise characterized by a reactance X_L.

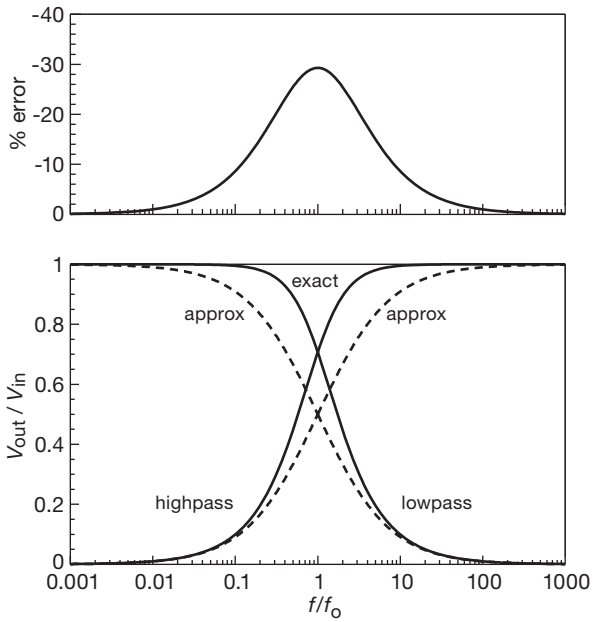


Figure 1.91. Frequency response of single-section *RC* filters, showing the results both of a simple approximation that ignores phase (dashed curve), and the exact result (solid curve). The fractional error (i.e., dashed/solid) is plotted above.

$$\frac{V_{out}}{V_{in}} \asymp \frac{X_C}{R+X_C} = \frac{1/\omega C}{R+1/\omega C} = \frac{1}{1+\omega RC} \quad \text{(approximate!)} \tag{1.27}$$

We've plotted that ratio in Figure 1.91 (and also that of its cousin, the *highpass filter*), along with their exact results that we'll understand soon enough in §1.7.8.

You can see that the circuit passes low frequencies fully (because at low frequencies the capacitor's reactance is very high, so it's like a divider with a smaller resistor atop a larger one) and that it blocks high frequencies. In particular, the crossover from "pass" to "block" (often called the *breakpoint*) occurs at a frequency ω_0 at which the capacitor's reactance ($1/\omega_0 C$) is equal to the resistance R: $\omega_0 = 1/RC$. At frequencies well beyond the crossover (where the product $\omega RC \gg 1$), the output decreases inversely with increasing frequency; that makes sense because the reactance of the capacitor, already much smaller than R, continues dropping as $1/\omega$. It's worth noting that, even with our "ignoration of phase shifts," the equation (and graph) for the ratio of voltages is quite accurate at both low and high frequencies and is only modestly in error around the crossover

frequency, where the correct ratio is $V_{out}/V_{in} = 1/\sqrt{2} \approx 0.7$, rather than the 0.5 that we got.[38]

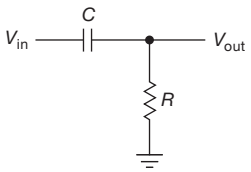


Figure 1.92. Highpass filter.

B. *RC* highpass filter (approximate)

You get the reverse behavior (pass high frequencies, block low) by interchanging R and C, as in Figure 1.92. Treating it as a frequency-dependent voltage divider, and ignoring phase shifts once again, we get (see Figure 1.91)

$$\frac{V_{out}}{V_{in}} \asymp \frac{R}{R+X_C} = \frac{R}{R+1/\omega C} = \frac{\omega RC}{1+\omega RC} \quad (approximate!) \tag{1.28}$$

High frequencies (above the same crossover frequency as before, $\omega \gg \omega_0 = 1/RC$) pass through (because the capacitor's reactance is much smaller than R), whereas frequencies well below the crossover are blocked (the capacitor's reactance is much larger than R). As before, the equation and graph are accurate at both ends, and only modestly in error at the crossover, where the correct ratio is, again, $V_{out}/V_{in} = 1/\sqrt{2}$.

C. Blocking capacitor

Sometimes you want to let some band of signal frequencies pass through a circuit, but you want to block any steady dc voltage that may be present (we'll see how this can happen when we learn about amplifiers in the next chapter). You can do the job with an *RC* highpass filter if you choose the crossover frequency correctly: a highpass filter always blocks dc, so what you do is choose component values so that the crossover frequency is *below* all frequencies of interest. This is one of the more frequent uses of a capacitor and is known as a dc *blocking capacitor*.

For instance, every stereo audio amplifier has all its inputs capacitively coupled, because it doesn't know what dc level its input signals might be riding on. In such a coupling application you always pick R and C so that all frequencies of interest (in this case, 20 Hz to 20 kHz) are passed without loss (attenuation). That determines the product RC: $RC > 1/\omega_{min}$, where for this case you might choose $f_{min} \approx 5$ Hz, and so $RC = 1/\omega_{min} = 1/2\pi f_{min} \approx 30$ ms.

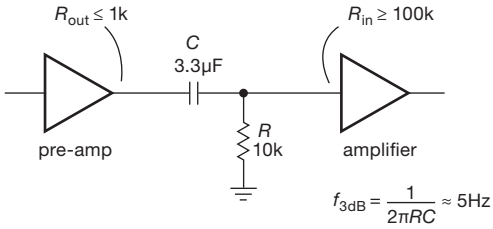


Figure 1.93. "Blocking capacitor": a highpass filter for which all signal frequencies of interest are in the passband.

You've got the product, but you still have to choose individual values for R and C. You do this by noticing that the input signal sees a load equal to R at signal frequencies (where C's reactance is small – it's just a piece of wire there), so you choose R to be a reasonable load, i.e., not so small that it's hard to drive, and not so large that the circuit is prone to signal pickup from other circuits in the box. In the audio business it's common to see a value of 10 kΩ, so we might choose that value, for which the corresponding C is 3.3 μF (Figure 1.93). The circuit connected to the output should have an input resistance much greater than 10 kΩ, to avoid loading effects on the filter's output; and the driving circuit should be able to drive a 10 kΩ load without significant attenuation (loss of signal amplitude) to prevent circuit loading effects by the filter on the signal source. It's worth noting that our approximate model, ignoring phase shifts, is perfectly adequate for blocking capacitor design; that is because the signal band is fully in the passband, where the effects of phase shifts are negligible.

In this section we've been thinking in the frequency domain (sinewaves of frequency f). But it's useful to think in the time domain, where, for example, you might use a blocking capacitor to couple pulses, or square waves. In such situations you encounter waveform *distortion*, in the form of "droop" and overshoot (rather than the simple amplitude attenuation and phase shift you get with sinusoidal waves). Thinking in the time domain, the criterion you use to avoid waveform distortion in a pulse of duration T is that the time constant $\tau = RC \gg T$. The resulting droop is approximately T/τ (followed by a comparable overshoot at the next transition).

You often need to know the reactance of a capacitor at a given frequency (e.g., for design of filters). Figure 1.100 in §1.7.8 provides a very useful graph covering large ranges

[38] Of course, it fails to predict anything about phase shifts in this circuit. As we'll see later, the output signal's phase lags the input by 90° at high frequencies, going smoothly from 0° at low frequencies, with a 45° lag at ω_0 (see Figure 1.104 in §1.7.9).

of capacitance and frequency, giving the value of $X_C = 1/2\pi fC$.

D. Driving and loading *RC* filters

This example of an audio-blocking capacitor raised the issue of driving and loading *RC* filter circuits. As we discussed in §1.2.5A, in the context of voltage dividers, you generally like to arrange things so that the circuit being driven does not significantly load the driving resistance (Thévenin equivalent source resistance) of the signal source.

The same logic applies here, but with a generalized kind of resistance that includes the reactance of capacitors (and inductors), known as *impedance*. So a signal source's impedance should generally be small compared with the impedance of the thing being driven.[39] We'll have a precise way of talking about impedance shortly, but it's correct to say that, apart from phase shifts, the impedance of a capacitor is equal to its reactance.

What we want to know, then, are the input and output impedances of the two simple *RC* filters (lowpass and highpass). This sounds complicated, because there are four impedances, and they all vary with frequency. However, if you ask the question the right way, the answer is simple, and the same in all cases!

First, assume that in each case the right thing is being done to the other end of the filter: when we ask the input impedance, we assume the output drives a high impedance (compared with its own); and when we ask the output impedance, we assume the input is driven by a signal source of low internal (Thévenin) impedance. Second, we dispose of the variation of impedances with frequency by asking only for the *worst-case* value; that is, we care what only the *maximum* output impedance of a filter circuit may be (because that is the worst for driving a load), and we care about only the *minimum* input impedance (because that is the hardest to drive).

Now the answer is astonishingly simple: in all cases the worst-case impedance is just R.

Exercise 1.23. Show that the preceding statement is correct.

So, for example, if you want to hang an *RC* lowpass filter onto the output of an amplifier whose output resistance is 100 Ω, start with $R = 1$k, then choose C for the breakpoint you want. Be sure that whatever loads the output has an input impedance of at least 10k. You can't go wrong.

Exercise 1.24. Design a two-stage "bandpass" *RC* filter, in which the first stage is highpass with a breakpoint of 100 Hz, and the second stage is lowpass with a breakpoint of 10 kHz. Assume the input signal source has an impedance of 100Ω. What is the worst-case output impedance of your filter, and therefore what is the minimum recommended load impedance?

[39] With two important exceptions, namely, transmission lines and current sources.

1.7.2 Reactance of inductors

Before we embark on a fully correct treatment of impedance, replete with complex exponentials and the like, let's use our approximation tricks to figure out the reactance of an inductor.

It goes as before: we imagine an inductor L driven by a sinusoidal voltage source of angular frequency ω such that a current $I(t) = I_0 \sin \omega t$ is flowing.[40] Then the voltage across the inductor is

$$V(t) = L\frac{dI(t)}{dt} = L\omega I_0 \cos \omega t.$$

And so the ratio of *magnitudes* of voltage to current – the resistance-like quantity called *reactance* – is just

$$\frac{|V|}{|I|} = \frac{L\omega I_0}{I_0} = \omega L.$$

So. for an inductor,

$$X_{\mathrm{L}} = \omega L. \tag{1.29}$$

Inductors, like capacitors, have a frequency-dependent reactance; however, here the reactance *increases* with increasing frequency (the opposite of capacitors, where it *decreases* with increasing frequency). So, in the simplest view, a series inductor can be used to pass dc and low frequencies (where its reactance is small) while blocking high frequencies (where its reactance is high). You often see inductors used this way, particularly in circuits that operate at radio frequencies; in that application they're sometimes called *chokes*.

1.7.3 Voltages and currents as complex numbers

At this point it is necessary to get into some complex algebra; you may wish to skip over the math in some of the following sections, taking note of the results as we derive them. A knowledge of the detailed mathematics is not necessary for understanding the remainder of the book. Very little mathematics will be used in later chapters. The section ahead is easily the most difficult for the reader with little mathematical preparation. *Don't be discouraged!*

As we've just seen, there can be phase shifts between

[40] We take the easy path here by specifying the *current*, rather than the voltage; we are rewarded with a simple derivative (rather than a simple integral!).

the voltage and current in an ac circuit being driven by a sinewave at some frequency. Nevertheless, as long as the circuit contains only *linear* elements (resistors, capacitors, inductors), the magnitudes of the currents everywhere in the circuit are still proportional to the magnitude of the driving voltage, so we might hope to find some generalization of voltage, current, and resistance in order to rescue Ohm's law. Evidently a single number won't suffice to specify the current, say, at some point in the circuit, because we must somehow have information about both the magnitude and phase shift.

Although we can imagine specifying the magnitudes and phase shifts of voltages and currents at any point in the circuit by writing them out explicitly, e.g., $V(t) = 23.7\sin(377t + 0.38)$, it turns out that we can meet our requirements more simply by using the algebra of complex numbers to *represent* voltages and currents. Then we can simply add or subtract the complex number representations, rather than laboriously having to add or subtract the actual sinusoidal functions of time themselves. Because the actual voltages and currents are real quantities that vary with time, we must develop a rule for converting from actual quantities to their representations, and vice versa. Recalling once again that we are talking about a single sinewave frequency, ω, we agree to use the following rules.

1. Voltages and currents are *represented* by the complex quantities $\mathbf{V}$ and $\mathbf{I}$. The voltage $V_0\cos(\omega t + \phi)$ is to be represented by the complex number $V_0 e^{j\phi}$. Recall that $e^{j\phi} = \cos\phi + j\sin\phi$, where $j = \sqrt{-1}$.
2. We obtain *actual* voltages and currents by multiplying their complex number representations by $e^{j\omega t}$ and then taking the real part: $V(t) = \mathscr{R}e(\mathbf{V}e^{j\omega t})$, $I(t) = \mathscr{R}e(\mathbf{I}e^{j\omega t})$.

In other words,

circuit voltage versus time		complex number representation
$V_0\cos(\omega t + \phi)$	$\longrightarrow$ $\longleftarrow$ multiply by $e^{j\omega t}$ and take real part	$V_0 e^{j\phi} = a + jb$

(In electronics, the symbol j is used instead of i in the exponential in order to avoid confusion with the symbol i meaning small-signal current.) Thus, in the general case, the actual voltages and currents are given by

$$\begin{aligned} V(t) &= \mathscr{R}e(\mathbf{V}e^{j\omega t}) \\ &= \mathscr{R}e(\mathbf{V})\cos\omega t - \mathscr{I}m(\mathbf{V})\sin\omega t \\ I(t) &= \mathscr{R}e(\mathbf{I}e^{j\omega t}) \\ &= \mathscr{R}e(\mathbf{I})\cos\omega t - \mathscr{I}m(\mathbf{I})\sin\omega t. \end{aligned}$$

For example, a voltage whose complex representation is

$$\mathbf{V} = 5j$$

corresponds to a (real) voltage versus time of

$$\begin{aligned} V(t) &= \mathscr{R}e[5j\cos\omega t + 5j(j)\sin\omega t] \\ &= -5\sin\omega t \text{ volts.} \end{aligned}$$

1.7.4 Reactance of capacitors and inductors

With this convention we can apply complex Ohm's law correctly to circuits containing capacitors and inductors, just as for resistors, once we know the reactance of a capacitor or inductor. Let's find out what these are. We begin with a simple (co)sinusoidal voltage $V_0\cos\omega t$ applied across a capacitor:

$$V(t) = \mathscr{R}e(V_0 e^{j\omega t}).$$

Then, using $I = C(dV(t)/dt)$, we obtain

$$\begin{aligned} I(t) &= -V_0 C\omega\sin\omega t = \mathscr{R}e\left(\frac{V_0 e^{j\omega t}}{-j/\omega C}\right), \\ &= \mathscr{R}e\left(\frac{V_0 e^{j\omega t}}{\mathbf{Z}_C}\right) \end{aligned}$$

i.e., for a capacitor

$$\mathbf{Z}_C = -j/\omega C \quad (= -jX_C);$$

$\mathbf{Z}_C$ is the impedance of a capacitor at frequency ω; it is equal in magnitude to the reactance $X_C = 1/\omega C$ that we found earlier, but with a factor of $-j$ that accounts for the 90° leading phase shift of current versus voltage. As an example, a 1 μF capacitor has an impedance of $-2653j\,\Omega$ at 60 Hz, and $-0.16j\,\Omega$ at 1 MHz. The corresponding reactances are 2653 Ω and 0.16 Ω.[41] Its reactance (and also its impedance) at dc is infinite.

If we did a similar analysis for an inductor, we would find

$$\mathbf{Z}_L = j\omega L \quad (= jX_L).$$

A circuit containing only capacitors and inductors always has a purely imaginary impedance, meaning that the voltage and current are always 90° out of phase – it is purely reactive. When the circuit contains resistors, there is also

[41] Note the convention that the reactance X_C is a real number (the 90° phase shift is implicit in the term "reactance"), but the corresponding impedance is purely imaginary: $\mathbf{Z} = R - jX$.

a real part to the impedance. The term "reactance" in that case means the imaginary part only.

1.7.5 Ohm's law generalized

With these conventions for representing voltages and currents, Ohm's law takes a simple form. It reads simply

$$\mathbf{I} = \mathbf{V}/\mathbf{Z},$$
$$\mathbf{V} = \mathbf{I}\mathbf{Z},$$

where the voltage represented by $\mathbf{V}$ is applied across a circuit of impedance $\mathbf{Z}$, giving a current represented by $\mathbf{I}$. The complex impedance of devices in series or parallel obeys the same rules as resistance:

$$\mathbf{Z} = \mathbf{Z_1} + \mathbf{Z_2} + \mathbf{Z_3} + \cdots \quad \text{(series)}, \tag{1.30}$$

$$\mathbf{Z} = \frac{1}{\frac{1}{\mathbf{Z_1}} + \frac{1}{\mathbf{Z_2}} + \frac{1}{\mathbf{Z_3}} + \cdots} \quad \text{(parallel)}. \tag{1.31}$$

Finally, for completeness we summarize here the formulas for the impedance of resistors, capacitors, and inductors:

$$\begin{aligned} \mathbf{Z}_\mathrm{R} &= R && \text{(resistor)}, \\ \mathbf{Z}_\mathrm{C} &= -j/\omega C = 1/j\omega C && \text{(capacitor)}, \\ \mathbf{Z}_\mathrm{L} &= j\omega L && \text{(inductor)}. \end{aligned} \tag{1.32}$$

With these rules we can analyze many ac circuits by the same general methods we used in handling dc circuits, i.e., application of the series and parallel formulas and Ohm's law. Our results for circuits such as voltage dividers will look nearly the same as before. For multiply-connected networks we may have to use Kirchhoff's laws, just as with dc circuits, in this case using the complex representations for V and I: the sum of the (complex) voltage drops around a closed loop is zero, and the sum of the (complex) currents into a point is zero. The latter rule implies, as with dc circuits, that the (complex) current in a series circuit is the same everywhere.

Exercise 1.25. Use the preceding rules for the impedance of devices in parallel and in series to derive the formulas (1.17) and (1.18) for the capacitance of two capacitors (a) in parallel and (b) in series. *Hint*: in each case, let the individual capacitors have capacitances C_1 and C_2. Write down the impedance of the parallel or series combination; then equate it to the impedance of a capacitor with capacitance C. Then find C.

Let's try out these techniques on the simplest circuit imaginable, an ac voltage applied across a capacitor, which we looked at earlier, in §1.7.1. Then, after a brief look at power in reactive circuits (to finish laying the groundwork), we'll analyze (correctly, this time) the simple but extremely important and useful *RC* lowpass and highpass filter circuits.

Imagine putting a $1\,\mu$F capacitor across a 115 volts (rms) 60 Hz powerline. What current flows? Using complex Ohm's law, we have

$$\mathbf{Z} = -j/\omega C.$$

Therefore, the current is given by

$$\mathbf{I} = \mathbf{V}/\mathbf{Z}.$$

The phase of the voltage is arbitrary, so let us choose $\mathbf{V} = A$, i.e., $V(t) = A\cos\omega t$, where the amplitude $A = 115\sqrt{2} \approx 163$ volts. Then

$$\mathbf{I} = j\omega CA \approx -0.061\sin\omega t.$$

The resulting current has an amplitude of 61 mA (43 mA rms) and leads the voltage by 90°. This agrees with our previous calculation. More simply, we could have noticed that the impedance of the capacitor is negative imaginary, so whatever the absolute phase of V, the phase of I_cap must lead by 90°. And in general the phase angle between current and voltage, for any two-terminal *RLC* circuit, is equal to the angle of the (complex) impedance of that circuit.

Note that if we wanted to know just the magnitude of the current, and didn't care what the relative phase was, we could have avoided doing any complex algebra: if

$$\mathbf{A} = \mathbf{B}/\mathbf{C},$$

then

$$A = B/C,$$

where A, B, and C are the magnitudes of the respective complex numbers; this holds for multiplication, also (see Exercise 1.18). Thus, in this case,

$$I = V/Z = \omega CV.$$

This trick, which we used earlier (because we didn't know any better), is often useful.

Surprisingly, there is no power dissipated by the capacitor in this example. Such activity won't increase your electric bill; you'll see why in the next section. Then we will go on to look at circuits containing resistors and capacitors with our complex Ohm's law.

Exercise 1.26. Show that, if $\mathbf{A} = \mathbf{BC}$, then $A = BC$, where A, B, and C are magnitudes. *Hint*: represent each complex number in polar form, i.e., $\mathbf{A} = Ae^{i\theta}$.

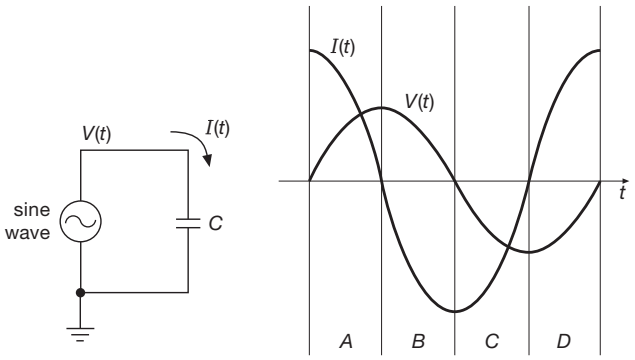


Figure 1.94. The power delivered to a capacitor is zero over a full sinusoidal cycle, owing to the 90° phase shift between voltage and current.

1.7.6 Power in reactive circuits

The instantaneous power delivered to any circuit element is always given by the product $P = VI$. However, in reactive circuits where V and I are not simply proportional, you can't just multiply their amplitudes together. Funny things can happen; for instance, the sign of the product can reverse over one cycle of the ac signal. Figure 1.94 shows an example. During time intervals A and C, power is being delivered to the capacitor (albeit at a variable rate), causing it to charge up; its stored energy is increasing (power is the rate of change of energy). During intervals B and D, the power delivered to the capacitor is negative; it is discharging. The average power over a whole cycle for this example is in fact exactly zero, a statement that is always true for any purely reactive circuit element (inductors, capacitors, or any combination thereof). If you know your trigonometric integrals, the next exercise will show you how to prove this.

Exercise 1.27. Optional exercise: prove that a circuit whose current is 90° out of phase with the driving voltage consumes no power, averaged over an entire cycle.

How do we find the average power consumed by an arbitrary circuit? In general, we can imagine adding up little pieces of VI product, then dividing by the elapsed time. In other words,

$$P = \frac{1}{T}\int_0^T V(t)I(t)\,dt, \tag{1.33}$$

where T is the time for one complete cycle. Luckily, that's almost never necessary. Instead, it is easy to show that the average power is given by

$$P = \mathscr{R}e(\mathbf{V}\mathbf{I}^*) = \mathscr{R}e(\mathbf{V}^*\mathbf{I}), \tag{1.34}$$

where $\mathbf{V}$ and $\mathbf{I}$ are complex rms amplitudes (and an asterisk means *complex conjugate* – see the math review, Appendix A, if this is unfamiliar).

Let's take an example. Consider the preceding circuit, with a 1 volt (rms) sinewave driving a capacitor. We'll do everything with rms amplitudes, for simplicity. We have

$$\mathbf{V} = 1,$$

$$\mathbf{I} = \frac{\mathbf{V}}{-j/\omega C} = j\omega C,$$

$$P = \mathscr{R}e(\mathbf{V}\mathbf{I}^*) = \mathscr{R}e(-j\omega C) = 0.$$

That is, the average power is zero, as stated earlier.

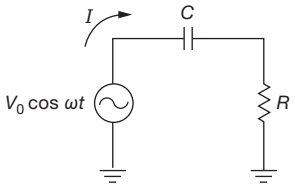


Figure 1.95. Power and power factor in a series *RC* circuit.

As another example, consider the circuit shown in Figure 1.95. Our calculations go like this:

$$\mathbf{Z} = R - \frac{j}{\omega C},$$
$$\mathbf{V} = V_0,$$
$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{V_0}{R-(j/\omega C)} = \frac{V_0\,[R+(j/\omega C)]}{R^2+(1/\omega^2C^2)},$$
$$P = \mathscr{R}e(\mathbf{V}\mathbf{I}^*) = \frac{V_0^2 R}{R^2+(1/\omega^2C^2)}.$$

(In the third line we multiplied numerator and denominator by the complex conjugate of the denominator in order to make the denominator real.) The calculated power[42] is less than the product of the magnitudes of $\mathbf{V}$ and $\mathbf{I}$. In fact, their ratio is called the *power factor*:

$$|\mathbf{V}|\,|\mathbf{I}| = \frac{V_0^2}{[R^2+(1/\omega^2C^2)]^{1/2}},$$
$$\text{power factor} = \frac{\text{power}}{|\mathbf{V}|\,|\mathbf{I}|}$$
$$= \frac{R}{[R^2+(1/\omega^2C^2)]^{1/2}}$$

[42] It's always a good idea to check limiting values: here we see that $P \to V^2/R$ for large C; and for small C the magnitude of the current $|I| \to V_0/X_C$, or $V_0\omega C$, thus $P \to I^2R = V_0^2\omega^2C^2R$, in agreement at both limits.

in this case. The power factor is the cosine of the phase angle between the voltage and the current, and it ranges from 0 (purely reactive circuit) to 1 (purely resistive). A power factor of less than 1 indicates some component of reactive current.[43] It's worth noting that the power factor goes to unity, and the dissipated power goes to V^2/R, in the limit of large capacitance (or of high frequency), where the reactance of the capacitor becomes much less than R.

Exercise 1.28. Show that all the average power delivered to the preceding circuit winds up in the resistor. Do this by computing the value of V_R^2/R. What is that power, in watts, for a series circuit of a 1 μF capacitor and a 1.0k resistor placed across the 115 volt (rms), 60 Hz powerline?

Power factor is a serious matter in large-scale electrical power distribution, because reactive currents don't result in useful power being delivered to the load, but cost the power company plenty in terms of I^2R heating in the resistance of generators, transformers, and wiring. Although residential users are billed only for "real" power $[\mathscr{R}e(\mathbf{VI}^*)]$, the power company charges industrial users according to the power factor. This explains the capacitor yards that you see behind large factories, built to cancel the inductive reactance of industrial machinery (i.e., motors).

Exercise 1.29. Show that adding a series capacitor of value $C = 1/\omega^2 L$ makes the power factor equal 1.0 in a series RL circuit. Now do the same thing, but with the word "series" changed to "parallel."

1.7.7 Voltage dividers generalized

Our original voltage divider (Figure 1.6) consisted of a pair of resistors in series to ground, input at the top and output at the junction. The generalization of that simple resistive divider is a similar circuit in which either or both resistors are replaced with a capacitor or inductor (or a more complicated network made from R, L, and C), as in Figure 1.96. In general, the division ratio $V_{\text{out}}/V_{\text{in}}$ of such a divider is not constant, but depends on frequency (as we have already seen, in our approximate treatment of the lowpass and highpass filters in §1.7.1). The analysis is straightforward:

$$\mathbf{I} = \frac{\mathbf{V}_{\text{in}}}{\mathbf{Z}_{\text{total}}},$$

$$\mathbf{Z}_{\text{total}} = \mathbf{Z}_1 + \mathbf{Z}_2$$

[43] Or, for nonlinear circuits, it indicates that the current waveform is not proportional to the voltage waveform. More on this in §9.7.1.

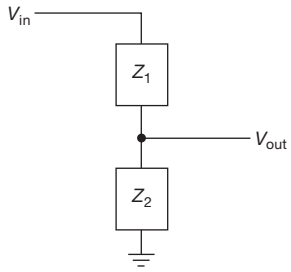


Figure 1.96. Generalized voltage divider: a pair of arbitrary impedances.

$$\mathbf{V}_{\text{out}} = \mathbf{I}\mathbf{Z}_2 = \mathbf{V}_{\text{in}} \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}.$$

Rather than worrying about this result in general, let's look at some simple (but very important) examples, beginning with the RC highpass and lowpass filters we approximated earlier.

1.7.8 *RC* highpass filters

We've seen that by combining resistors with capacitors it is possible to make frequency-dependent voltage dividers, owing to the frequency dependence of a capacitor's impedance $\mathbf{Z}_{\text{C}} = -j/\omega C$. Such circuits can have the desirable property of passing signal frequencies of interest while rejecting undesired signal frequencies. In this subsection and the next we revisit the simple lowpass and highpass RC filters, correcting the approximate analysis of §1.7.1; though simple, these circuits are important and widely used. Chapter 6 and Appendix E describe filters of greater sophistication.

Referring back to the classic RC highpass filter (Figure 1.92), we see that the complex Ohm's law (or the complex voltage-divider equation) gives

$$\mathbf{V}_{\text{out}} = \mathbf{V}_{\text{in}} \frac{R}{R - j/\omega C} = \mathbf{V}_{\text{in}} \frac{R(R + j/\omega C)}{R^2 + (1/\omega^2 C^2)}.$$

(For the last step, multiply top and bottom by the complex conjugate of the denominator.) Most often we don't care about the phase of V_{out}, just its amplitude:

$$\begin{aligned} V_{\text{out}} &= (\mathbf{V}_{\text{out}}\mathbf{V}_{\text{out}}^*)^{1/2} \\ &= \frac{R}{[R^2 + (1/\omega^2 C^2)]^{1/2}} V_{\text{in}}. \end{aligned}$$

Note the analogy to a resistive divider, where

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V_{\text{in}}.$$

Here the impedance of the series RC combination

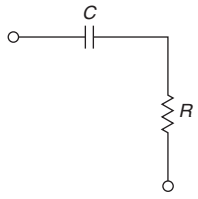


Figure 1.97. Input impedance of unloaded highpass filter.

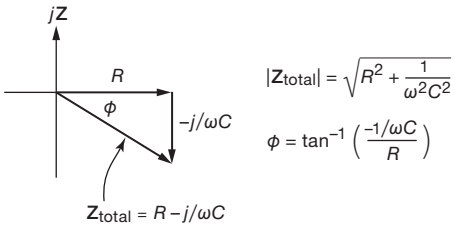


Figure 1.98. Impedance of series *RC*.

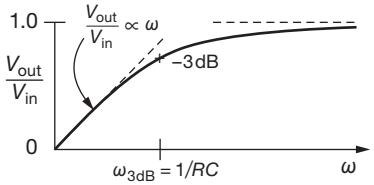


Figure 1.99. Frequency response of highpass filter. The corresponding phase shift goes smoothly from $+90^\circ$ (at $\omega = 0$), through $+45^\circ$ (at $\omega_{3\,\mathrm{dB}}$), to 0° (at $\omega = \infty$), analogous to the lowpass filter's phase shift (Figure 1.104).

(Figure 1.97) is as shown in Figure 1.98. So the "response" of this circuit, ignoring phase shifts by taking magnitudes of the complex amplitudes, is given by

$$V_{\mathrm{out}} = \frac{R}{[R^2 + (1/\omega^2 C^2)]^{1/2}} V_{\mathrm{in}} = \frac{2\pi f RC}{[1 + (2\pi f RC)^2]^{1/2}} V_{\mathrm{in}} \qquad (1.35)$$

and looks as shown in Figure 1.99 (and earlier in Figure 1.91).

Note that we could have gotten this result immediately by taking the ratio of the *magnitudes* of impedances, as in Exercise 1.26 and the example immediately preceding it; the numerator is the magnitude of the impedance of the lower leg of the divider (R), and the denominator is the magnitude of the impedance of the series combination of R and C.

As we noted earlier, the output is approximately equal to the input at high frequencies (how high? $\omega \gtrsim 1/RC$) and goes to zero at low frequencies. The highpass filter is

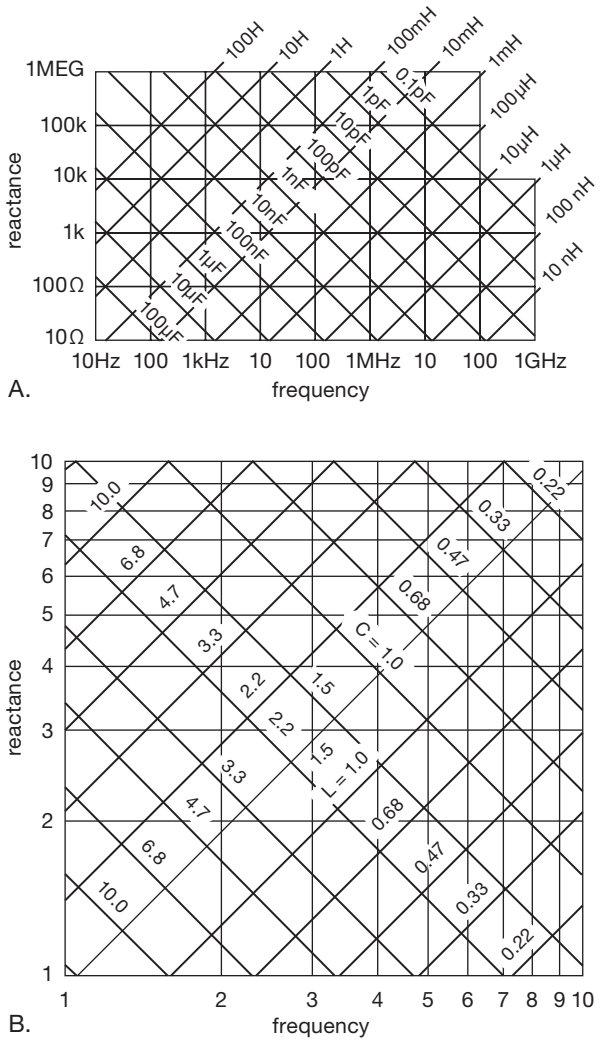


Figure 1.100. A: Reactance of inductors and capacitors versus frequency; all decades are identical, except for scale. B: A single decade from part A expanded, with standard 20% component values (EIA "E6") shown.

very common; for instance, the input to the oscilloscope can be switched to "ac coupling." That's just an *RC* highpass filter with the bend at about 10 Hz (you would use ac coupling if you wanted to look at a small signal riding on a large dc voltage). Engineers like to refer to the -3 dB "breakpoint" of a filter (or of any circuit that behaves like a filter). In the case of the simple *RC* high-pass filter, the -3 dB breakpoint is given by

$$f_{3\,\mathrm{dB}} = 1/2\pi RC.$$

You often need to know the impedance of a capacitor at a given frequency (e.g., for the design of filters).

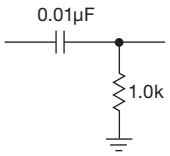


Figure 1.101. Highpass filter example.

Figure 1.100 provides a very useful graph covering large ranges of capacitance and frequency, giving the value of $|\mathbf{Z}| = 1/2\pi fC$.

As an example, consider the filter shown in Figure 1.101. It is a highpass filter with the 3 dB point[44] at 15.9 kHz. The impedance of a load driven by it should be much larger than 1.0k in order to prevent circuit loading effects on the filter's output, and the driving source should be able to drive a 1.0k load without significant attenuation (loss of signal amplitude) in order to prevent circuit loading effects by the filter on the signal source (recall §1.7.1D for worst-case source and load impedances of *RC* filters).

1.7.9 *RC* lowpass filters

Revisiting the lowpass filter, in which you get the opposite frequency behavior by interchanging *R* and *C* (Figure 1.90, repeated here as Figure 1.102), we find the accurate result

$$V_{\text{out}} = \frac{1}{(1+\omega^2R^2C^2)^{1/2}}\,V_{\text{in}} \qquad (1.36)$$

as seen in Figure 1.103 (and earlier in Figure 1.91). The 3 dB point is again at a frequency[45] $f = 1/2\pi RC$. Lowpass filters are quite handy in real life. For instance, a lowpass filter can be used to eliminate interference from nearby radio and television stations (0.5–800 MHz), a problem that plagues audio amplifiers and other sensitive electronic equipment.

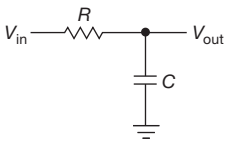


Figure 1.102. Lowpass filter.

Exercise 1.30. Show that the preceding expression for the response of an *RC* lowpass filter is correct.

[44] One often omits the minus sign when referring to the −3 dB point.

[45] As mentioned in §1.7.1A, we often like to define the breakpoint frequency $\omega_0 = 1/RC$, and work with frequency ratios ω/ω_0. Then a useful form for the denominator in eq'n 1.36 is $\sqrt{1+(\omega/\omega_0)^2}$. The same applies to eq'n 1.35, where the numerator becomes ω/ω_0.

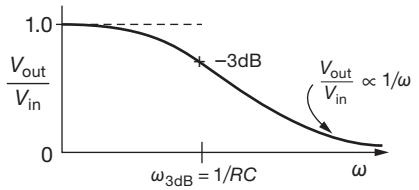


Figure 1.103. Frequency response of lowpass filter.

The lowpass filter's output can be viewed as a signal source in its own right. When driven by a perfect ac voltage (zero source impedance), the filter's output looks like *R* at low frequencies (the perfect signal source can be replaced with a short, i.e., by its small-signal source impedance, for the purpose of impedance calculations). It drops to zero impedance at high frequencies, where the capacitor dominates the output impedance. The signal driving the filter sees a load of *R* plus the load resistance at low frequencies, dropping to just *R* at high frequencies. As we remarked in §1.7.1D, the worst-case source impedance and the worst-case load impedance of an *RC* filter (lowpass or highpass) are both equal to *R*.

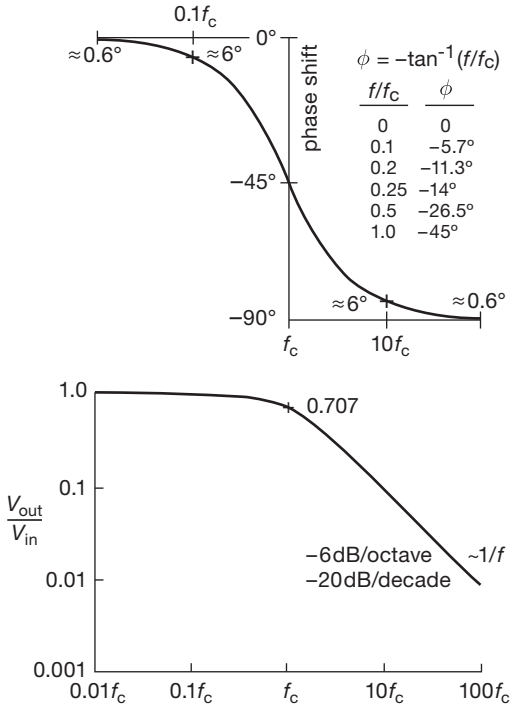


Figure 1.104. Frequency response (phase and amplitude) of lowpass filter plotted on logarithmic axes. Note that the phase shift is −45° at the −3 dB point and is within 6° of its asymptotic value for a decade of frequency change.

In Figure 1.104, we've plotted the same lowpass filter response with logarithmic axes, which is a more common way that it's done. You can think of the vertical axis as decibels, and the horizontal axis as octaves (or decades). On such a plot, equal distances correspond to equal ratios. We've also plotted the phase shift, using a linear vertical axis (degrees) and the same logarithmic frequency axis. This sort of plot is good for seeing the detailed response even when it is greatly attenuated (as at right); we'll see a number of such plots in Chapter 6, when we treat active filters. Note that the filter curve plotted here becomes a straight line at large attenuations, with a slope of -20 dB/decade (engineers prefer to say "-6 dB/octave"). Note also that the phase shift goes smoothly from $0°$ (at frequencies well below the breakpoint) to $-90°$ (well above it), with a value of $-45°$ at the -3 dB point. A rule of thumb for single-section *RC* filters is that the phase shift is $\approx 6°$ from its asymptotic value at $0.1 f_{3\,\mathrm{dB}}$ and at $10 f_{3\,\mathrm{dB}}$.

Exercise 1.31. Prove the last assertion.

An interesting question is the following: is it possible to make a filter with some arbitrary specified amplitude response and some other arbitrary specified phase response? Surprisingly, the answer is no: the demands of causality (i.e., that response must follow cause, not precede it) force a relationship between phase and amplitude response of realizable analog filters (known officially as the Kramers–Kronig relation).

1.7.10 *RC* differentiators and integrators in the frequency domain

The *RC* differentiator that we saw in §1.4.3 is exactly the same circuit as the highpass filter in this section. In fact, it can be considered as either, depending on whether you're thinking of waveforms in the time domain or response in the frequency domain. We can restate the earlier time-domain condition for its proper operation ($dV_{\mathrm{out}} \ll dV_{\mathrm{in}}$) in terms of the frequency response: for the output to be small compared with the input, the signal frequency (or frequencies) must be well below the 3 dB point. This is easy to check: suppose we have the input signal $V_{\mathrm{in}} = \sin \omega t$. Then, using the equation we obtained earlier for the differentiator output, we have

$$V_{\mathrm{out}} = RC \frac{d}{dt} \sin \omega t = \omega RC \cos \omega t,$$

and so $dV_{\mathrm{out}} \ll dV_{\mathrm{in}}$ if $\omega RC \ll 1$, i.e., $RC \ll 1/\omega$. If the input signal contains a range of frequencies, this must hold for the highest frequencies present in the input.

The *RC* integrator (§1.4.4) is the same circuit as the lowpass filter; by similar reasoning, the criterion for a good integrator is that the lowest signal frequencies must be well above the 3 dB point.

1.7.11 Inductors versus capacitors

Instead of capacitors, inductors can be combined with resistors to make lowpass (or highpass) filters. In practice, however, you rarely see *RL* lowpass or highpass filters. The reason is that inductors tend to be more bulky and expensive and perform less well (i.e., they depart further from the ideal) than capacitors (see Chapter *1x*). If you have a choice, use a capacitor. One important exception to this general statement is the use of ferrite beads and chokes in high-frequency circuits. You just string a few beads here and there in the circuit; they make the wire interconnections slightly inductive, raising the impedance at very high frequencies and preventing oscillations, without the added series resistance you would get with an *RC* filter. An *RF choke* is an inductor, usually a few turns of wire wound on a ferrite core, used for the same purpose in RF circuits. Note, however, that inductors are *essential* components in (a) *LC* tuned circuits (§1.7.14), and (b) switch-mode power converters (§9.6.4).

1.7.12 Phasor diagrams

There's a nice graphical method that can be helpful when we are trying to understand reactive circuits. Let's take an example, namely, the fact that an *RC* filter attenuates 3 dB at a frequency $f = 1/2\pi RC$, which we derived in §1.7.8. This is true for both highpass and lowpass filters. It is easy to get a bit confused here, because at that frequency the reactance of the capacitor equals the resistance of the resistor; so you might at first expect 6 dB attenuation (a factor of 1/2 in voltage). That is what you would get, for example, if you were to replace the capacitor with a resistor of the same impedance magnitude. The confusion arises because the capacitor is reactive, but the matter is clarified by a phasor diagram (Figure 1.105). The axes are the real (resistive) and imaginary (reactive) components of the impedance. In a series circuit like this, the axes also represent the (complex) voltage, because the current is the same everywhere. So for this circuit (think of it as an *RC* voltage divider) the input voltage (applied across the series *RC* pair) is proportional to the length of the hypotenuse, and the output voltage (across *R* only) is proportional to the length of the *R* leg of the triangle. The diagram represents the situation at the frequency where the capacitor's reactance equals *R*,

i.e., $f = 1/2\pi RC$, and shows that the ratio of output voltage to input voltage is $1/\sqrt{2}$, i.e., -3 dB.

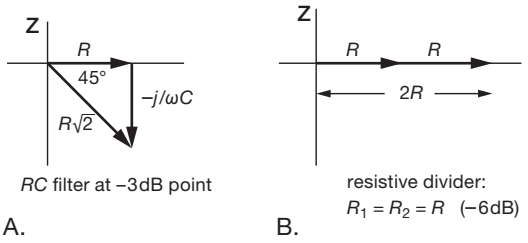


Figure 1.105. Phasor diagram for lowpass filter at 3 dB point.

The angle between the vectors gives the phase shift from input to output. At the 3 dB point, for instance, the output amplitude equals the input amplitude divided by the square root of 2, and it leads by 45° in phase. This graphical method makes it easy to read off amplitude and phase relationships in *RLC* circuits. You can use it, for example, to get the response of the highpass filter that we previously derived algebraically.

Exercise 1.32. Use a phasor diagram to derive the response of an *RC* high-pass filter: $V_{\text{out}} = V_{\text{in}}R/\sqrt{R^2 + (1/\omega^2C^2)}$.

Exercise 1.33. At what frequency does an *RC* lowpass filter attenuate by 6 dB (output voltage equal to half the input voltage)? What is the phase shift at that frequency?

Exercise 1.34. Use a phasor diagram to obtain the lowpass filter response previously derived algebraically.

In the next chapter (§2.2.8) we'll see a nice example of phasor diagrams in connection with a constant-amplitude phase-shifting circuit.

1.7.13 "Poles" and decibels per octave

Look again at the response of the *RC* lowpass filter (Figures 1.103 and 1.104). Far to the right of the "knee" the output amplitude is dropping proportional to $1/f$. In one octave (as in music, one octave is twice the frequency) the output amplitude will drop to half, or -6 dB; so a simple *RC* filter has a 6 dB/octave falloff. You can make filters with several *RC* sections; then you get 12 dB/octave (two *RC* sections), 18 dB/octave (three sections), and so on. This is the usual way of describing how a filter behaves beyond the cutoff. Another popular way is to say a "three-pole filter," for instance, meaning a filter with three *RC* sections (or one that behaves like one). (The word "pole" derives from a method of analysis that is beyond the scope of this book and that involves complex transfer functions in the complex frequency plane, known by engineers as the "*s*-plane." This is discussed in the advanced volume, in Chapter *1x*.)

A caution on multistage filters: you can't simply cascade several identical filter sections in order to get a frequency response that is the concatenation of the individual responses. The reason is that each stage will load the previous one significantly (since they're identical), changing the overall response. Remember that the response function we derived for the simple *RC* filters was based on a zero-impedance driving source and an infinite-impedance load. One solution is to make each successive filter section have much higher impedance than the preceding one. A better solution involves active circuits like transistor or operational amplifier (op-amp) interstage "buffers," or active filters. These subjects will be treated in Chapters 2–4, 6, and 13.

1.7.14 Resonant circuits

When capacitors are combined with inductors or are used in special circuits called active filters, it is possible to make circuits that have very sharp frequency characteristics (e.g., a large peak in the response at a particular frequency), compared with the gradual characteristics of the *RC* filters we've seen so far. These circuits find applications in various audio and RF devices. Let's now take a quick look at *LC* circuits (there will be more on them, and active filters, in Chapter 6 and in Appendix E).

A. Parallel and series *LC* circuits

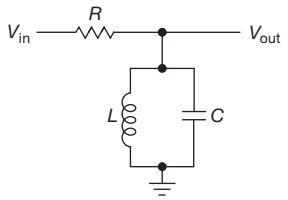


Figure 1.106. *LC* resonant circuit: bandpass filter.

First, consider the circuit shown in Figure 1.106. The impedance of the *LC* combination at frequency f is just

$$\frac{1}{\mathbf{Z}_{\text{LC}}} = \frac{1}{\mathbf{Z}_{\text{L}}} + \frac{1}{\mathbf{Z}_{\text{C}}} = \frac{1}{j\omega L} - \frac{\omega C}{j}$$
$$= j\left(\omega C - \frac{1}{\omega L}\right),$$

i.e.,

$$\mathbf{Z}_{\text{LC}} = \frac{j}{(1/\omega L) - \omega C}.$$

In combination with R it forms a voltage divider. Because of the opposite behaviors of inductors and capacitors, the impedance of the parallel LC goes to infinity at the *resonant frequency*

$$f_0 = 1/2\pi\sqrt{LC} \tag{1.37}$$

(i.e., $\omega_0 = 1/\sqrt{LC}$), giving a peak in the response there. The overall response is as shown in Figure 1.107.

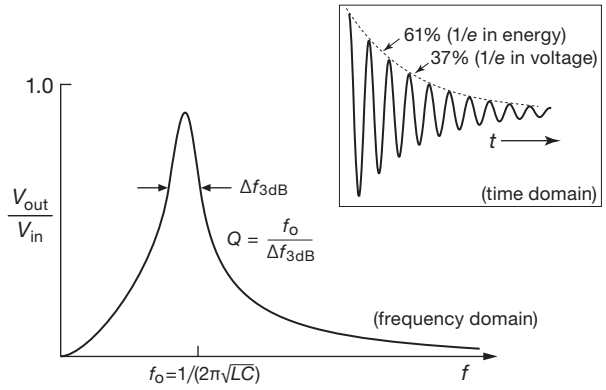


Figure 1.107. Frequency response of parallel LC "tank" circuit. The inset shows the time-domain behavior: a damped oscillation ("ringing") waveform following an input voltage step or pulse.

In practice, losses in the inductor and capacitor limit the sharpness of the peak, but with good design these losses can be made very small. Conversely, a Q-spoiling resistor is sometimes added intentionally to reduce the sharpness of the resonant peak. This circuit is known simply as a parallel LC resonant circuit (or "tuned circuit," or "tank") and is used extensively in RF circuits to select a particular frequency for amplification (the L or C can be variable, so you can tune the resonant frequency). The higher the driving impedance, the sharper the peak; it is not uncommon to drive them with something approaching a current source, as you will see later. The *quality factor* Q is a measure of the sharpness of the peak. It equals the resonant frequency divided by the width at the -3 dB points. For a parallel RLC circuit, $Q = \omega_0 RC$.[46]

Another variety of LC circuit is the series LC (Figure 1.108). By writing down the impedance formulas involved, and assuming that both the capacitor and inductor are ideal, i.e., that they have no resistive losses,[47] you can convince yourself that the impedance of the LC goes to zero

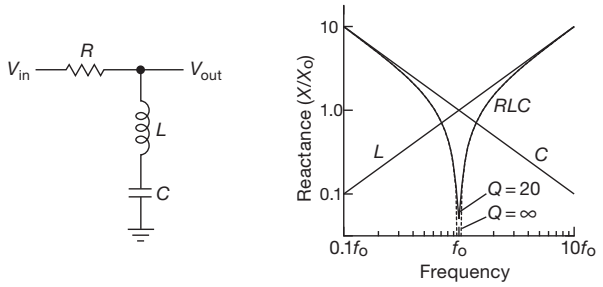


Figure 1.108. LC notch filter ("trap"). The inductive and capacitive reactances behave as shown, but the opposite sign of their complex impedances causes the series impedance to plummet. For ideal components the reactance of the series LC goes completely to zero at resonance; for real-world components the minimum is non-zero, and usually dominated by the inductor.

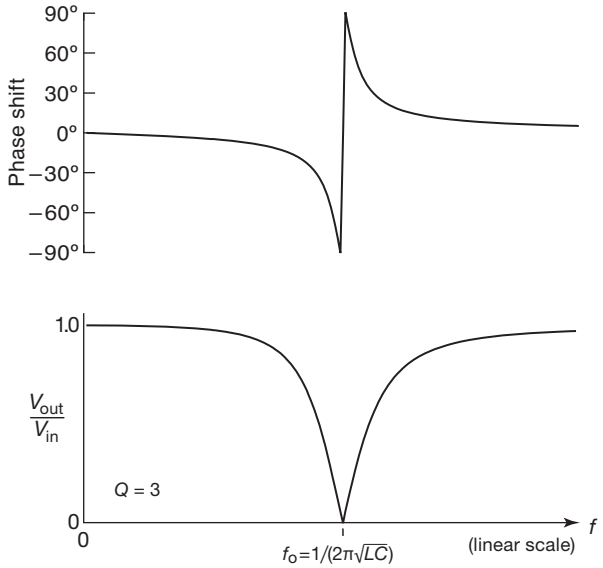


Figure 1.109. Frequency and phase response of the series LC trap. The phase changes abruptly at resonance, an effect seen in other resonator types (see for example Figure 7.36).

at resonance ($f_0 = 1/2\pi\sqrt{LC}$). Such a circuit is a "trap" for signals at or near the resonant frequency, shorting them to ground. Again, this circuit finds application mainly in RF circuits. Figure 1.109 shows what the response looks like. The Q of a series RLC circuit is $Q = \omega_0 L/R$.[48] To see the impact of increasing Q, look at the accurate plots of tank and notch response in Figure 1.110.

[46] Or, equivalently, $Q = R/X_C = R/X_L$, where $X_L = X_C$ are the reactances at ω_0.

[47] We'll see in Chapter *1x* that real components depart from the ideal, often expressed in terms of an effective series resistance, ESR.

[48] Or, equivalently, $Q = X_L/R = X_C/R$, where $X_L = X_C$ are the reactances at ω_0.

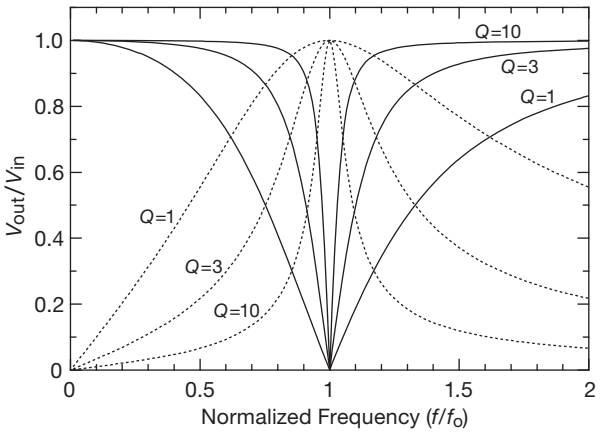


Figure 1.110. Response of *LC* tank (dotted curves) and trap (solid curves) for several values of quality factor, Q.

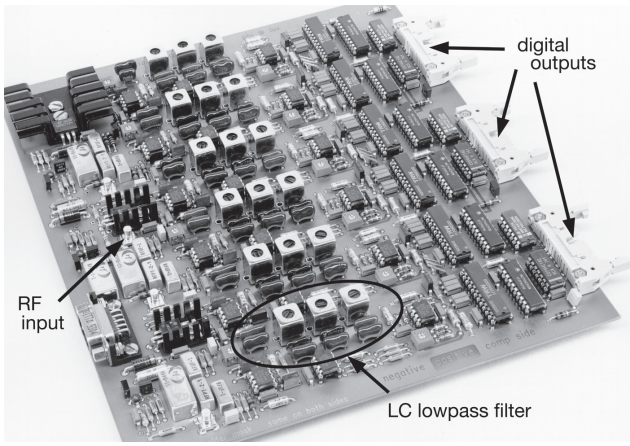


Figure 1.111. There are six *LC* lowpass filters on this circuit board, part of the process of frequency conversion and digitizing for which this "mixer-digitizer" was designed.

Exercise 1.35. Find the response (V_{out}/V_{in} versus frequency) for the series *LC* trap circuit in Figure 1.108.

These descriptions of *LC* resonant circuits are phrased in terms of frequency response, i.e., in the frequency domain. In the time domain you're generally interested in a circuit's response to pulses, or steps; there you see the sort of behavior shown in the inset of Figure 1.107, an *LC* circuit with Q=20. The signal voltage falls to $1/e$ (37%) in Q/π cycles; the stored *energy* (proportional to v^2) falls to $1/e$ (61% in amplitude) in $Q/2\pi$ cycles. You may prefer to think in radians: the energy falls to $1/e$ in Q radians, and the voltage falls to $1/e$ in $2Q$ radians. *LC* resonant circuits are not unique in providing highly frequency-selective circuit behavior; alternatives include quartz-crystal, ceramic, and surface acoustic-wave (SAW) resonators; transmission lines; and resonant cavities.

1.7.15 *LC* filters

By combining inductors with capacitors you can produce filters (lowpass, highpass, bandpass) with far sharper behavior in frequency response than you can with a filter made from a simple *RC*, or from any number of cascaded *RC* sections. We'll see more of this, and the related topic of active filters, in Chapter 6. But it's worth admiring now how well this works, to appreciate the virtue of the humble inductor (an often-maligned circuit component).

As an example, look at Figure 1.111, a photograph of a "mixer-digitizer" circuit board that we built for a project some years back (specifically, a radio receiver with 250 million simultaneous channels). There's lots of stuff on the board, which has to frequency-shift and digitize three RF bands; its design could occupy a book chapter. For now just gaze at the lumpy filter in the oval (there are five more on the board), comprising three inductors (the square metal cans) and four capacitors (the pairs of shiny oblongs). It's a lowpass filter, designed to cut off at 1.0 MHz; it prevents 'aliases" in the digitized output, a subject we'll visit in Chapter 13.

How well does it work? Figure 1.112 shows a "frequency sweep," in which a sinewave input goes from 0 Hz to 2 MHz as the trace goes from left to right across the screen. The sausage shapes are the "envelope" of the sinewave output, here comparing the *LC* filter with an *RC* lowpass filter with the same 1 MHz cutoff (1 kΩ and 160 pF). The *LC* wins, hands down. The *RC* is pathetic by comparison. It's not even good English to call 1 MHz its "cutoff": it hardly cuts anything off.

1.7.16 Other capacitor applications

In addition to their uses in filters, resonant circuits, differentiators, and integrators, capacitors are needed for several other important applications. We treat these in detail later in the book, mentioning them here only as a preview.

A. Bypassing

The impedance of a capacitor decreases with increasing frequency. This is the basis of another important application: *bypassing*. There are places in circuits where you want to allow a dc voltage, but you don't want signals present. Placing a capacitor across that circuit element (usually a resistor) will help to kill any signals there. You choose the (noncritical) capacitor value so that its

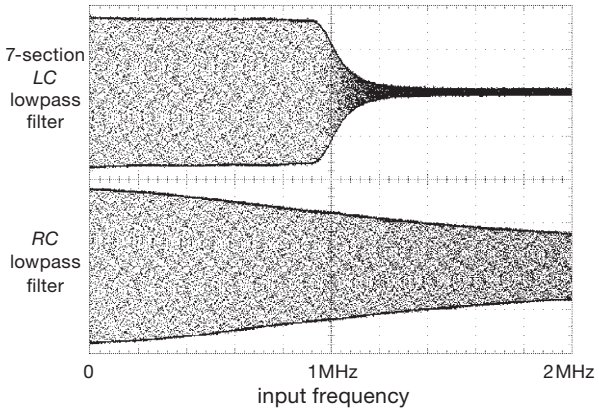


Figure 1.112. Frequency sweep of the *LC* lowpass filter shown in Figure 1.111 compared with an *RC* lowpass filter with the same 1 MHz cutoff frequency. The dark outline is the amplitude envelope of the fast swept sinewave, which achieves a sandpaper appearance in this digital 'scope capture.

impedance at signal frequencies is small compared with what it is bypassing. You will see much more of this in later chapters.

B. Power-supply filtering

We saw this application in §1.6.3, to filter the ripple from rectifier circuits. Although circuit designers often call them *filter* capacitors, this is really a form of bypassing, or energy storage, with large-value capacitors; we prefer the term *storage* capacitor. And these capacitors really are large – they're the big shiny round things you see inside most electronic instruments. We'll get into dc power-supply design in detail in Chapter 9.

C. Timing and waveform generation

As we've seen, a capacitor supplied with a constant current charges up with a ramp waveform. This is the basis of ramp and sawtooth generators, used in analog function generators, oscilloscope sweep circuits, analog–digital converters, and timing circuits. *RC* circuits are also used for timing, and they form the basis of delay circuits (monostable multivibrators). These timing and waveform applications are important in many areas of electronics and will be covered in Chapters 3, 6, 10, and 11.

1.7.17 Thévenin's theorem generalized

When capacitors and inductors are included, Thévenin's theorem must be restated: any two-terminal network of resistors, capacitors, inductors, and signal sources is equivalent to a single complex impedance in series with a single signal source. As before, you find the (complex) impedance and the signal source (waveform, amplitude, and phase) from the open-circuit output voltage and the short-circuit output current.

1.8 Putting it all together – an AM radio

In our circuit course we tie together the topics of this chapter by hooking up a simple AM radio. The signal that's transmitted is a sinewave at the station's frequency in the AM band (520–1720 kHz), with its amplitude varied ("modulated") according to the audio waveform (Figure 1.113). In other words, an audio waveform described by some function $f(t)$ would be transmitted as a RF signal $[A + f(t)]\sin 2\pi f_c t$; here f_c is the station's "carrier" frequency, and the constant A is added to the audio waveform so that the coefficient $[A + f(t)]$ is never negative.

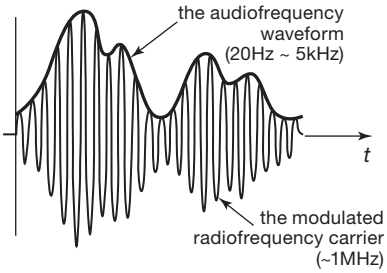


Figure 1.113. An AM signal consists of an RF carrier (~1 MHz) whose amplitude is varied by the audio-frequency signal (speech or music; audible frequencies up to ~5 kHz). The audio waveform is dc offset so that the envelope does not cross zero.

At the receiver end (that's *us*!) the task is to select this station (among many) and somehow extract the modulating *envelope*, which is the desired audio signal. Figure 1.114 shows the simplest AM radio; it is the "crystal set" of yesteryear. It's really quite straightforward: the parallel *LC* resonant circuit is tuned to the station's frequency by the variable capacitor C_1 (§1.7.14); the diode D is a half-wave rectifier (§1.6.2), which (if ideal) would pass only the positive half-cycles of the modulated carrier; and R_1 provides a light load, so that the rectified output follows the half-cycles back down to zero. We're almost done. We add small capacitor C_2 to prevent the output from following the fast half-cycles of carrier (it's a storage capacitor, §1.7.16B), choosing the time constant R_1C_2 to be long compared with a carrier period (~1 μs), but short compared with the period of the highest audio frequency (~200 μs).

Figures 1.115 and 1.116 show what you see when you probe around with a 'scope. The *bare* antenna shows plenty of low-frequency pickup (mostly 60 Hz ac powerline), and a tiny bit of signal from all the AM stations at once. But

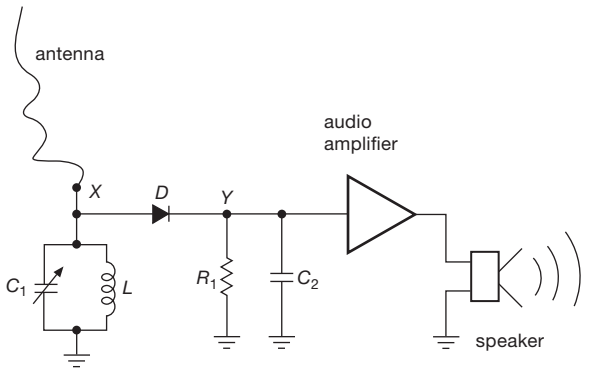


Figure 1.114. The simplest AM receiver. Variable capacitor C_1 tunes the desired station, diode D picks off the positive envelope (smoothed by R_1C_2), and the resulting weak audio signal is amplified to drive the loudspeaker, loudly.

when you connect it to the *LC* resonant circuit, all the low-frequency stuff disappears (because the *LC* looks like a very low impedance, Figure 1.107) and it sees only the selected AM station. What's interesting here is that the amplitude of the selected station is much larger with the *LC* attached than with nothing connected to the antenna: that's because the resonant circuit's high Q is storing energy from multiple cycles of the signal.[49]

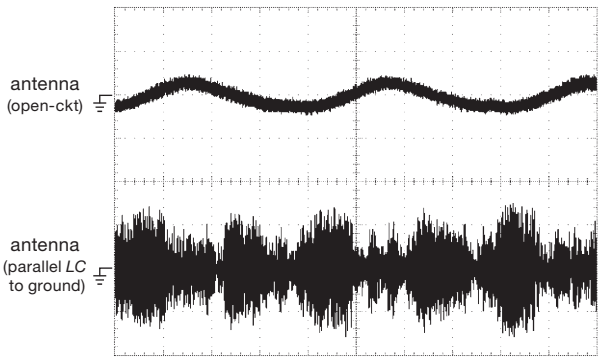


Figure 1.115. Observed waveforms at point "X" from the bare antenna (top) and with the *LC* connected. Note that the low-frequency junk disappears and that the radio signal gets *larger*. These are single-shot traces, in which the ∼1 MHz radiofrequency carrier appears as a solid filled area. Vertical: 1 V/div; horizontal: 4 ms/div.

The audio amplifier is fun, too, but we're not ready for it. We'll see how to make one of those in Chapter 2 (with discrete transistors), and again in Chapter 4 (with operational amplifiers, the Lego™ block of analog design).

[49] There are more complicated ways of framing this, but you don't really want to know just yet...

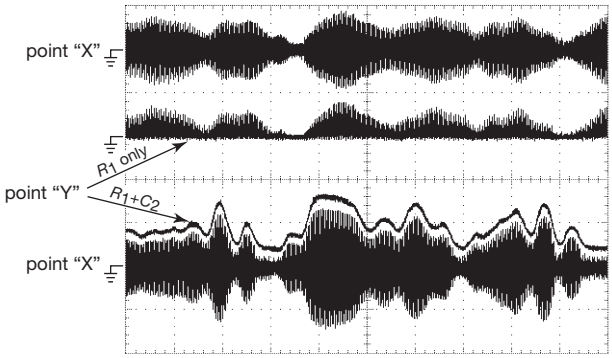


Figure 1.116. Observed waveforms at point "Y" with R_1 only (top) and with smoothing capacitor C_2 included (bottom). The upper pair is a single-shot capture (with the ∼1 MHz carrier appearing as solid black), and the lower pair is a separate single-shot capture, in which we have offset the rectified wave for clarity. Vertical: 1 V/div; horizontal: 1 ms/div.

And one amusing final note: in our class, we like to show the effect of probing point "X" with a length of BNC (bayonet Neill–Concelman) cable going to a 'scope input (that's how we start out, in the first week). When we do that, the cable's capacitance (about 30 pF/ft) adds to C_1, lowering the resonant frequency and so tuning to a different station. If we choose right, it changes *languages* (from English to Spanish)! The students howl with laughter – a language-translating electronic component. Then we use an ordinary 'scope probe, with its ∼10 pF of capacitance: no change of station, nor of language.

1.9 Other passive components

In the following subsections we would like to introduce briefly an assortment of miscellaneous but essential components. If you are experienced in electronic construction, you may wish to proceed to the next chapter.

1.9.1 Electromechanical devices: switches

These mundane but important devices seem to wind up in most electronic equipment. It is worth spending a few paragraphs on the subject (and there's more in Chapter *1x*). Figures 1.117 and 1.118 show some common switch types.

A. Toggle switches

The simple toggle switch is available in various configurations, depending on the number of poles; Figure 1.119 shows the usual ones (SPST indicates a single-pole single-throw switch, SPDT indicates a single-pole double-throw

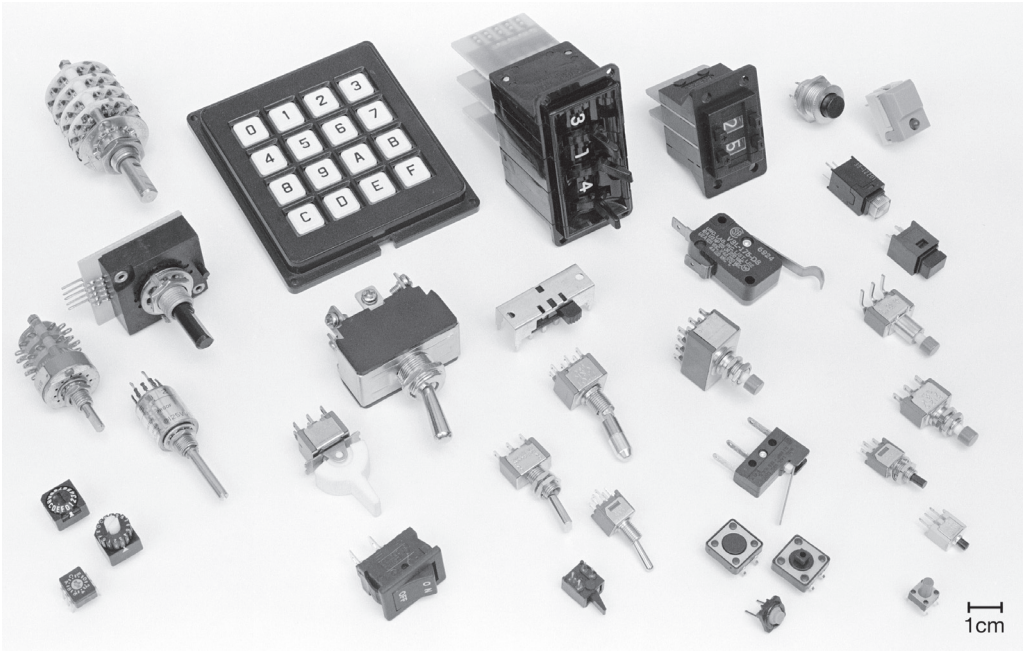


Figure 1.117. Switch Smorgasbord. The nine switches at right are momentary-contact ("pushbutton") switches, including both panel-mounting and PCB-mounting types (PCB, printed-circuit board). To their left are additional types, including lever-actuated and multipole styles. Above them are a pair of panel-mounting binary-coded *thumbwheel* switches, to the left of which is a matrix-encoded hexadecimal keypad. The switches at center foreground are toggle switches, in both panel-mounting and PCB-mounting varieties; several actuator styles are shown, including a locking variety (fourth from front) that must be pulled before it will switch. The rotary switches in the left column illustrate binary-coded types (the three in front and the larger square one), and the traditional multipole–multiposition configurable wafer switches.

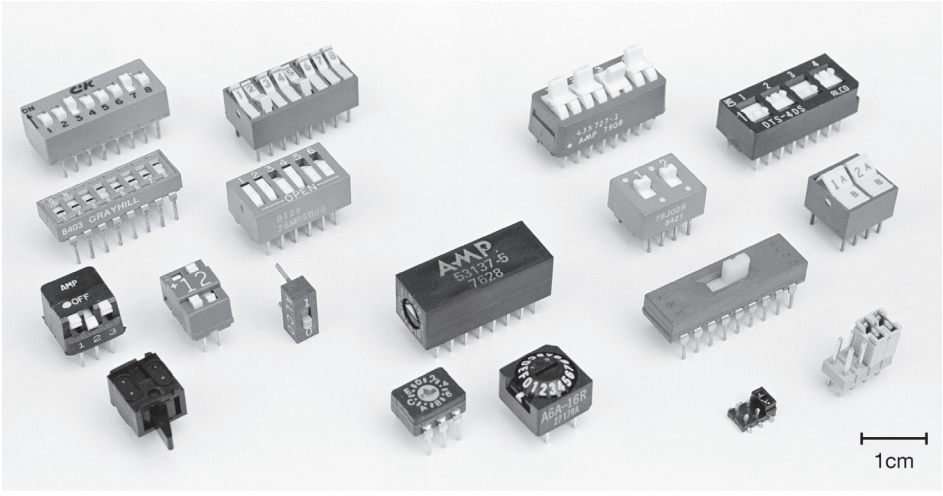


Figure 1.118. Board-mounted "DIP switches." Left group, front to back and left to right (all are SPST): single station side-action toggle; three-station side-action, two-station rocker, and single-station slide; eight-station slide (low-profile) and six-station rocker; eight-station slide and rocker. Middle group (all are hexadecimal coded): six-pin low-profile, six-pin with top or side adjust; 16-pin with true and complement coding. Right group: 2 mm×2 mm surface-mount header block with movable jumper ("shunt"), 0.1″×0.1″ (2.54 mm×2.54 mm) through-hole header block with shunts; 18-pin SPDT (common actuator); eight-pin dual SPDT slide and rocker; 16-pin quad SPDT slide (two examples).

switch, and DPDT indicates a double-pole double-throw switch). Toggle switches are also available with "center OFF" positions and with up to four poles switched simultaneously. Toggle switches are always "break before make," e.g., the moving contact never connects to both terminals in an SPDT switch.

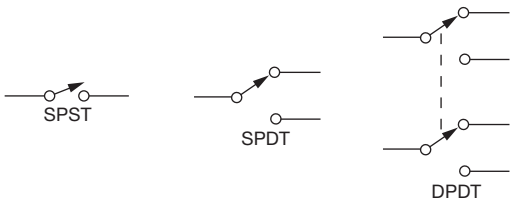


Figure 1.119. Fundamental switch types.

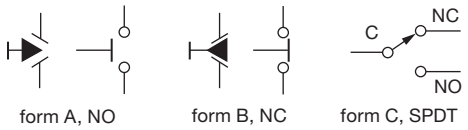


Figure 1.120. Momentary-contact (pushbutton) switches.

B. Pushbutton switches

Pushbutton switches are useful for momentary-contact applications; they are drawn schematically as shown in Figure 1.120 (NO and NC mean normally open and normally closed). For SPDT momentary-contact switches, the terminals must be labeled NO and NC, whereas for SPST types the symbol is self-explanatory. Momentary-contact switches are always "break before make." In the electrical (as opposed to electronic) industry, the terms form A, form B, and form C are used to mean SPST (NO), SPST (NC), and SPDT, respectively.

C. Rotary switches

Rotary switches are available with many poles and many positions, often as kits with individual wafers and shaft hardware. Both *shorting* (make-before-break) and *nonshorting* (break-before-make) types are available, and they can be mixed on the same switch. In many applications the shorting type is useful to prevent an open circuit between switch positions, because circuits can go amok with unconnected inputs. Nonshorting types are necessary if the separate lines being switched to one common line must not ever be connected to each other.

Sometimes you don't really want all those poles, you just want to know how many clicks (detents) the shaft has been turned. For that a common form of rotary switch encodes its position as a 4-bit binary quantity, thereby saving lots of wires (only five are needed: the four bits, and a common line). An alternative is the use of a *rotary encoder*, an electromechanical panel-mounting device that creates a sequence of *N* pulse pairs for each full rotation of the knob. These come in two flavors (internally using either mechanical contacts or electro-optical methods), and typically provide from 16 to 200 pulse pairs per revolution. The optical varieties cost more, but they last forever.

D. PC-mounting switches

It's common to see little arrays of switches on printed-circuit (PC) boards, like the ones shown in Figure 1.118. They're often called *DIP switches*, referring to the integrated circuit dual in-line package that they borrow, though contemporary practice increasingly uses the more compact *surface-mount technology* (SMT) package. As the photograph illustrates, you can get coded rotary switches; and because these are used for set-and-forget internal settings, you can substitute a multipin *header* block, with little slide-on "shunts" to make the connections.

E. Other switch types

In addition to these basic switch types, there are available various exotic switches such as Hall-effect switches, reed switches, proximity switches, etc. All switches carry maximum current and voltage ratings; a small toggle switch might be rated at 150 volts and 5 amps. Operation with inductive loads drastically reduces switch life because of arcing during turn-off. It's always OK to operate a switch *below* its maximum ratings, with one notable exception: since many switches rely on substantial current flow to clean away contact oxides, it's important to use a switch that is designed for "dry switching" when switching low-level signals;[50] otherwise you'll get noisy and intermittent operation (see Chapter *1x*).

F. Switch examples

As an example of what can be done with simple switches, let's consider the following problem: suppose you want to sound a warning buzzer if the driver of a car is seated and one of the car doors is open. Both doors and the driver's seat have switches, all normally open. Figure 1.121 shows a circuit that does what you want. If one OR the other door is open (switch closed) AND the seat switch is closed, the buzzer sounds. The words OR and AND are used in a logic sense here, and we will see this example again in

[50] These use gold contact plating.

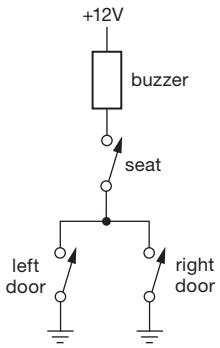


Figure 1.121. Switch circuit example: open door warning.

Chapters 2, 3, and 10 when we talk about transistors and digital logic.

Figure 1.122 shows a classic switch circuit used to turn a ceiling lamp on or off from a switch at either of two entrances to a room.

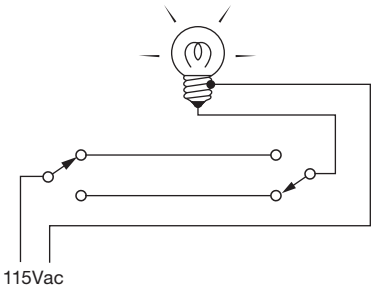


Figure 1.122. Electrician's "three-way" switch wiring.

Exercise 1.36. Although few electronic circuit designers know how, every *electrician* can wire up a light fixture so that any of N switches can turn it on or off. See if you can figure out this generalization of Figure 1.122. It requires two SPDT switches and $N-2$ DPDT switches.

1.9.2 Electromechanical devices: relays

Relays are electrically controlled switches. In the traditional electromechanical relay, a coil pulls in an armature (to close the contacts) when sufficient coil current flows. Many varieties are available, including "latching" and "stepping" relays.[51] Relays are available with dc or ac excitation, and coil voltages from 3 volts up to 115 volts (ac or dc) are common. "Mercury-wetted" and "reed" relays are intended for high-speed ($\sim$1 ms) applications, and giant relays intended to switch thousands of amps are used by power companies.

The *solid-state relay* (SSR) – consisting of a semiconductor electronic switch that is turned on by a LED – provides better performance and reliability than mechanical relays, though at greater cost. SSRs operate rapidly, without contact "bounce," and usually provide for smart switching of ac power (they turn on at the moment of zero voltage, and they turn off at the moment of zero current). Much more on these useful devices in Chapter 12.

As we'll learn, electrically controlled switching of signals within a circuit can be accomplished with transistor switches, without having to use relays of any sort (Chapters 2 and 3). The primary uses of relays are in remote switching and high-voltage (or high-current) switching, where it is important to have complete electrical isolation between the control signal and the circuit being switched.

1.9.3 Connectors

Bringing signals in and out of an instrument, routing signal and dc power around between the various parts of an instrument, providing flexibility by permitting circuit boards and larger modules of the instrument to be unplugged (and replaced) – these are the functions of the *connector*, an essential ingredient (and usually the most unreliable part) of any piece of electronic equipment. Connectors come in a bewildering variety of sizes and shapes.[52] Figures 1.123, 1.124, and 1.125 give some idea of the variety.

A. Single-wire connectors

The simplest kind of connector is the simple pin jack or banana jack used on multimeters, power supplies, etc. It is handy and inexpensive, but not as useful as the shielded-cable or multiwire connectors you often need. The humble binding post is another form of single-wire connector, notable for the clumsiness it inspires in those who try to use it.

B. Shielded-cable connectors

To prevent capacitive pickup, and for other reasons we'll go into in Appendix H, it is usually desirable to pipe signals around from one instrument to another in shielded coaxial cable. The most popular connector is the BNC type that

[51] In an amusing historical footnote, the stepping relay used for a century as the cornerstone of telephone exchanges (the "Strowger selector") was invented by a Topeka undertaker, Almon Strowger, evidently because he suspected that telephone calls intended for his business were being routed (by the switchboard operators in his town) to a funeral home competitor.

[52] A search for "connector" on the DigiKey website returns 116 categories, with approximately 43,000 individual varieties in stock.

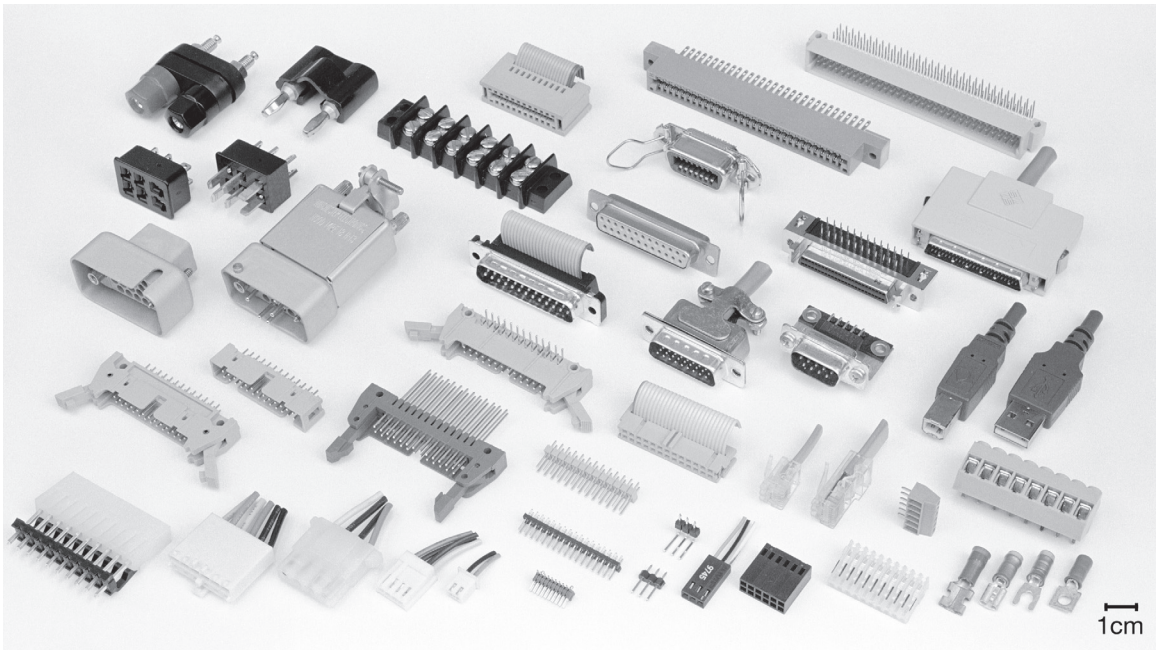


Figure 1.123. Rectangular connectors. The variety of available multipin connectors is staggering. Here is a collection of common specimens: the five connectors at lower left are multipin nylon power connectors (sometimes called *Molex-type* for historical reasons). Above them are four dual-row box headers (0.1″ spacing, shown with and without latch ejectors, and also with Wire-Wrap® and right-angle tails), and to their right an open ("unshrouded") 0.1″ dual-row header, along with a pair of dual-row headers of finer pitch (2 mm and 1.27 mm). These dual-row male connectors mate with *insulation displacement* connectors (IDC) such as the one shown attached to a short length of ribbon cable (just above the unshrouded header). Just below the ribbon are shown single-row 0.1″ headers, with mating shells (AMP MODU) that accept individual wire leads. At bottom right are several terminal blocks used for power wiring, and four "Faston"-type crimpable spade lugs. Above them are USB connectors, and to their left are the common RJ-45 and RJ-11 modular telephone/data connectors. The popular and reliable D-subminiature connectors are at center, including (right to left) a pair of 50-pin micro-D (cable plug, PCB socket), the 9-pin D-sub, 26-pin high-density, and a pair of 25-pin D-subs (one IDC). Above them are (right to left) a 96-pin VME backplane connector, a 62-pin card-edge connector with solder tails, a "Centronics-type" connector with latching bail, and a card-edge connector with ribbon IDC. At top left is a miscellany – a mating pair of "GR-type" dual banana connectors, a mating pair of Cinch-type connectors, a mating pair of shrouded Winchester-type connectors with locking jackscrews, and (to their right) a screw-terminal barrier block. Not shown here are the *really* tiny connectors used in small portable electronics (smartphones, cameras, etc); you can see a fine example in Figure 1.131.

adorns most instrument front panels. It connects with a quarter-turn twist and completes both the shield (ground) circuit and inner conductor (signal) circuit simultaneously. Like all connectors used to mate a cable to an instrument, it comes in both panel-mounting and cable-terminating varieties.

Among the other connectors for use with coaxial cable are the TNC ("threaded Neill–Concelman," a close cousin of the BNC, but with threaded outer shell), the high-performance but bulky type N, the miniature SMA and SMB, the subminiature LEMO and SMC, and the high-voltage MHV and SHV. The so-called phono jack used in audio equipment is a nice lesson in bad design, because the inner (signal) conductor mates *before* the shield (ground) when you plug it in; furthermore, the design of the connector is such that both shield and center conductor tend to make poor contact. You've undoubtedly *heard* the results! Not to be outdone, the television industry has responded with its own bad standard, the type-F coax "connector," which uses the unsupported inner wire of the coax as the pin of the male plug, and a shoddy arrangement to mate the shield.[53]

We hereby induct these losers into the Electronic

[53] Advocates of each would probably reply "This is our most *modestly* priced receptacle."

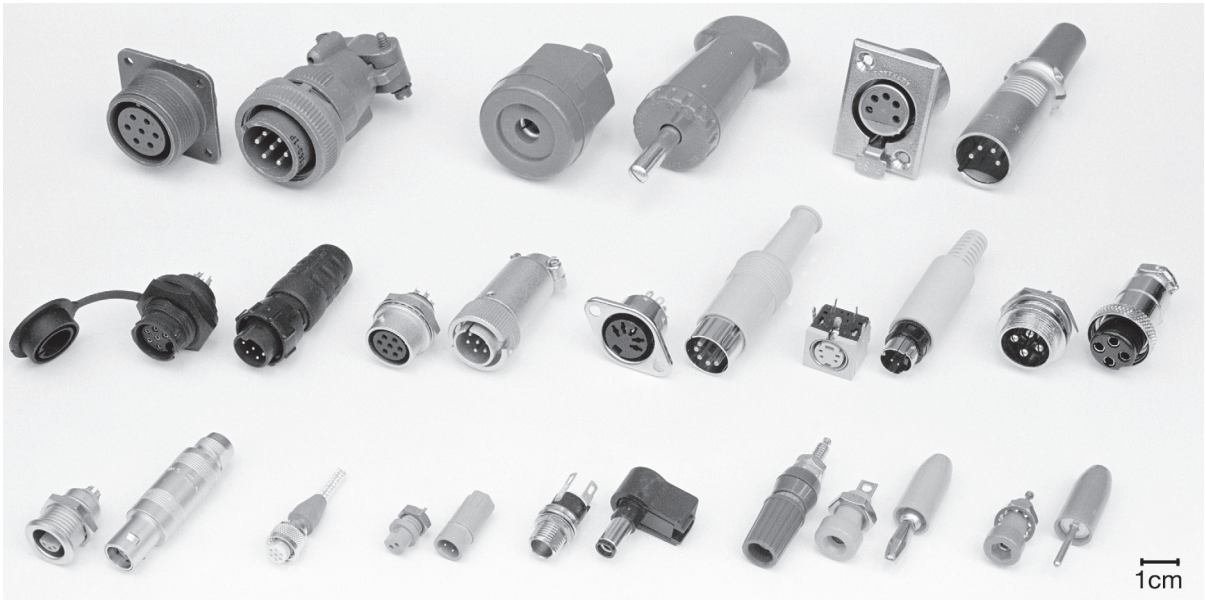


Figure 1.124. Circular connectors. A selection of multipin and other "non-RF" connectors; the panel-mounting receptacle is shown to the left of each cable-mounted plug. Top row, left to right: "MS"-type (MIL-C-5015) rugged connector (available in hundreds of configurations), high-current (50 A) "Supericon," multipin locking XLR. Middle row: weatherproof (Switchcraft EN3), 12 mm video (Hirose RM), circular DIN, circular mini-DIN, 4-pin microphone connector. Bottom row: locking 6-pin (Lemo), microminiature 7-pin shielded (Microtech EP-7S), miniature 2-pin shrouded (Litton SM), 2.5 mm power, banana, pin jack.

Components Hall of Infamy, some charter members of which are shown in Figure 1.126.

C. Multipin connectors

Very frequently electronic instruments demand multiwire cables and connectors. There are literally dozens of different kinds. The simplest example is a three-wire "IEC" powerline cord connector. Among the more popular are the excellent type-D subminiature, the Winchester MRA series, the venerable MS type, and the flat ribbon-cable mass-termination connectors. These and others are shown in Figure 1.123.

Beware of connectors that can't tolerate being dropped on the floor (the miniature hexagon connectors are classic) or that don't provide a secure locking mechanism (e.g., the Jones 300 series).

D. Card-edge connectors

The most common method used to make connection to printed-circuit cards is the card-edge connector, which mates to a row of gold-plated contacts at the edge of the card; common examples are the motherboard connectors that accept plug-in computer memory modules. Card-edge connectors may have from 15 to 100 or more connections, and they come with different lug styles according to the method of connection. You can solder them to a "motherboard" or "backplane," which is itself just another PCB containing the interconnecting wiring between the individual circuit cards. Alternatively, you may want to use edge connectors with standard solder-lug terminations, particularly in a system with only a few cards. A more reliable (though more costly) solution is the use of "two-part" PCB connectors, in which one part (soldered onto the board) mates with the other part (on a backplane, etc); an example is the widely used VME (VersaModule Eurocard) connector (upper right-hand corner of Figure 1.123).

1.9.4 Indicators

A. Meters

To read out the value of some voltage or current, you have a choice between the time-honored moving-pointer type of meter and digital-readout meters. The latter are more expensive and more accurate. Both types are available in a variety of voltage and current ranges. There are, in addition, exotic panel meters that read out such things as VUs (volume units, an audio dB scale), expanded-scale ac volts (e.g., 105 to 130 V), temperature (from a thermocouple),



Figure 1.125. RF and shielded connectors. The panel-mounting receptacle is shown to the left of each cable-mounted plug. Top row, left to right: stereo phone jack, audio "XLR" type; N and UHF (RF connectors). Second row down: BNC, TNC, type F; MHV and SHV (high voltage). Third row down: 2.5 mm (3/32″) audio, 3.5 mm stereo, improved 3.5 mm stereo, phono ("RCA type"), LEMO coaxial. Bottom row: SMA (panel jack, flexible coax plug), SMA (board-mount jack, rigid coax plug), SMB; SC and ST (optical fiber).

percentage motor load, frequency, etc. Digital panel meters often provide the option of logic-level outputs, in addition to the visible display, for internal use by the instrument.

As a substitute for a dedicated meter (whether analog or digital), you increasingly see an LCD (liquid-crystal display) or LED panel with a meter-like pattern. This is flexible and efficient: with a graphic LCD display module (§12.5.3) you can offer the user a choice of "meters," according to the quantity being displayed, all under the control of an embedded controller (a built-in microprocessor; see Chapter 15).

B. Lamps, LEDs, and displays

Flashing lights, screens full of numbers and letters, eerie sounds – these are the stuff of science fiction movies, and except for the last, they form the subject of lamps and displays (see §12.5.3). Small incandescent lamps used to be standard for front-panel indicators, but they have been replaced with LEDs. The latter behave electrically like ordinary diodes, but with a forward voltage drop in the range of 1.5 to 2 volts (for red, orange, and some green LEDs; 3.6 V for blue[54] and high-brightness green; see Figure 2.8). When current flows in the forward direction, they light up. Typically, 2 mA to 10 mA produces adequate brightness. LEDs are cheaper than incandescent lamps, they last pretty much forever, and they come in four standard colors as well as "white" (which is usually a blue LED with a yellow fluorescent coating). They come in convenient panel-mounting packages; some even provide built-in current limiting.[55]

LEDs can also be used for digital displays, for example

[54] The invention of the gallium nitride blue LED was the breakthrough product of a lone and unappreciated employee of Nichia Chemical Industries, Shuji Nakamura.

[55] And of course, for both residential and commercial area lighting, LEDs have now largely relegated to the dustbin of history the century-old hot-filament incandescent lamp.

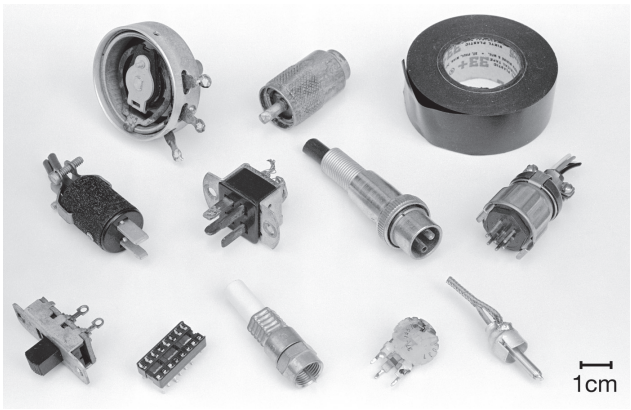


Figure 1.126. Components to avoid. We advise against using components like these, if you have a choice (see text if you need convincing!). Top row, left to right: low-value wirewound pot, type UHF connector, electrical tape ("just say no!"). Middle row: "cinch-type" connectors, microphone connector, hexagon connectors. Bottom row: slide switch, cheap IC socket (not "screw-machined"), type-F connector, open-element trimmer pot, phono connector.

as 7-segment numeric displays or (for displaying letters as well as numbers – "alphanumeric") 16-segment displays or dot-matrix displays. However, if more than a few digits or characters need to be displayed, LCDs are generally preferred. These come in line-oriented arrays (e.g., 16 characters by 1 line, up to 40 characters by 4 lines), with a simple interface that permits sequential or addressable entry of alphanumeric characters and additional symbols. They are inexpensive, low power, and visible even in sunlight. Back-lighted versions work well even in subdued light, but are not low power. Much more on these (and other) *optoelectronic* devices in §12.5.

1.9.5 Variable components

A. Variable resistors

Variable resistors (also called volume controls, potentiometers, pots, or trimmers) are useful as panel controls or internal adjustments in circuits. A classic panel type is the 2-watt-type AB potentiometer; it uses the same basic material as the fixed carbon-composition resistor, with a rotatable "wiper" contact. Other panel types are available with ceramic or plastic resistance elements, with improved characteristics. Multiturn types (3, 5, or 10 turns) are available, with counting dials, for improved resolution and linearity. "Ganged" pots (several independent sections on one shaft) are also manufactured, although in limited variety, for applications that demand them. Figure 1.8 shows a representative selection of pots and trimmers.

For use inside an instrument, rather than on the front panel, *trimmer pots* come in single-turn and multiturn styles, most intended for printed-circuit mounting. These are handy for calibration adjustments of the "set-and-forget" type. Good advice: resist the temptation to use lots of trimmers in your circuits. Use good design instead.

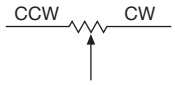


Figure 1.127. Potentiometer (three-terminal variable resistor).

The symbol for a variable resistor, or pot, is shown in Figure 1.127. Sometimes the symbols CW and CCW are used to indicate the clockwise and counterclockwise ends.

An all-electronic version of a potentiometer can be made with an array of electronic (transistor) switches that select a tap in a long chain of fixed resistors. As awkward as that may sound, it is a perfectly workable scheme when implemented as an IC. For example, Analog Devices, Maxim/Dallas Semiconductor, and Xicor make a series of "digital potentiometers" with up to 1024 steps; they come as single or dual units, and some of them are "nonvolatile," meaning that they remember their last setting even if power has been turned off. These find application in consumer electronics (televisions, stereos) where you want to adjust the volume from your infrared remote control, rather than by turning a knob; see §3.4.3E.

One important point about variable resistors: don't attempt to use a potentiometer as a substitute for a precise resistor value somewhere within a circuit. This is tempting, because you can trim the resistance to the value you want. The trouble is that potentiometers are not as stable as good (1%) resistors, and in addition they may not have good resolution (i.e., they can't be set to a precise value). If you must have a precise and settable resistor value somewhere, use a combination of a 1% (or better) precision resistor and a potentiometer, with the fixed resistor contributing most of the resistance. For example, if you need a 23.4k resistor, use a 22.6k 1% fixed resistor (a standard value) in series with a 2k trimmer pot. Another possibility is to use a series combination of several precision resistors, selecting the last (and smallest) resistor to give the desired series resistance.

As we'll see later (§3.2.7), it is possible to use FETs as voltage-controlled variable resistors in some applications. Another possibility is an "optophotoresistor" (§12.7). Transistors can be used as variable-gain amplifiers, again controlled by a voltage. Keep an open mind when design brainstorming.



Figure 1.128. Variable capacitor.

B. Variable capacitors

Variable capacitors are primarily confined to the smaller capacitance values (up to about 1000 pF) and are commonly used in RF circuits. Trimmers are available for in-circuit adjustments, in addition to the panel type for user tuning. Figure 1.128 shows the symbol for a variable capacitor.

Diodes operated with applied reverse voltage can be used as voltage-variable capacitors; in this application they're called *varactors*, or sometimes *varicaps* or *epicaps*. They're very important in RF applications, especially phase-locked loops, automatic frequency control (AFC), modulators, and parametric amplifiers.

C. Variable inductors

Variable inductors are usually made by arranging to move a piece of core material in a fixed coil. In this form they're available with inductances ranging from microhenrys to henrys, typically with a 2:1 tuning range for any given inductor. Also available are rotary inductors (coreless coils with a rolling contact).[56]

D. Variable transformers

Variable transformers are handy devices, especially the ones operated from the 115 volt ac line. They're usually configured as "autotransformers," which means that they have only one winding, with a sliding contact. They're also commonly called Variacs (the name given to them by General Radio), and they are made by Technipower, Superior Electric, and others. Figure 1.129 shows a classic unit from General Radio. Typically they provide 0 to 135 volts ac output when operated from 115 volts, and they come in current ratings from 1 amp to 20 amps or more. They're good for testing instruments that seem to be affected by powerline variations, and in any case to verify worst-case performance. *Important Warning*: don't forget that the output is not electrically isolated from the powerline, as it would be with a transformer!

[56] An interesting form of variable inductor of yesteryear was the *variometer*, a rotatable coil positioned within a fixed outer coil and connected in series with it. As the inner coil was rotated, the total inductance went from maximum (four times the inductance of either coil alone) all the way down to zero. These things were *consumer* items, listed for example in the 1925 Sears Roebuck catalog.



Figure 1.129. A powerline variable transformer ("Variac") lets you adjust the ac input voltage to something you are testing. Here a 5 A unit is shown, both clothed and undressed.

1.10 A parting shot: confusing markings and itty-bitty components

In our electronics course,[57] and indeed in day-to-day electronics on the bench, we encounter a wonderful confusion of component markings. Capacitors in particular are just, well, perverse: they rarely bother specifying *units* (even though they span 12 orders of magnitude, picofarads to farads), and for ceramic SMT varieties they dispense with any markings whatsoever! Even worse, they are still caught up in the transition from printing the value as an integer (e.g., "470" meaning 470 pF) versus using exponent notation (e.g., "470" meaning 47×10^0, i.e., 47 pF). Figure 1.130 shows exactly that case! Another trap for the unwary (and sometimes the wary, as well) is the date-code *gotcha*: the 4-digit code (yydd) can masquerade as a part number, as in the four examples in the photo. And, as components become smaller and smaller, there's precious little room for all but the briefest of markings; so, following the pharmaceutical industry, manufacturers invent a short

[57] Physics 123 ("Laboratory Electronics") at Harvard University: "Half course (fall term; repeated spring term). A lab-intensive introduction to electronic circuit design. Develops circuit intuition and debugging skills through daily hands-on lab exercises, each preceded by class discussion, with minimal use of mathematics and physics. Moves quickly from passive circuits, to discrete transistors, then concentrates on operational amplifiers, used to make a variety of circuits including integrators, oscillators, regulators, and filters. The digital half of the course treats analog–digital interfacing, emphasizing the use of microcontrollers and programmable logic devices (PLDs)." See `http://webdocs.registrar.fas.harvard.edu/courses/Physics.html`.

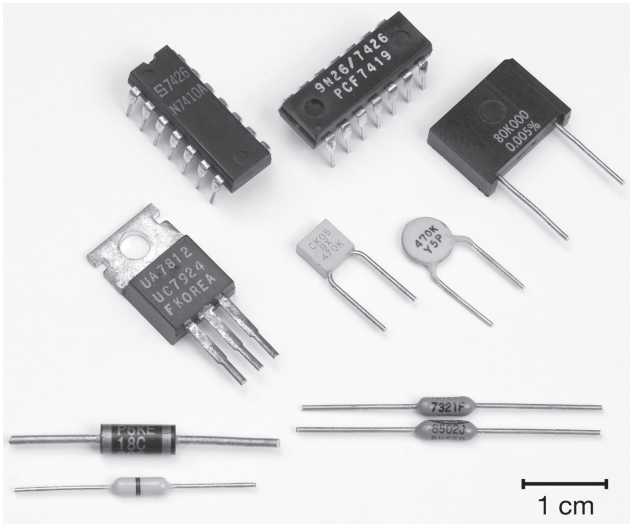


Figure 1.130. Confusion Central! The three ICs are each marked with both a part number (e.g., UA7812) and a "date code" (e.g., UC7924, signifying the 24th week of 1979). Unfortunately, both are perfectly valid part numbers (a +12 V or a −24 V regulator). The resistor pair (actually two views of identically marked resistors) suffers from the same problem: it could be 7.32 KΩ±1%, or it could be 85.0 kΩ±5% (it's the former, but who would know?). The pair of ceramic capacitors are both marked 470K (470,000 of something?), but, surprise, the "K" means 10% tolerance; and, bigger surprise, the square cap is 47 pF, the round one is 470 pF. And what is one to make of a black box labeled 80K000 (pronounced "eighty-koooh"), or a diode with two cathodes (and no anode?), or a resistor with a single black band in the center?

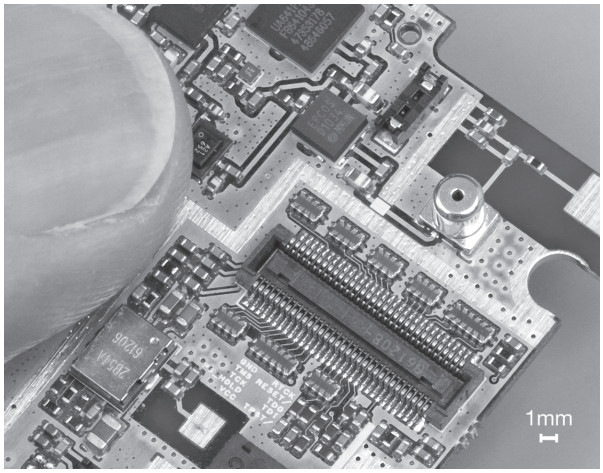


Figure 1.131. We're "all thumbs" when working with surface-mount technology (SMT). This is a corner of a cellphone circuit board, showing small ceramic resistors and capacitors, integrated circuits with ball-grid connecting dots on their undersides, and the Lilliputian connectors for the antenna and display panel. See also Figure 4.84.

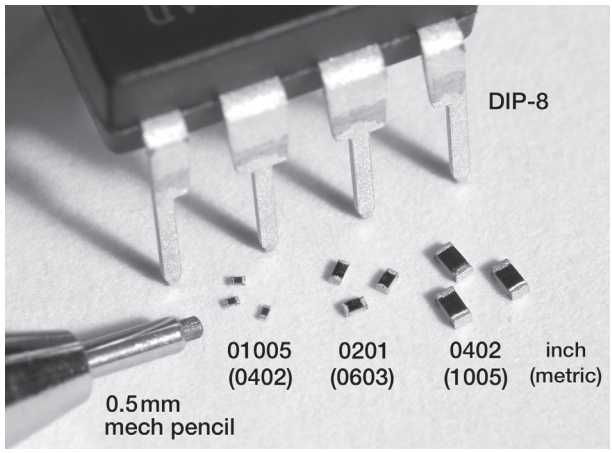


Figure 1.132. How small can these things get?! The "01005"-size SMT (0.016″×0.008″, or 0.4mm×0.2mm) represents the industry's greatest insult to the experimenter.

alphanumeric code for each component. And that's all you get. For example, National's LMV981 op-amp comes in several 6-pin packages: the SOT23 is marked "A78A," the smaller SC70 says "A77," and the really tiny microSMD blurts out a single letter "A" (or "H" if it's free of lead). Not much to go on.

1.10.1 Surface-mount technology: the joy and the pain

While we're complaining, let's whine just a bit about the difficulty of prototyping circuits with tiny surface-mount components. From an *electrical* point of view they are excellent: low inductance, and compact. But they are nearly impossible to wire up in prototype breadboard fashion, in the way that was easy with "through-hole" (or "leaded" – pronounced lee′-ded) components, such as resistors with axial leads (a wire sticking out each end), or integrated circuits in DIP (dual in-line) cases. Figure 1.131 gives a sense of the scale of these little components, and Figure 1.132 displays the true horror of the tiniest of these – the "01005"-size chip components (0402 metric) that measure 200μm×400μm: not much thicker than a human hair, and indistinguishable from dust!

Sometimes you can use little adapter carriers (from companies like Bellin Dynamic Systems, Capital Advanced Technologies, or Aries) to convert an SMT integrated circuit to a fake DIP. But the densest surface-mount

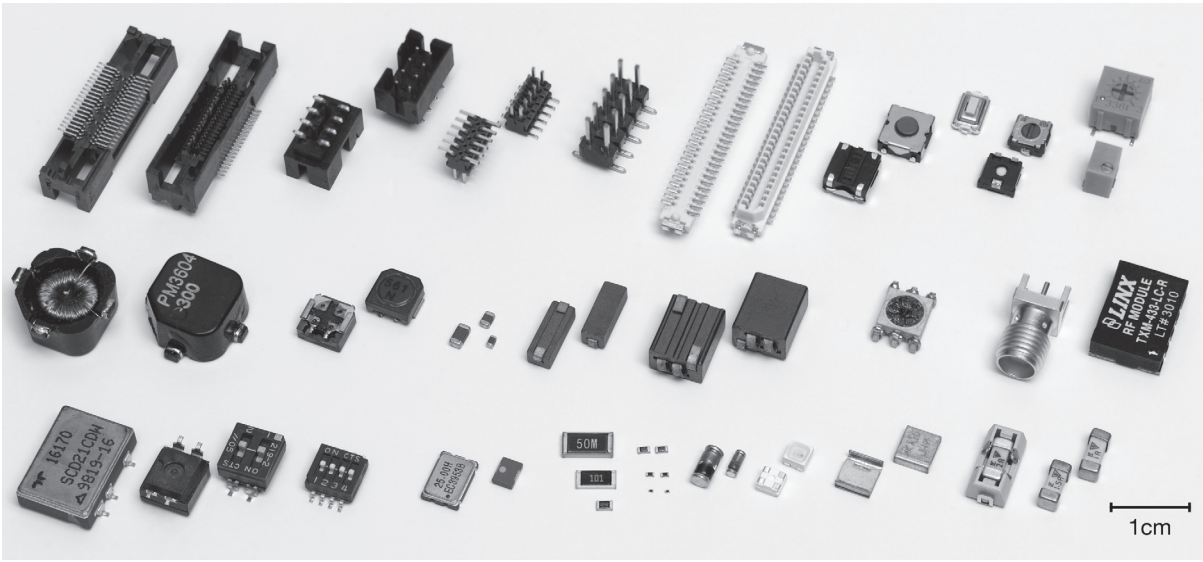


Figure 1.133. A taste of the world of passive components in surface-mount packages: connectors, switches, trimmer pots, inductors, resistors, capacitors, crystals, fuses. . . . If you can name it, you can probably get it in SMT.

packages have no leads at all, just an array of bumps (up to several thousand!) on the underside; and these require serious "reflow" equipment before you can do anything with them. Sadly, we cannot ignore this disturbing trend, because the majority of new components are offered only in surface-mount packages. Woe to the lone basement experimenter–inventor! Figure 1.133 give a sense of the variety of passive component types that come in surface-mount configurations.

Additional Exercises for Chapter 1

Exercise 1.37. Find the Norton equivalent circuit (a current source in parallel with a resistor) for the voltage divider in Figure 1.134. Show that the Norton equivalent gives the same output voltage as the actual circuit when loaded by a 5k resistor.

Exercise 1.38. Find the Thévenin equivalent for the circuit shown in Figure 1.135. Is it the same as the Thévenin equivalent for Exercise 1.37?

Exercise 1.39. Design a "rumble filter" for audio. It should pass frequencies greater than 20 Hz (set the -3 dB point at 10 Hz). Assume zero source impedance (perfect voltage source) and 10k (minimum) load impedance (that's important so that you can choose R and C such that the load doesn't affect the filter operation significantly).

Exercise 1.40. Design a "scratch filter" for audio signals (3 dB down at 10 kHz). Use the same source and load impedances as in Exercise 1.39.

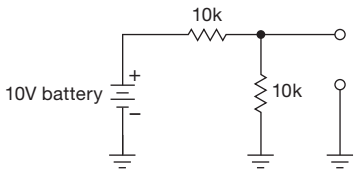


Figure 1.134. Example for Norton equivalent circuit.

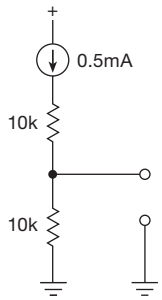


Figure 1.135. Example for Thévenin equivalent circuit.

Exercise 1.41. How would you make a filter with R's and C's to give the response shown in Figure 1.136?

Exercise 1.42. Design a bandpass RC filter (as in Figure 1.137);

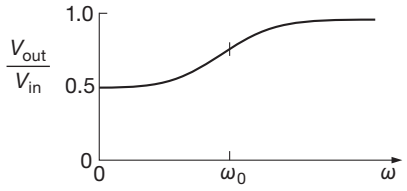


Figure 1.136. High-emphasis filter response.

f_1 and f_2 are the 3 dB points. Choose impedances so that the first stage isn't much affected by the loading of the second stage.

Exercise 1.43. Sketch the output for the circuit shown in Figure 1.138.

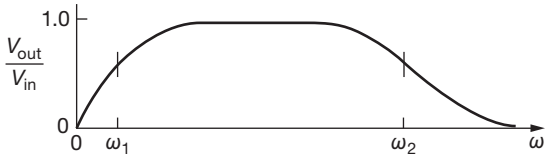


Figure 1.137. Bandpass filter response.

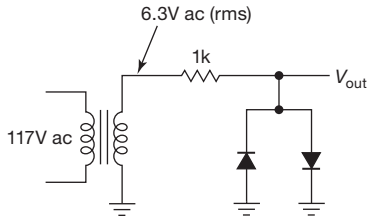


Figure 1.138. Circuit for Exercise 1.43.

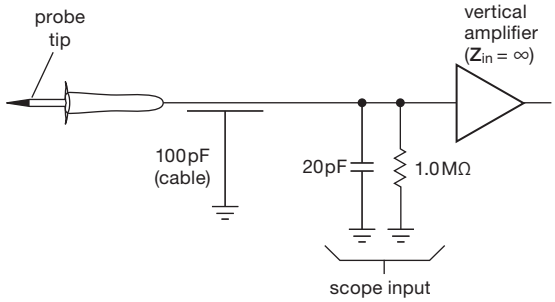


Figure 1.139. Oscilloscope $\times 10$ probe.

Exercise 1.44. Design an oscilloscope "$\times 10$ probe" to use with a scope whose input impedance is 1 MΩ in parallel with 20 pF by figuring out what goes inside the probe handle in Figure 1.139. Assume that the probe cable adds an additional 100 pF and that the probe components are placed at the tip end (rather than at the scope end) of the cable. The resultant network should have 20 dB ($\times 10$ voltage division ratio) attenuation at all frequencies, including dc. The reason for using a $\times 10$ probe is to increase the load impedance seen by the circuit under test, which reduces loading effects. What input impedance (R in parallel with C) does your $\times 10$ probe present to the circuit under test when used with the scope?

Review of Chapter 1

An A-to-H summary of what we have learned in Chapter 1. This summary reviews basic principles and facts in Chapter 1, but it does not cover application circuit diagrams and practical engineering advice presented there.

¶A. Voltage and Current.

Electronic circuits consist of components connected together with wires. *Current* (I) is the rate of flow of charge through some point in these connections; it's measured in amperes (or milliamps, microamps, etc.). *Voltage* (V) between two points in a circuit can be viewed as an applied driving "force" that causes currents to flow between them; voltage is measured in volts (or kilovolts, millivolts, etc.); see §1.2.1. Voltages and currents can be steady (dc), or varying. The latter may be as simple as the sinusoidal alternating voltage (ac) from the wallplug, or as complex as a high-frequency modulated communications waveform, in which case it's usually called a *signal* (see ¶B below). The algebraic sum of currents at a point in a circuit (a *node*) is zero (Kirchhoff's current law, KCL, a consequence of conservation of charge), and the sum of voltage drops going around a closed loop in a circuit is zero (Kirchhoff's voltage law, KVL, a consequence of the conservative nature of the electrostatic field).

¶B. Signal Types and Amplitude.

See §1.3. In digital electronics we deal with *pulses*, which are signals that bounce around between two voltages (e.g., +5 V and ground); in the analog world it's *sinewaves* that win the popularity contest. In either case, a periodic signal is characterized by its frequency f (units of Hz, MHz, etc.) or, equivalently, period T (units of ms, μs, etc.). For sinewaves it's often more convenient to use *angular* frequency (radians/s), given by $\omega = 2\pi f$.

Digital amplitudes are specified simply by the HIGH and LOW voltage levels. With sinewaves the situation is more complicated: the amplitude of a signal $V(t) = V_0 \sin \omega t$ can be given as (a) *peak* amplitude (or just "amplitude") V_0, (b) *root-mean-square* (rms) amplitude $V_{\text{rms}} = V_0/\sqrt{2}$, or (c) peak-to-peak amplitude $V_{\text{pp}} = 2V_0$. If unstated, a sinewave amplitude is usually understood to be V_{rms}. A signal of rms amplitude V_{rms} delivers power $P = V_{\text{rms}}^2/R_{\text{load}}$ to a resistive load (regardless of the signal's waveform), which accounts for the popularity of rms amplitude measure.

Ratios of signal amplitude (or power) are commonly expressed in *decibels* (dB), defined as $\text{dB} = 10\log_{10}(P_2/P_1)$ or $20\log_{10}(V_2/V_1)$; see §1.3.2. An amplitude ratio of 10 (or power ratio of 100) is 20 dB; 3 dB is a doubling of power; 6 dB is a doubling of amplitude (or quadrupling of power). Decibel measure is also used to specify amplitude (or power) directly, by giving a reference level: for example, −30 dBm (dB relative to 1 mW) is 1 microwatt; +3 dBVrms is a signal of 1.4 V rms amplitude (2 Vpeak, 4 Vpp).

Other important waveforms are square waves, triangle waves, ramps, noise, and a host of *modulation* schemes by which a simple "carrier" wave is varied in order to convey information; some examples are AM and FM for analog communication, and PPM (pulse-position modulation) or QAM (quadrature-amplitude modulation) for digital communication.

¶C. The Relationship Between Current and Voltage.

This chapter concentrated on the fundamental, essential, and ubiquitous *two-terminal linear devices*: resistors, capacitors, and inductors. (Subsequent chapters deal with *transistors* – three-terminal devices in which a signal applied to one terminal controls the current flow through the other pair – and their many interesting applications. These include amplification, filtering, power conversion, switching, and the like.) The simplest linear device is the *resistor*, for which $I = V/R$ (Ohm's Law, see §1.2.2A). The term "linear" means that the response (e.g., current) to a combined sum of inputs (i.e., voltages) is equal to the sum of the responses that each input would produce: $I(V_1 + V_2) = I(V_1) + I(V_2)$.

¶D. Resistors, Capacitors, and Inductors.

The resistor is clearly linear. But it is not the only linear two-terminal component, because linearity does not require $I \propto V$. The other two linear components are *capacitors* (§1.4.1) and *inductors* (§1.5.1), for which there is a time-dependent relationship between voltage and current: $I = C\,dV/dt$ and $V = L\,dI/dt$, respectively. These are the *time domain* descriptions. Thinking instead in the *frequency domain*, these components are described by their *impedances*, the ratio of voltage to current (as a function of frequency) when driven with a sinewave (§1.7). A linear device, when driven with a sinusoid, responds with a sinusoid of the same frequency, but with changed amplitude and phase. Impedances are therefore complex, with the real part representing the amplitude of the response that is in-phase, and the imaginary part representing the amplitude of the response that is in quadrature (90° out of phase). Alternatively, in the polar representation of complex impedance ($Z = |Z|e^{i\theta}$), the magnitude $|Z|$ is the ratio of magnitudes ($|Z| = |V|/|I|$) and the quantity θ is the phase shift between V and I. The impedances of the three

linear 2-terminal components are $Z_R=R$, $Z_C=-j/\omega C$, and $Z_L=j\omega L$, where (as always) $\omega=2\pi f$; see §1.7.5. Sinewave current through a resistor is in phase with voltage, whereas for a capacitor it leads by 90°, and for an inductor it lags by 90°.

¶E. Series and Parallel.

The impedance of components connected in series is the sum of their impedances; thus $R_{series}=R_1+R_2+\cdots$, $L_{series}=L_1+L_2+\cdots$, and $1/C_{series}=1/C_1+1/C_2+\cdots$. When connected in parallel, on the other hand, it's the *admittances* (inverse of impedance) that add. Thus the formula for capacitors in parallel looks like the formula for resistors in series, $C_{parallel}=C_1+C_2+\cdots$; and vice versa for resistors and inductors, thus $1/R_{parallel}=1/R_1+1/R_2+\cdots$. For a pair of resistors in parallel this reduces to $R_{parallel}=(R_1R_2)/(R_1+R_2)$. For example, two resistors of value R have resistance $R/2$ when connected in parallel, or resistance $2R$ in series.

The power dissipated in a resistor R is $P=I^2R=V^2/R$. There is no dissipation in an ideal capacitor or inductor, because the voltage and current are 90° out of phase. See §1.7.6.

¶F. Basic Circuits with *R*, *L*, and *C*.

Resistors are everywhere. They can be used to set an operating current, as for example when powering an LED or biasing a zener diode (Figure 1.16); in such applications the current is simply $I=(V_{supply}-V_{load})/R$. In other applications (e.g., as a transistor's load resistor in an amplifier, Figure 3.29) it is the *current* that is known, and a resistor is used to convert it to a voltage. An important circuit fragment is the *voltage divider* (§1.2.3), whose unloaded output voltage (across R_2) is $V_{out}=V_{in}R_2/(R_1+R_2)$.

If one of the resistors in a voltage divider is replaced with a capacitor, you get a simple *filter*: lowpass if the lower leg is a capacitor, highpass if the upper leg is a capacitor (§§1.7.1 and 1.7.7). In either case the −3 dB transition frequency is at $f_{3dB}=1/2\pi RC$. The ultimate rolloff rate of such a "single-pole" lowpass filter is −6 dB/octave, or −20 dB/decade; i.e., the signal amplitude falls as $1/f$ well beyond f_{3dB}. More complex filters can be created by combining inductors with capacitors, see Chapter 6. A capacitor in parallel with an inductor forms a *resonant circuit*; its impedance (for ideal components) goes to infinity at the resonant frequency $f=1/(2\pi\sqrt{LC})$. The impedance of a *series LC* goes to zero at that same resonant frequency. See §1.7.14.

Other important capacitor applications in this chapter (§1.7.16) include (a) *bypassing*, in which a capacitor's low impedance at signal frequencies suppresses unwanted signals, e.g., on a dc supply rail; (b) *blocking* (§1.7.1C), in which a highpass filter blocks dc, but passes all frequencies of interest (i.e., the breakpoint is chosen below all signal frequencies); (c) *timing* (§1.4.2D), in which an *RC* circuit (or a constant current into a capacitor) generates a sloping waveform used to create an oscillation or a timing interval; and (d) *energy storage* (§1.7.16B), in which a capacitor's stored charge $Q=CV$ smooths out the ripples in a dc power supply.

In later chapters we'll see some additional applications of capacitors: (e) *peak detection* and *sample-and-hold* (§§4.5.1 and 4.5.2), which capture the voltage peak or transient value of a waveform, and (f) the *integrator* (§4.2.6), which performs a mathematical integration of an input signal.

¶G. Loading; Thévenin Equivalent Circuit.

Connecting a load (e.g., a resistor) to the output of a circuit (a "signal source") causes the unloaded output voltage to drop; the amount of such *loading* depends on the load resistance, and the signal source's ability to drive it. The latter is usually expressed as the *equivalent source impedance* (or *Thévenin impedance*) of the signal. That is, the signal source is modeled as a perfect voltage source V_{sig} in series with a resistor R_{sig}. The output of the resistive voltage divider driven from an input voltage V_{in}, for example, is modeled as a voltage source $V_{sig}=V_{in}R_2/(R_1+R_2)$ in series with a resistance $R_{sig}=R_1R_2/(R_1+R_2)$ (which is just $R_1\|R_2$). So the output of a 1kΩ–1kΩ voltage divider driven by a 10 V battery looks like 5 V in series with 500 Ω.

Any combination of voltage sources, current sources, and resistors can be modeled perfectly by a single voltage source in series with a single resistor (its "Thévenin equivalent circuit"), or by a single current source in parallel with a single resistor (its "Norton equivalent circuit"); see Appendix D. The Thévenin equivalent source and resistance values are found from the open-circuit voltage and short-circuit current as $V_{Th}=V_{oc}$, $R_{Th}=V_{oc}/I_{sc}$; and for the Norton equivalent they are $I_N=I_{sc}$, $R_N=V_{oc}/I_{sc}$.

Because a load impedance forms a voltage divider with the signal's source impedance, it's usually desirable for the latter to be small compared with any anticipated load impedance (§1.2.5A). However, there are two exceptions: (a) a *current source* has a high source impedance (ideally infinite), and should drive a load of much lower impedance; and (b) signals of *high frequency* (or fast risetime), traveling through a length of cable, suffer reflections unless the load impedance equals the so-called "characteristic impedance" Z_0 of the cable (commonly 50 Ω), see Appendix H.

¶H. The Diode, a Nonlinear Component.

There are important two-terminal devices that are not linear, notably the *diode* (or *rectifier*), see §1.6. The ideal diode conducts in one direction only; it is a "one-way valve." The onset of conduction in real diodes is roughly at 0.5 V in the "forward" direction, and there is some small leakage current in the "reverse" direction, see Figure 1.55. Useful diode circuits include power-supply *rectification* (conversion of ac to dc, §1.6.2), signal rectification (§1.6.6A), *clamping* (signal limiting, §1.6.6C), and *gating* (§1.6.6B). Diodes are commonly used to prevent polarity reversal, as in Figure 1.84; and their exponential current versus applied voltage can be used to fashion circuits with logarithmic response (§1.6.6E).

Diodes specify a maximum safe reverse voltage, beyond which avalanche breakdown (an abrupt rise of current) occurs. You don't go there! But you can (and should) with a *zener diode* (§1.2.6A), for which a reverse breakdown voltage (in steps, going from about 3.3 V to 100 V or more) is specified. Zeners are used to establish a voltage within a circuit (Figure 1.16), or to limit a signal's swing.